

SMITHIAN FOLD THEORY - MATERIALS SCIENCE BRANCH PAPER 001

# From Fold to Materials

An Exact, Parameter-Free and Machine-Closed Reconstruction of Materials Science  
from Smithian Fold Theory

---

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

Version 1.2 - current-evidence closed, extension-open foundation

92 required laws - 23,552 exact candidates

27 July 2026

DOI: 10.5281/zenodo.21629306

Paper: CC BY 4.0 - Code: Apache-2.0

**PUBLISHED OPEN-ACCESS BRANCH PAPER**

# Abstract

This paper reports the Materials Science foundation of the third clean-room reconstruction of Smithian Fold Theory at its current-evidence-closed, extension-open boundary. The frozen successor inventory contains 92 obligations in ten ordered subbranches. Their exact grammars execute 23,552 candidates and 23,552 one-for-one decisions, yielding 92 unique survivors, 92 depth-independent closure certificates, 368 passing adverse controls, 92 implementation-distinct reconstructions and 92 post-seal NIST/BIPM correspondence checks. All 92 dependency graphs reach the single premise-free root theorem. The original 84-law surface and the subsequent eight-law V1/V2 atomic-reconciliation surface were each sealed in full before their external target identities were selected. Headline exact results include six simple-cubic neighbours; periodic rotation orders one, two, three, four and six with five excluded; seven crystal systems; fourteen Bravais classes; three acoustic displacement orientations; exact two-label aperiodic substitution without a positive rational fixed scale; shared acoustic and opposed optical basis modes; rank-three cube-count thermal support; p-n rectification without signed proof values; superconducting isotope-response provenance; ferrimagnetic unequal-sublattice residual support; integer and primary reduced odd-denominator Hall classes with even-denominator observations explicitly retained outside that primary hierarchy; exact positive-whole topological edge-class gaps; and a source-bound water bulk-property ledger including boiling, liquid/solid density and heat capacity.

## Results first: exact materials classifications

| Headline result                       | Exact result                                                                                                                     | Empirical and scientific meaning                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simple-cubic nearest adjacency        | 3 spatial generators x 2 held orientations = 6 neighbours                                                                        | The full adjacency count is forced without signed coordinates and agrees with the registered crystallographic source boundary.                                         |
| Crystallographic rotation restriction | Allowed positive orders are exactly 1, 2, 3, 4, 6; 5 is the least excluded order                                                 | A depth-independent factor certificate replaces trigonometric approximation; the standard crystallographic restriction is recovered after sealing.                     |
| Crystal systems                       | Exactly 7 rotation-compatible rank-three metric quotient classes                                                                 | The complete 3 x 4 length/angle grammar is enumerated and quotient compatibility retains seven.                                                                        |
| Bravais classes                       | Exactly 14 survivors from 7 x 5 = 35 system/centering candidates                                                                 | Basis and centering compatibility eliminate the other 21 candidates; the standard classification is a post-derivation correspondence.                                  |
| Acoustic branches                     | Exactly 3: one longitudinal and two transverse                                                                                   | A held propagation orientation consumes one of the three spatial orientations and leaves two independent cross-axis orientations.                                      |
| Complete branch evidence              | 92 laws, 23,552 candidates, 92 survivors, 368 adverse controls, 92 independent reconstructions and 92 post-seal NIST/BIPM checks | Structural results and specimen-dependent measurements are not mixed. Each external comparison preserves method, scale, condition, uncertainty and falsification rows. |

## Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Materials Science, an attractive specimen, vendor datasheet or selected favorable measurement cannot stand for a material law. Carrier, composition, phase, microstructure, preparation, direction, scale, method and condition remain in the observation record. Exact structural counts are stated as exact; variable magnitudes remain bound to their measured specimen rather than being promoted into fictitious universal constants.

Maria Smith retains copyright and scientific authorship. Papers and documentation are licensed CC BY 4.0 and code is licensed Apache-2.0, permitting inspection, copying, criticism, reproduction, modification and redistribution with attribution. “Ernos Labs” is a separate, revocable standards-conformance designation. A reuse or fork may use the open work under its licences, but may describe itself as Ernos Labs only while it preserves the public empirical constitution, complete adverse evidence, unchanged admission route, critical review and community standards.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com), submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

## 1. Publication, authorship and open-science boundary

**PUBLISHED OPEN-ACCESS BRANCH PAPER.** DOI: [10.5281/zenodo.21629306](https://doi.org/10.5281/zenodo.21629306).

Maria Smith, independent researcher and founder of Ernos Labs. Contact: [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com). Reproducibility reports and submissions: <https://discord.gg/ucwGryVxGr>. GitHub: <https://github.com/MettaMazza>.

The paper is released under CC BY 4.0 and code under Apache-2.0. Copyright preserves authorship while the licenses preserve open inspection, reuse, criticism and independent replication. The Ernos Labs designation identifies conformance to the published methodological and community standards; copying alone does not confer that designation.

## 2. Exact scope and meaning of closure

Closure means every obligation in the frozen Materials question surface has one engine-admitted theorem within its exact generated grammar, full dependency trace, adverse-control boundary and registered empirical protocol. It does not mean that every possible material, specimen, processing history or future observation has been enumerated. Biology beyond the material interface, clinical outcomes, Earth-system fate and application engineering remain later or external validation domains.

## 3. Constitutional arithmetic and prohibited premises

Structural absence is empty One, never numerical zero. Counts are generated positive wholes; exact parts are positive held fractions. Opposition and direction are held fibres, never negative proof quantities. Irrational and imaginary proof values, floating proof arithmetic, a completed infinity, ungenerated continuum, axioms, free or fitted parameters, imported constitutive equations, pretrained models, consensus selection, target-derived rules and application results are prohibited. Host counters can enumerate artifacts but cannot act as scientific values.

## 4. Dependency spine to the root theorem

Each Materials registration has `root_theorems=(SFT-ROOT-THERE-IS-NO-NOTHING, )`, zero axioms and zero free parameters. Dependencies are accepted only when their immutable model-admission receipts already exist. A separate graph traversal over the materialized registrations confirms 92 of 92 paths reach the root. The path runs through the independently admitted Foundation, Mathematics, Information Science, Physics and Chemistry receipts named in each claim section; no branch name substitutes for an actual dependency identity.

## 5. Complete pre-source seal

Before NIST or BIPM source identities were chosen, V3 froze the original 84 obligations, exact counting module and target-blind blueprints. That seal records 21,504 candidates and remains `sha256:da97a6cb6a001964a069b45a5a3698e7ea90f334a08d69c62bd09c46d8112035`. The later atomic V1/V2 audit exposed eight nonduplicative Materials-owned omissions. Before selecting any new source identity, V3 froze all eight successor obligations together, 2,048 candidates and their complete prediction set under `sha256:0e8d7f14a7389b7ec44a37205ce2c9074db65f7b6ed5a466b97f2a01418ef331`. Both seals expressly record that source identities were unselected and target content unopened.

## 6. Single fail-closed engine

For each claim, the engine verifies registration, source identity, admitted dependencies, exact candidate identity and cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four adverse controls and implementation-distinct reconstruction. Empirical evidence additionally requires an evaluator-verified derivation seal, capability isolation, target absence before seal, custody release only after a matching prediction seal, all-row retention, an explicit falsification condition and a tampered unfavorable control. A violation halts and preserves a rejection receipt; it cannot enter the model census.

## 7. Empirical method

Forty-two registered byte-frozen records from NIST and BIPM/JCGM supply the independent external boundary across the original and successor seals. Each claim requires at least two claim-specific discriminators. The predictor cannot read filesystem, network, clock, environment, subprocess, dynamic import or foreign functions. A distinct custodian opens source content only after sealing, verifies every snapshot hash and required feature, constructs the registered observation record and releases it through the portable custody exchange. External authority may falsify a correspondence but cannot alter the already sealed grammar or survivor.

This branch distinguishes exact structural counts from specimen-dependent magnitudes. Six neighbours, allowed rotation orders, seven systems, fourteen Bravais classes and three acoustic branches are exact count/classification claims. Elastic moduli, conductivities, strengths, transition temperatures and analogous quantities depend on material identity, preparation, direction, scale, method and condition; the Fold laws force those dependencies and ledgers, not one fictitious number for every material.

## 8. Exact structural count certificates

### 8.1 Six simple-cubic neighbours

Three independently admitted stable spatial generators each carry exactly two held Fold orientations. Their literal product contains six distinct nearest-adjacency positions. No signed coordinate or numerical zero is needed.

### 8.2 Rotation restriction

The rank-two integral rotation boundary requires Euler-totient degree no greater than two. The factor certificate proves depth independence: if odd prime  $p$  divides an order,  $p-1$  divides its degree, leaving only odd prime three; the remaining powers satisfy  $a \leq 2$ ,  $b \leq 1$  and exclude simultaneous  $a=2$ ,  $b=1$ . Orders one, two, three, four and six survive; five is the least excluded positive order.

### 8.3 Seven crystal systems and fourteen Bravais classes

The rank-three metric grammar enumerates three length-equivalence classes times four angle-equivalence classes and retains seven rotation-compatible quotient classes. Crossing those seven classes with five centering forms yields thirty-five candidates; basis and symmetry compatibility retain fourteen. The complete named survivors and exact claim receipts appear in their sections below.

### 8.4 Three acoustic branches

For one held propagation orientation in rank three, displacement has one propagation-aligned orientation and two independent transverse orientations. The complete displacement census therefore contains three acoustic branches.

## 9. V1/V2 reconciliation without answer import

The V2 corpus is used only as a question-level completeness obligation. Steps 47, 49, 52, 54, 72, 74, 75, 133, 137, 143, 193 and the Materials portion of 291 are mapped to newly executed V3 claims. None of their prior values, equations or certificates is exposed to the V3 derivation. Step 291's planetary and Tully-Fisher content is explicitly routed to Astronomy and Cosmology rather than hidden inside Materials closure.

## 10. Reading the exhaustive ledger

Every claim section below records the theorem, exact dependency identities, grammar and boundary, all eight binary axes, the 256-candidate census, sole survivor, base/successor certificate, witnesses, meaning, controls, independent implementation, post-seal source custody, falsifier and immutable evidence identities. The machine-readable JSON remains authoritative and is bundled with this paper.

## 11. Measurement Identity

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

### 1. Material system and retained identity

Claim identity: SFT-MAT-MEAS-MATERIAL-001

**Question and exact theorem.** A material is the least complete constituent-and-relation carrier whose composition, bonding organization and declared observation boundary remain distinguishable.

complete-constituent-carrier\_\_composition-and-bonding-retained\_\_material-boundary-held\_\_declared-scale-identity\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-constituent-carrier             | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | composition-and-bonding-retained         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | material-boundary-held                   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | declared-scale-identity                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-constituent-carrier\_\_composition-and-bonding-retained\_\_material-boundary-held\_\_declared-scale-identity\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:5fd1f0e5902535142cb31d865d5ea64e83b9e08cd7f630cac919c3a01e84d840.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:0bbc08685cb1726579e93191fb6f6bde56681445d2d229aa315724a6eb6ade94.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:a401a612b8a1d870368aa2d6a487a31c012370a228b62c70915c9c6914706a23.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:b2b3e2c6051ce0f6601c2466140f40ceeda0b4cb823d5e91c18daebcf3c5fa50.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:9a39682a714c1e73b5df88972c201028726440e1bf460169e557a45de1513cd5; certificate  
sha256:bbe90ba7d0b746808b9a3dbe8e33c1c7ef995b34f1db2b994dcc8810c85f6ce; external-validation  
hash sha256:f5880932fc39e96a61053ce252e6ffa011365bb09e677042321c321c9247ef32.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:683ba4d17cac39004f018001749fe7c52b9a376f80659e6bca9ecdabdebf44cf. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-REFERENCE-MATERIALS-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/reference-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-reference-materials.html;  
sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; scope: material  
identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- **sft-mat-meas-material-001-authority-correspondence**: predicted complete-constituent-carrier\_\_composition-and-bonding-retained\_\_material-boundary-held\_\_declared-scale-identity\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-constituent-carrier\_\_composition-and-bonding-retained\_\_material-boundary-held\_\_declared-scale-identity\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material system and retained identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule



- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3e6c63ac46ba6fa3a33796b77a14d8e56af8461367e178d8e4f7133e5691f15a; derivation seal sha256:03747f2235206b6276b4df2c863ec64edb6581fe3f24c6cd674312492061e806; engine receipt sha256:100e32f861c1188720ddb10a07f5fa28b9a94d7650c859382865519fb64e1be6 at receipts/engine/model\_admitted/SFT-MAT-MEAS-MATERIAL-001-100e32f861c11887.json; empirical validation sha256:02058a82c7880fbce8194cc5b4834494351c29d565fe93525deabe152b873680; measurement receipt sha256:179da930e46c4b46067ebf11bd97d1a18acda644cb9cc6930c57c5e6e178f935; isolation sha256:88f0cd6f457584283c4106d0765538acfd07856df2cdba70cd8d43e5127c8da6; custody sha256:9586bdfec7797dd283c176a2c49ac47e128f15b36d4c3a52b4a31b1011477fa2.

## 2. Specimen and sampling relation

Claim identity: SFT-MAT-MEAS-SPECIMEN-001

**Question and exact theorem.** A specimen is a source-bound finite part of a material with an explicit sampling map, retained preparation history and declared representativeness boundary.

source-bound-specimen-carrier\_\_specimen-to-material-provenance\_\_sampling-boundary-retained\_\_representativeness-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.



Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-bound-specimen-carrier            | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | specimen-to-material-provenance          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | sampling-boundary-retained               | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | representativeness-declared              | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: source-bound-specimen-carrier\_\_specimen-to-material-provenance\_\_sampling-boundary-retained\_\_representativeness-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 9c35918d4fd63e38482a40eb699e350a2ee519aa20cf3e4da387a1c8afe55fd9.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 88c6c3fff183744cea93e433facfb571d89ebed74760801f5f079da131b8e6e3.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 63c1304a41770cfadab42af9fb67769ed75aaea6637ccef8bc2320976630e09b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 90dbb259614c0b3e1e2f7c8918b80aaf091a937d31290d4872e84c9b43ee85ee.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : f58c2ea33d59fda86da1273df9a7eda41e54ca393804aa6349eb3f8e3140be34; certificate  
sha256 : 23109cf272ed2894bb8ede6176ff82975653df8ff6a282aa3d2b31fedf0bdcad; external-validation  
hash sha256 : ea630c1ae5ba9b7c2b8b409f95a58ea69bcd82f8ea2dba1e62b98b4d681311aa.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 3a695fe79bc147ca2cae3cc48766472922fb8ad57766bc9f0d92fd979e449ac0. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TEXTURE - PHASE - FRACTION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot  
experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html;  
sha256 : b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase  
fractions, texture, deformation and uncertainty.

Comparison and custody records:

- **sft-mat-meas-specimen-001-authority-correspondence:** predicted source-bound-specimen-carrier\_\_specimen-to-material-provenance\_\_sampling-boundary-retained\_\_representativeness-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived source-bound-specimen-carrier\_\_specimen-to-material-provenance\_\_sampling-boundary-retained\_\_representativeness-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered specimen and sampling relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:eb6cb5d9d6699b6fa8c5d5a1b80d08a30e5340ee165f372adb13a378fc1c4c9d; derivation seal sha256:de5d12aa6b9d38df4dd41e838be989f5efdaa3c54b549b8eda98543e04d1c829; engine receipt sha256:64d737e3f4c0fc5a33a13a5af86282e4e74420b103aba81e067d3baa1b55bbac at receipts/engine/model\_admitted/SFT-MAT-MEAS-SPECIMEN-001-64d737e3f4c0fc5a.json; empirical validation sha256:fc00c8fadea60629d2db8350091da01088f9de13941e22ca9288749e6c0cb1bd; measurement receipt sha256:8b2eda391cb0daa95d3a916046d65defafa288867eda601ac9119b008d44878f; isolation sha256:6582228d88ed2f11aece7eb4830965c3a7cd5b5f166c50c7eaba4e4280125636; custody sha256:19398416e13e841a09e8a15c06fb484d7eb5911ea03afcef1ef03a01cd5e28aa.

### 3. Material composition record

Claim identity: SFT-MAT-MEAS-COMPOSITION-001

**Question and exact theorem.** Material composition is the exact part-to-whole record of every retained component on one declared amount or support basis.

complete-component-support\_\_exact-part-to-whole-composition\_\_component-identities-retained\_\_basis-and-uncertainty-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and

basis-and-uncertainty-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-component-support               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | exact-part-to-whole-composition          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | component-identities-retained            | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | basis-and-uncertainty-declared           | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:1495e5970a1ecdcd9d4372587b7b84cd29df2e68eb10bc9924511e2258d147b.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8a163f92ab47dd17cee0a3c65a94c64a6eafced530cc6453a2784f2177cd17b6.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:09fd0a2fc18530f0598ae2dcbf496100a032e91d27f614edab80f0a72aac8539.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:ee80dc3cb72bf4024bf186ddbc6b8334e46b57e03c145399957f2b9d82451f92.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:1c370302c7b1df34e6027d61af197f02a43ec583c4b066fe47287bca0deddddfc; certificate  
sha256:ab5e53b673a0f24c340e78feed5d77ad4fdcc972a291abcc90cd54195a54b571; external-validation  
hash sha256:d7de9d562123cd8b519c68f77809dc68f4ce4fbd53724a2b4f4ed888b238f21.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:ed1f6ea00fbb6f3c4efccac54ff08f15e6cf8eb6a86628d65ea245838980eb27. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MATERIALS-DIVISION-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot  
experiments/external\_sources/materials/snapshots/nist-materials-division.html;  
sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; scope: materials  
measurement, composition, interfaces, processing, damage, classes and lifecycle.

Comparison and custody records:

- **sft-mat-meas-composition-001-authority-correspondence:** predicted complete-component-support\_\_exact-part-to-whole-composition\_\_component-identities-retained\_\_basis-and-uncertainty-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-component-support\_\_exact-part-to-whole-composition\_\_component-identities-retained\_\_basis-and-uncertainty-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material composition record observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and basis-and-uncertainty-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:cdcbc439c4cfe1e162e40a6b8bc72068658e552c4e4a03c16c6984d5dd8f8340; derivation seal sha256:5767520188e33d726e0ef14871b31adc74a3418ab648a59016e22946abc54465; engine receipt sha256:b172a8dfe74dc3bb3d5159178251f4b5a4c6de311ac14a084fe99ba9f87bf1a6 at receipts/engine/model\_admitted/SFT-MAT-MEAS-COMPOSITION-001-b172a8dfe74dc3bb.json; empirical validation sha256:ff62f3576785762465d2a17af0abc019afb29cf2e5e89f883176c02f23eb5f1; measurement receipt sha256:6879a614933015d04fd3da7337ddce7118ae2db28bb23b7ed439645af4394a62; isolation sha256:a133e07a5e3cb3a00fa8d464abe63e91a2c1dc18f20be9c5fd070e6cc73ef503; custody sha256:8de12da81639f5178c647e8da843491a125d1a038e8a92f7c88882f323409f6a.

## 4. Material phase identity

Claim identity: SFT-MAT-MEAS-PHASE-001

**Question and exact theorem.** A material phase is a maximal connected support whose constituents share one stable recurrence and symmetry observation class at declared conditions.

complete-phase-support\_\_shared-recurrence-equivalence\_\_phase-boundary-retained\_\_condition-bounded-phase-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.



The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-phase-support                   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | shared-recurrence-equivalence            | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | phase-boundary-retained                  | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | condition-bounded-phase-class            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt



sha256:1ac7de1af45e524b8c9b0da7b3f8f877fba4a1b78e7a2f01c44861bb0d2901f3.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:77b4f455bc737b32694908d129af28b0479a19481f4c990307d06cbc4a071c00.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:25fc762e494c9db8e616dca4b2a87b825f2d18221bb4b9de422f5e343895e0af.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8fac8c656448533cc88dc6c3a67e5b772dd899d27934e94ed86ceda602fa3351.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:dbb3ede679d2df917b3b1d3925870abcd640c7ac5b6bef94da3b36f5ad31bb09; certificate

sha256:9b3957f05da97dff4942b040c1abb90b95d8a5e7bc4a8d7656b4ccbf48f3c04; external-validation hash sha256:60dc5f40aab324386373a4beebe4ed741c3e27f21624dfe65830c09f7f607c2a.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:14b6f6f9b39bcb2995bc42d4164785b635c0a872b167d05174462ef7f62a4a06. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- **sft-mat-meas-phase-001-authority-correspondence:** predicted complete-phase-support\_\_shared-recurrence-equivalence\_\_phase-boundary-retained\_\_condition-bounded-phase-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-phase-support\_\_shared-recurrence-equivalence\_\_phase-boundary-retained\_\_condition-bounded-phase-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material phase identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:8a93bd0b5a0225fb750f1e4251ae057e9ba3c649e1e767ba04f9b9ae00b3da99; derivation seal sha256:ad68cff33969540719db7d2b44560a371ddaa5bd4de6645407eaff0e0e21db79; engine receipt sha256:820c18b5745c2eb4d0f4dd9fb15f1c243d4f34d2c1c7b87d421316ff4426f1ee at receipts/engine/model\_admitted/SFT-MAT-MEAS-PHASE-001-820c18b5745c2eb4.json; empirical validation sha256:3217c3fcbef32eaab16508631f6a9200e0536e889546fb08ca0b278e7d225a9; measurement receipt sha256:40605d06c8c56d5d1cf3d24e347e2f925be621d0d0b1755d25c887234d0af836; isolation sha256:7f5971803416f24c7c627e35f7ad99fc03e6b838442f461e642f65437aea1736; custody sha256:68aa9c9ad49548681df4ac2ca134e5cae64aa76b8e217219cb3affd796e11a10.

## 5. Microstructure as mesoscale organization

Claim identity: SFT-MAT-MEAS-MICROSTRUCTURE-001

**Question and exact theorem.** Microstructure is the complete declared-scale arrangement of phases, grains, interfaces and defects between constituent and bulk observation levels.

complete-mesoscale-carrier\_\_constituent-arrangement-relation\_\_interfaces-and-defects-retained\_\_resolution-bounded-record\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-mesoscale-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | constituent-arrangement-relation         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | interfaces-and-defects-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | resolution-bounded-record                | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-mesoscale-carrier\_\_constituent-arrangement-t-relation\_\_interfaces-and-defects-retained\_\_resolution-bounded-record\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 4d9da1817be3190c573b9b4258210ab512e0d2c2e6fbc8e28e954970f97378ae.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f9e6509348500baddf4417a261a58945012068b796a21f30110cb7079c17d2ce.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: e7c3258831f4ada06f194a610d6b18187859f9ddb3d850ccf651c1241220632c.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 124c0a150b4c3c2041ea25e07f9a87131a8ee452c614b442f8bf200cf41b0fd9.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 5e3d4ac0287c0720de1b47e900c43f267b74a387e9ca2f75328eb09130bd251f; certificate

sha256: f076e7ddf7a8650c7b13642e5bae9cb217f53a23479887a911e2b00883c47195; external-validation

hash sha256: ee24cfe1b083142c03c82b0c868bf8c5112305f35839a1fff8a5f4204e76f52d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: 53a34e89965e7c97cfc0780ecddb9937112caa7dc849483c429122346ce0f93. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256: 2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- **sft-mat-meas-microstructure-001-authority-correspondence**: predicted complete-mesoscale-carrier\_\_constituent-arrangement-relation\_\_interfaces-and-defects-retained\_\_resolution-bounded-record\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-mesoscale-carrier\_\_constituent-arrangement-relation\_\_interfaces-and-defects-retained\_\_resolution-bounded-record\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered microstructure as mesoscale organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest

sha256:8e536fdd3d154712fc1b447a39f6c62acaa780b4680effe34a6c5b81c3de1e3c; derivation seal  
 sha256:86d7179cfce073c0669747b856b118570b7f56bdb77f9049719a61d9479239b9; engine receipt  
 sha256:2bd2dd39260bb2c978e97390f519d99570ec14a1ed71f2438040642a7a39fb2a at  
 receipts/engine/model\_admitted/SFT-MAT-MEAS-MICROSTRUCTURE-001-2bd2dd39260bb2c9.json;  
 empirical validation sha256:d1d475bce79364060bd227a00440231974a97b9a698adcb5aa593114f145c1fe;  
 measurement receipt sha256:687767a7a15260acce4b221cf3dcae4c7de94a93c6de310d64d9ea0763e65430;  
 isolation sha256:86f3233608ea9a812c9e9b5f8ceee097f36d34afd46309e024c52f2c0f659472; custody  
 sha256:51b87a705394f2363ee5beb0436787fdfaa257ee17792af99a438512bfb55e70.

## 6. Material property relation

Claim identity: SFT-MAT-MEAS-PROPERTY-001

**Question and exact theorem.** A material property is a reproducible relation between a declared material state and stimulus/response records, conditional on direction, scale, history and method.

material-state-carrier\_\_stimulus-response-relation\_\_condition-and-direction-retained\_\_method-bounded-property\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-state-carrier                   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | stimulus-response-relation               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | condition-and-direction-retained         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | method-bounded-property                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: material-state-carrier\_\_stimulus-response-relation\_\_condition-and-direction-retained\_\_method-bounded-property\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 9d5f10ef32de296ff8fd844e2d412fc5a4fa61e6b436547719a742ceae9254f6.



- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: eee60e6d8c42048b68a43ec3d3460d5ab672fe4c623828350fe25db95e1e6b0b.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 5ba5573122548c6b1fbc93f566e3201eeb7d0a7862100fded5a2eb25342d30d5.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 7a63bce6b4e5bcf0be1bec3e56ba88f15183e47d97e62c5be2ba087f4d9c2c66.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: c102a26be454aff89e7e294cb42102b49138996753da2059894e887ce197b1fd; certificate

sha256: a8cb28c31491b1f1a5a7f43ee96c82ef56e8f5ebf213bef1a308b35284d49599; external-validation

hash sha256: 8d89cd0128a033bc11d35c7516a69ed90d3575999a75632bee9b7aed56a4b3bd.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: 80cd9c287b41551e990818ae07a98fdf4cdd4528a814d66ff7bdd8f8ae2c4a0f. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-REFERENCE-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/reference-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-reference-materials.html; sha256: 2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; scope: material identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- **sft-mat-meas-property-001-authority-correspondence**: predicted material-state-carrier\_\_stimulus-response-relation\_\_condition-and-direction-retained\_\_method-bounded-property\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-state-carrier\_\_stimulus-response-relation\_\_condition-and-direction-retained\_\_method-bounded-property\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material property relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest



sha256:80fd697248d6321e6b90d9794452f6b9fd387ea961db765471df7548a94761d7; derivation seal  
 sha256:1d1f8fd9378247d9df92613f2b5fec6cb4c9c3909dc46815a7f416d4db2bdc; engine receipt  
 sha256:d13a1540d3b9d11dd631f188e97f4096b58dcc42f19ad0dae0b4d7ad1eb60857 at  
 receipts/engine/model\_admitted/SFT-MAT-MEAS-PROPERTY-001-d13a1540d3b9d11d.json; empirical  
 validation sha256:13e08bdc1143aa9d50e2532125113b4afabc13db1c34e8d4079f2f91bb00e27c;  
 measurement receipt sha256:7b6727c92a907c19971097f1e19c34a2064840a58b1892285497d238cfff014;  
 isolation sha256:fadc55d82e80bdb6922f6cf2b8e4dc9e0b233a27f6e4324a1db1b348cba2c95d; custody  
 sha256:a75d0c32265d651a4565cb6f302dae96060190de2899773c0b4fcb191e5981ee.

## 7. Materials uncertainty and metrological traceability

Claim identity: SFT-MAT-MEAS-TRACEABILITY-001

**Question and exact theorem.** A materials measurement is admissible only with a complete result, uncertainty, method, specimen, condition, calibration and unbroken reference chain.

complete-measurement-record\_\_result-to-reference-chain\_\_uncertainty-components-r  
 etained\_\_calibration-and-condition-bounded\_\_complete-state-transition-boundary-t  
 race\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-ru  
 le

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-SUBSTANCE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-measurement-record              | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | result-to-reference-chain                | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | uncertainty-components-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | calibration-and-condition-bounded        | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-measurement-record\_\_result-to-reference-chain\_\_uncertainty-components-retained\_\_calibration-and-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 06fa01f03b954ef5285336e526a90aed8ebffbc55a443245ef359e1ca0562e4.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:00c29832548d03a45275175072a6636fe6db09d58277db2f6b593eb666bb770b.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:664397bfce016fac6e18934501599f9cd4d638f486cf68fccd6a6e85a98e4e09.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:816bbf074bc4def5e8a11de819f574d9f945a4197e01ebce28f46f3b6c1fa869.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:f70bd0aa9ed0e1a4e81927798902ca76757e566963e1a41c71ec14bf9b9f06d8; certificate

sha256:93eb0026b754f4235aed15c8d33ccada191ea0ddfe7879abed1bee7aa0b05f57; external-validation

hash sha256:67f1a8e737f8f5da46378dcebd60f9ee9adce75c99f638dc9252cc1ee114c5b4.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:da9c91f2ca355e816db0feb24791e6b9e2f07fd54d1ae885376692c784507db3. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **BIPM - JCGM - VIM - TRACEABILITY - 2026-07-24** - Joint Committee for Guides in Metrology / Bureau International des Poids et Mesures; <https://jcgm.bipm.org/vim/en/2.41.html>; snapshot experiments/external\_sources/materials/snapshots/bipm-vim-traceability.html; sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4; scope: measurement result, reference chain, calibration and uncertainty.

Comparison and custody records:

- **sft-mat-meas-traceability-001-authority-correspondence**: predicted complete-measurement-record\_\_result-to-reference-chain\_\_uncertainty-components-retained\_\_calibration-and-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-measurement-record\_\_result-to-reference-chain\_\_uncertainty-components-retained\_\_calibration-and-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered materials uncertainty and metrological traceability observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement\_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:0331cf713780ef603d89ab128524abbac7f2778801318dd0b8fd437d65b9ca0e; derivation seal sha256:827cfa9bc16633bc9e45d63d74209619bcfd40942f632c5bcd6d0cf83da8098; engine receipt sha256:b0bf205091f5c263309fad96e87e51572c4a3f60f40d61263205778c4509f299 at receipts/engine/model\_admitted/SFT-MAT-MEAS-TRACEABILITY-001-b0bf205091f5c263.json; empirical validation sha256:7929a09e55e51aa0aa0d30d435bfa2a2755c8f23d5951673931c13c94d98e889; measurement receipt sha256:075d6446b70e416f2a1961f712d1b26ac009326b0eec2664ac4e1edb2398ad06; isolation sha256:e79a2ff20201393fcff90b3a3fa417752ae743648e9b0cc3f7eb73989252a169; custody sha256:16ebf8cce951c7bf7d037eaa5be0825ab8703ad75184e4291b569d394fb4dff3.

## 12. Crystal Quasicrystal

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

### 8. Generated lattice organization

Claim identity: SFT-MAT-CRYST-LATTICE-001

**Question and exact theorem.** A lattice is a completely generated site-and-adjacency network whose local word recurs under retained translations.

complete-site-carrier\_\_translation-recurrence-relation\_\_adjacency-word-repeated\_\_finite-support-lattice-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-site-carrier                    | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | translation-recurrence-relation          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | adjacency-word-repeated                  | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | finite-support-lattice-class             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-site-carrier\_\_translation-recurrence-relation\_\_adjacency-word-repeated\_\_finite-support-lattice-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : c9646ff6ef73ed550d9dd57d3bd3376e1f2b6d1505670ff5b473a2ecc1f16dc5.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 4b1c36d0e527ba983cfb859c1ee7030c18d678f7766b0b00850691b28a340ce7.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : d9773d30b11ed06149d52d2aaade731e6df807698242f853bb43b56cad410efe.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 3fffc29ee144f3e4f558b5a20eed2e6c152a3d985a503c009f484019144df44f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 45dc456bf89de22438417c63625a628d65c2cfed0f0a944c50fb3e1b2f26e5cb; certificate  
sha256 : 4daea4fea105991ed38bba1541939504c775085e2bc923e88d0c73fbdc4374dc; external-validation  
hash sha256 : b25db7f0facb055cb4017842334dd5509e5a9064e847f2b0cafb42796cf683ec.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 79e502524e622bf1c982a0b73cbb777eb604a59fdc448a82abdc3696c033cabf. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - WULFFMAN - CRYSTALLOGRAPHY - 2026 - 07 - 24** - National Institute of Standards and Technology;  
[https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot  
experiments/external\_sources/materials/snapshots/nist-wulffman-crystallography.html;  
sha256 : 52558a4ed54b8ecc464b0ba83d879a0cbbf7ed25385f7742d021e87b43cb6; scope: lattice  
generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- **sft-mat-cryst-lattice-001-authority-correspondence**: predicted complete-site-carrier\_\_translation-recurrence-relation\_\_adjacency-word-repeated\_\_finite-support-lattice-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-site-carrier\_\_translation-recurrence-relation\_\_adjacency-word-repeated\_\_finite-support-lattice-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered generated lattice organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:953d8aba886fe20066df18e9a13f9d6984043da3ff9379cd8346719abac8a594; derivation seal sha256:8de7be19a12b78faf87780e5d8b279fb8c8e7ba7e9a087d8adb7f85a14bde14a; engine receipt sha256:29a30c7d21227319c53ac130261c6ef9c7619d92e841bef00c3bc3f2f8813209 at receipts/engine/model\_admitted/SFT-MAT-CRYST-LATTICE-001-29a30c7d21227319.json; empirical validation sha256:53c0b57c7770e4ba2e65a1ded9835b42fcf0fa452df42f4323d3f4cabac1b40b; measurement receipt sha256:5f15d36c9fc8bcd15bcaddbcaa0a7f8868ebf2f9bcee0598548c35f646f00bd3; isolation sha256:76de27215d27a3fdf1c148dce465d257b19dc89c1c9cae51905839acfd2d905b; custody sha256:ffa931b5491e2c28acb540ecd2204c2262a022b75ecf8e633df32ad8b3791436.

## 9. Primitive unit-cell identity

Claim identity: SFT-MAT-CRYST-UNIT-CELL-001

**Question and exact theorem.** A primitive unit cell is the least complete labelled cell whose generated translations reproduce every lattice site and relation once.

minimal-cell-carrier\_\_translation-generates-whole\_\_no-redundant-site-copy\_\_basis-choice-canonicalized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.



**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | minimal-cell-carrier                     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | translation-generates-whole              | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | no-redundant-site-copy                   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | basis-choice-canonicalized               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 95bee5e1bc790838af7571ea763454bb76b0b4cfd29afc969b46e5b894b1773a.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 655991b393c1177811a27031f45194e75585bd304199e7024f505539a7895dd8.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 52f038e5dccc305e1b474d16809934ad563804788d0130f0902e8985630f440e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: a0d6eaaecef72a08b0d991621e1f772a0c5b2da721de6f518654924f3b606102.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: 0154a35feeac91950959dc4366d8f1f7fbc32bdd85bba4a14373ce7828cc2de1; certificate  
sha256: 07c7eb750e1bfdc7075611d7232bff0c62de4bedbe700286e4d62936adec502a; external-validation  
hash sha256: 85798ec472c45e4f89fdcf19152f5e5835f6dbbbd8af31fdda44cfc5c1ec8d11.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: 7df3d5e1106dd9be2853c3ceff4a96c592603507e4909374b63428425490fbd0. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - WULFFMAN - CRYSTALLOGRAPHY - 2026 - 07 - 24 -** National Institute of Standards and Technology;  
[https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot  
[experiments/external\\_sources/materials/snapshots/nist-wulffman-crystallography.html](https://www.ctcms.nist.gov/wulffman/docs_1.2/experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html);  
sha256: 52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice  
generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- **sft-mat-cryst-unit-cell-001-authority-correspondence:** predicted minimal-cell-carrier\_\_translation-generates-whole\_\_no-redundant-site-copy\_\_basis-choice-canonicalized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived minimal-cell-carrier\_\_translation-generates-whole\_\_no-redundant-site-copy\_\_basis-choice-canonicalized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered primitive unit-cell identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:0ee1191c7afc84e64e0772d8b89185e5f90e00eae010409b4705ada433fe51f1; derivation seal sha256:1d98de21f5d7e99681d01977448efd664603dde6233b1a0a289b6339db5dae3c; engine receipt sha256:062606db8c6509ec2a58c31f865d99c78fe46b8a407e9a0bf823b7abbc04e868 at receipts/engine/model\_admitted/SFT-MAT-CRYST-UNIT-CELL-001-062606db8c6509ec.json; empirical validation sha256:0f5045547d470ffff7c7f7a76b331bcc87321b131e410d01dfedc202fef8b7fd; measurement receipt sha256:370d8291daf74de34741f463bb14759e3ae77c4e70542c12bb03fce0e4694a0c; isolation sha256:b54849491c03c4b7826fcc79b350bbff9e6969437a8a7f6ebd296f1e84f0de55; custody sha256:6b4d5550b99b1ece2327e422d03af3ab252fd537e9bcd3b86868b07fdce0cfb7.

## 10. Crystalline translational order

Claim identity: SFT-MAT-CRYST-TRANSLATION-001

**Question and exact theorem.** Crystalline translational order is invariance of the complete local constituent word and adjacency class under a generated integer translation basis.

complete-periodic-support\_\_integer-translation-action\_\_local-word-and-adjacency-invariant\_\_observation-depth-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

**Boundary:** Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-periodic-support                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | integer-translation-action               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | local-word-and-adjacency-invariant       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | observation-depth-declared               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-periodic-support\_\_integer-translation-action\_\_local-word-and-adjacency-invariant\_\_observation-depth-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

**Base:** One complete material carrier retains its generated relation, organization, observation boundary and proof record.

**Successor:** Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

**Closure scope:** depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:2dc5baa4e9c0b21b287632d0b7879ad79b6a20ac0f10e9043a7a9fc7c91eeaec.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:699c736da5c750925561e80681e07c690f72b1d453701d89b034b39beec00b55.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:c59d80682ced4dc231ab5a5e2eee49f48e53e6d78753baad047450f99ab1a36b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:f7bde3998b13bbb464ee34cba1ecf270b66e6a5fe4a30c001e0b4b0784a4e4b7.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:cf1267c3e93cd30aadb138996e6f53303d50d7bb730c304a24acfd22e2b4d913; certificate  
sha256:31470cfc67cc99e83e12e9f967814d55ed8aeb6725c114e1be33cd919c8ee3a9; external-validation  
hash sha256:0232399b0b20900fae5f945a17c32ff14a146310471f3d8e1013d98dc9fdd112.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:9fc8e3d73650812afd6ca2ff83c2faaf5fedd96177422de98734a64c4236f2aa. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - WULFFMAN - CRYSTALLOGRAPHY - 2026 - 07 - 24** - National Institute of Standards and Technology;  
[https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot  
experiments/external\_sources/materials/snapshots/nist-wulffman-crystallography.html;  
sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice  
generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- **sft-mat-cryst-translation-001-authority-correspondence:** predicted complete-periodic-support\_\_integer-translation-action\_\_local-word-and-adjacency-invariant\_\_observation-depth-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-periodic-support\_\_integer-translation-action\_\_local-word-and-adjacency-invariant\_\_observation-depth-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered crystalline translational order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:82c21d5ad24be00ddd2907d027deee7e0332eef8d95834009a8623769d1e6667; derivation seal sha256:ed8de31a2f6de9736c9f8dde2f1add2bd74b1cd305d41a3f2252e2f2b3e3f759; engine receipt sha256:bf36a3534aab9194516ce002531273c8bd47093f8db5304d723bcb5179c8af3e at receipts/engine/model\_admitted/SFT-MAT-CRYST-TRANSLATION-001-bf36a3534aab9194.json; empirical validation sha256:d5820432483ce1f6db98ac5243feb7d9739305010874fb8312bedf5b1efa5fb2; measurement receipt sha256:d863a33e6f7b1f9df5b48c1ca7be3000834fe51921330c7db33edd8a06780fed; isolation sha256:4a2fcc8f791a7da72410ba1c5ab79d7ea3b06949abc307bcb34e0032e28236e3; custody sha256:31b05b66c269dfbe59d8109ccf02e598b00d239c8876be66387f24d6d6765ad0.

## 11. Simple-cubic nearest-neighbour coordination

Claim identity: SFT-MAT-CRYST-CUBIC-COORDINATION-001

**Question and exact theorem.** Three stable spatial generators with two held orientations per generator force six nearest neighbours for the simple-cubic adjacency class.

three-axis-site-carrier\_\_two-orientations-per-axis\_\_nearest-adjacency-complete\_\_coordination-count-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.



**Boundary:** Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | three-axis-site-carrier                  | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | two-orientations-per-axis                | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | nearest-adjacency-complete               | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | coordination-count-six                   | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: three-axis-site-carrier\_\_two-orientations-per-axis\_\_nearest-adjacency-complete\_\_coordination-count-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

**Base:** One complete material carrier retains its generated relation, organization, observation boundary and proof record.

**Successor:** Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

**Closure scope:** depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.
- **coordination-six:** three directions times two held orientations generate six distinct neighbours; passed **true**.

**Detailed mathematical and physical meaning.** The exact product is three independently admitted spatial generators times the two Fold fibre orientations. The six generated neighbours are: axis-one/first-fibre, axis-one/second-fibre, axis-two/first-fibre, axis-two/second-fibre, axis-three/first-fibre, axis-three/second-fibre. No coordinate subtraction is used; opposed directions are held labels.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.



**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : a557cd3bae435643b540f71a220d86dead2a0af33a4b9db33127a736db1746d4.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : ffd89973296587921dcf25a21228bea827ef70e3557d3b263e39c5b3c5ecf3d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 8836eb5156029bb0d05e87199dbd03ea5ca0cc32842231958a80aca8dd63da1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 559485317a0696c17c8bee2f0a302354a5738b655be68072aa8f84e6516ff1ea.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256 : 20be4530b1fbc3b79cce20b0c8a94ff0794a8ac42dc8d633c2228dd92dbe0d82; certificate

sha256 : 7705717fd0f2eb7831c6a4f828389f697c3b1c0a157da3d9d59907a04de53b48; external-validation

hash sha256 : 882c896ef2f8a38ebf582d61bf71dd229923e81adfc46d9474398cb812a2caa3.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256 : fa547f7fa1727d64e737ac3f8ebc4a041bd00f93f18621aebd87ac98e47da155. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SIX - NEIGHBOUR - MESH - 2026 - 07 - 24 -** National Institute of Standards and Technology;  
[https://math.nist.gov/oommf/doc/userguide/20a0/userguide/Standard\\_Oxs\\_Ext\\_Child\\_Clas.html](https://math.nist.gov/oommf/doc/userguide/20a0/userguide/Standard_Oxs_Ext_Child_Clas.html); snapshot  
experiments/external\_sources/materials/snapshots/nist-six-neighbour-mesh.html;  
sha256 : 505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2; scope: six nearest  
neighbours as forward/backward positions on three coordinate axes.

Comparison and custody records:

- **sft-mat-cryst-cubic-coordination-001-authority-correspondence:** predicted three-axis-site-carrier\_\_two-orientations-per-axis\_\_nearest-adjacency-complete\_\_coordination-count-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived three-axis-site-carrier\_\_two-orientations-per-axis\_\_nearest-adjacency-complete\_\_coordination-count-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered simple-cubic nearest-neighbour coordination observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:d9f52d70cc6c34ca39fc40035d128e1c8aa5334910900b03aab1998a85286082; derivation seal sha256:e42124c8f179997887b6b5e7ab012c111153998f111f7ba98c1ec85a2d82d83b; engine receipt sha256:b63b31c5bf7aea6cde9b7495c3aaed6f34fac4baf513c2824a18d4c42102155a at receipts/engine/model\_admitted/SFT-MAT-CRYST-CUBIC-COORDINATION-001-b63b31c5bf7aea6c.json; empirical validation sha256:8a82df98e9284baf366a3492530356dfb56430eafa1cfd56fb362f46856df7e0; measurement receipt sha256:be7fafca4dbef1c845769908689a1b966df05d448dd42a7b6f91a704d1258123; isolation sha256:1b7a43467ab4c4a739ade2cb74f2bb3d1aeda8ad1fbb25ed3adfe8a9cd5c1d10; custody sha256:5608949988ae3687ef60d5f1df5ed7441439153ecac0d84f2fb6d20b19b26046.

## 12. Crystallographic rotation restriction

Claim identity: SFT-MAT-CRYST-ROTATION-RESTRICTION-001

**Question and exact theorem.** A finite rotation preserves a periodic lattice only for generated orders 1, 2, 3, 4 and 6; five is the least excluded positive order.

integer-lattice-action-carrier\_\_finite-order-rotation-relation\_\_translation-compatibility-retained\_\_orders-one-two-three-four-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | integer-lattice-action-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | finite-order-rotation-relation           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | translation-compatibility-retained       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | orders-one-two-three-four-six            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: integer-lattice-action-carrier\_\_finite-order-rotation-relation\_\_translation-compatibility-retained\_\_orders-one-two-three-four-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.
- **rotation-orders:** the depth-independent factor certificate retains exactly orders 1, 2, 3, 4 and 6; passed **true**.

**Detailed mathematical and physical meaning.** The rank-two integral rotation boundary requires totient degree at most two. If an odd prime  $p$  divides the order,  $p-1$  divides the degree, so only odd prime three survives. Every order is therefore  $2^a \times 3^b$  with  $a \leq 2$ ,  $b \leq 1$ , not ( $a=2$  and  $b=1$ ). Exhaustion leaves (1, 2, 3, 4, 6); five is the least excluded positive order. This is a depth-independent factor certificate, not a depth-seven lookup and not a trigonometric approximation.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:8e118cd050ba1f91713c4bac85af97b055953558ef2a08ad26c17387a3abde50.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:903699ea736f5b69ac09a10ca848e32b3b7bf4ee50f11d913c7d630147c8f3a7.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:a9244da0719ec18d4622cf446c89292a3c2267c7bb08fa457861ad57cf3f2c31.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:f6f523f473b879608080940789f14fa8b6f533bc3ca361aafce15dfffb28763b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:0a7974f67520983c1fdd51fccf1df5b8e5802ebc0562dd9a4e97fc4778af4b0f; certificate  
sha256:ce3cc18e06f56be003ec62dea474e6214221da3577fe714dc39437d1d192a4e3; external-validation  
hash sha256:2231c2f7810cf10930fb666398885e031d743432e234512ad4452968d2844c38.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:5bbabc66fa0c30a2122ad479107cf8fdbb52650090cfff138262db47417d2bda1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SHECHTMAN - QUASICRYSTALS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/nist-and-nobel/dan-shechtman>; snapshot  
experiments/external\_sources/materials/snapshots/nist-shechtman-quasicrystals.html;  
sha256:911c06e466a95d58749422c56cd4e33fa1fbee04ec06977af428d7dac23ae0e; scope: periodic  
rotation orders and fivefold quasicrystal observation.

Comparison and custody records:

- **sft-mat-cryst-rotation-restriction-001-authority-correspondence**: predicted integer-lattice-action-carrier\_\_finite-order-rotation-relation\_\_translation-compatibility-retained\_\_orders-one-two-three-four-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived integer-lattice-action-carrier\_\_finite-order-rotation-relation\_\_translation-compatibility-retained\_\_orders-one-two-three-four-six\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered crystallographic rotation restriction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:14f138cda0fee15f74ec859ff51be2b51c67c810be00b72df700c3d8a14211ca; derivation seal sha256:4731ceaa58a721703393d44192a3a9c00d6b8a04efad601d81c99abff6efaab7; engine receipt sha256:8fe85fc4bd955efdb5fcc37b159fec38f73a027159ead37660749a04313cd465 at receipts/engine/model\_admitted/SFT-MAT-CRYST-ROTATION-RESTRICTION-001-8fe85fc4bd955efd.json; empirical validation sha256:e93f338ce62e1bdbbba29721ad38fa839cd63180536515cf90c50df1a4e6b463; measurement receipt sha256:2ff74beb5fbbcbdb98e4efb667c9107057fe01c33a592299c0ed2a9b5e4604d9d; isolation sha256:e6db0b5eb3ece993947ca064c20b18abdb90df5cc6c021dc2e2281cc38d72220; custody sha256:290d1315e496177b5646f2fbc830723be3df434fc37fc87de423ee348b62e35f.

### 13. Three-dimensional crystal-system classification

Claim identity: SFT-MAT-CRYST-SYSTEMS-001

**Question and exact theorem.** Complete rank-three metric and crystallographic-rotation equivalence partitions periodic crystals into seven crystal-system classes.

rank-three-metric-carrier\_\_axis-length-angle-equivalence\_\_rotation-compatible-classes\_\_seven-system-partition\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | rank-three-metric-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | axis-length-angle-equivalence            | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | rotation-compatible-classes              | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | seven-system-partition                   | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: rank-three-metric-carrier\_\_axis-length-angle-equivalence\_\_rotation-compatible-classes\_\_seven-system-partition\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.
- **seven-systems:** the complete rank-three length/angle quotient has seven survivors; passed **true**.

**Detailed mathematical and physical meaning.** The complete rank-three metric grammar is the product of three length-equivalence classes and four angle-equivalence classes. Quotienting by axis relabelling and rotation compatibility retains seven forms: cubic (all-equal; all-right), trigonal (all-equal; all-equal-nonright), tetragonal (two-equal; all-right), hexagonal



(two-equal; two-right), orthorhombic (all-distinct; all-right), monoclinic (all-distinct; two-right), triclinic (all-distinct; all-distinct).

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:d6d53473525caba97aac0eef120adbe564356c71e4097f541c0d987a977f95f7.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d5ecbdab14d6f4451eabd1972a86974801daa1b3f29fbc0d095e536484148aa9.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:89c2d1f3852452f2909a9e5f7bd2121e7df76e5501c07b86b1cb3253df66abf7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:ef648732ef475e15f7f44a87514589d7ad95c001bf0beb0fb102ffd1422f6f1e.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:d5514861f898ded8174cc4056d7825fd0e5b938747e48538eaebd3cc84f8f5b8; certificate  
sha256:0e7973bb51c318e20750b3f30a48728851edd1fc9f4b8a4303aae2f3d8841560; external-validation  
hash sha256:e71174f60ede1e2827735dbd37e20c4ab1938398f2be76698070fa1426b66624.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:d7a99045f74e35aa49659bcc6ecb8293ac686764c1a2e6a3f153593a1df6f002. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24** - National Institute of Standards and Technology;  
[https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot  
[experiments/external\\_sources/materials/snapshots/nist-wulffman-crystallography.html](https://www.ctcms.nist.gov/wulffman/docs_1.2/experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html);  
sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice  
generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- **sft-mat-cryst-systems-001-authority-correspondence:** predicted rank-three-metric-carrier\_\_axis-length-angle-equivalence\_\_rotation-compatible-classes\_\_seven-system-partition\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived rank-three-metric-carrier\_\_axis-length-angle-equivalence\_\_rotation-compatible-classes\_\_seven-system-partition\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered three-dimensional crystal-system classification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:e2966380d9a9fda51fc07e0b73d5afee4d49e91d5a0af0869a4cb930524697b9; derivation seal sha256:05c0736603eda6753dcaed3534e5a3f3e54e3ae4d3a62489c4fcaa0dcf19f335; engine receipt sha256:e09270361d9fcd5e3e2e58018dd2c46dace3a54af702d711d8b22b0e34919dd at receipts/engine/model\_admitted/SFT-MAT-CRYST-SYSTEMS-001-e09270361d9fcd5.json; empirical validation sha256:c4b816b9a81da48b1bc8d1a3cc70e41847ad2f22c86313a5106036f69a7322e7; measurement receipt sha256:bd488ae3983b3fa598b381a978e307b19aa3263489e578414e7e71d91ec6a9b9; isolation sha256:a068a2cff6c89dff9921d705b6a08c6eac4f097a5ba44a238ab72455d3a278be; custody sha256:27e2d5486e52bf5f47c1ae8bcad33818e99eea50d8b1efd4791f031a0c6e82c4.

## 14. Three-dimensional Bravais classification

Claim identity: SFT-MAT-CRYST-BRAVAIS-001

**Question and exact theorem.** Complete rank-three translation lattices modulo basis and centering equivalence form fourteen Bravais classes.

rank-three-translation-carrier\_\_centering-equivalence-relation\_\_primitive-lattice-classes\_\_fourteen-bravais-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | rank-three-translation-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | centering-equivalence-relation           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | primitive-lattice-classes                | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | fourteen-bravais-classes                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: rank-three-translation-carrier\_\_centering-equivalence-relation\_\_primitive-lattice-classes\_\_fourteen-bravais-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.
- **fourteen-bravais:** the complete system/centering quotient has fourteen survivors; passed **true**.

**Detailed mathematical and physical meaning.** The system/centering grammar contains seven system classes times five centering classes, or thirty-five candidates. Exact basis/centering compatibility retains fourteen: cubic/primitive, cubic/body, cubic/face, trigonal/rhombohedral, tetragonal/primitive, tetragonal/body, hexagonal/primitive, orthorhombic/primitive, orthorhombic/base,

orthorhombic/body, orthorhombic/face, monoclinic/primitive, monoclinic/base, triclinic/primitive. The paper claims this quotient only inside the registered rank-three translation and centering grammar.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 78101747be3a1f60452a2a41fef4bbceacc6e5337dcbc7573bf342ffcd82ed15.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 1ae5e1a2f713f1df2305360821d929b44dc0ff0a5d58e620083b46859e5a6c77.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: a636fbbebbfe7ea076b33edcc43949081d4b538dfbf0e6a76d84730ca300815c6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: e348fb47b5b11603b3bef45c7ed6a17b63d22d9d6ef59d63964b9ab0dbd010fb.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 5255faf88e98383237d2363fc087370a4d87195e1b2a7960052221bcf3b45d69; certificate

sha256: 2ca709648229e6c57487ee5fa522f233b1084b25fdce360491a2b8656d3f7032; external-validation

hash sha256: 0247320bda7feded0c92b38377a9b25f7c6959cf48051ef443a41a804b8a6237.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: 6232a8f37319ed3012b7aa4f9826ecf52c3b38617950fa4facfc4637cfadd968. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - WULFFMAN - CRYSTALLOGRAPHY - 2026 - 07 - 24** - National Institute of Standards and Technology;  
[https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot  
[experiments/external\\_sources/materials/snapshots/nist-wulffman-crystallography.html](https://www.ctcms.nist.gov/wulffman/docs_1.2/experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html);  
sha256: 52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.
- **NIST - BRAVAIS - CLASSIFICATION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot  
[experiments/external\\_sources/materials/snapshots/nist-bravais-classification.html](https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification/experiments/external_sources/materials/snapshots/nist-bravais-classification.html);  
sha256: 82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; scope: diffraction classification across all fourteen Bravais lattices.

Comparison and custody records:

- **sft-mat-cryst-bravais-001-authority-correspondence:** predicted rank-three-translation-carrier\_\_centering-equivalence-relation\_\_primitive-lattice-classes\_\_fourteen-bravais-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived rank-three-translation-carrier\_\_centering-equivalence-relation\_\_primitive-lattice-classes\_\_fourteen-bravais-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered three-dimensional bravais classification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1ba2a2e7694b4e6f545b596b6a15b62501b2a48434945e1c12cd508bc407a61e; derivation seal sha256:da91f3783dc2f1e6d6e61900a6e577dc770edb550cbf5e3eef3b8144d95a5a50; engine receipt sha256:402977c8ad6256e131a459efdf0762f6f572b919f6a35e60619b4acaf66ad684 at receipts/engine/model\_admitted/SFT-MAT-CRYST-BRAVAIS-001-402977c8ad6256e1.json; empirical validation sha256:333e6b1b653c99e953f630f7eb8804adce7dbdd7d3cf7d3f79293700b0f60d16; measurement receipt sha256:003021a8d559eb1b26af67c2f11c9ed26cf12f49becff5a2006773efb7a33977; isolation sha256:527f1213f3d7c8a54c383d938c9a3d8232e0db8136f64e8aed5995f316676a4a; custody sha256:567f6142968e0c21f7265d5e24402a84854192f9c201dae1a7319410381357bb.

**15. Reciprocal support and diffraction**

Claim identity: SFT-MAT-CRYST-RECIPROCAL-001

**Question and exact theorem.** Reciprocal support is the exact class of probe path labels whose phase returns coherently over the complete generated material organization; retained classes form diffraction peaks.

complete-scatterer-carrier\_\_path-phase-return-relation\_\_translation-dual-support  
\_\_peak-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-  
forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001

- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-scatterer-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | path-phase-return-relation               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | translation-dual-support                 | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | peak-resolution-bounded                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-scatterer-carrier\_\_path-phase-return-relation\_\_translation-dual-support\_\_peak-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.



- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:dbbd7f07ef61e5278af026e0de60234376a22109306b91f4d1bf17bf3fc92db7.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c3cecb0b4bea526fd329b1d5f1ddb4790afeef854b36f7c80a3b54d70d679c209.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:6e2d17ef89a20df4e99c1242813468012f08aabc1d0f82b92ef260085e73c59b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:c03422bc2ec42444b078c5e063341eaa36057cc69ec87b869a1fba8c8c0f016.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:7fde687d50176b337eb1d74f974b35828f044bea87496e0145f9ed715e7a66c7; certificate  
sha256:b79a9101b1cd1d53c7602769ae73503014124fe572a19913e5e2d32930989b98; external-validation  
hash sha256:d01489f20f1663afdd82a9d4e83200885117dfa5c34feeaa4a85ea9f15449730.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:1f65c1f62878d5c4562fa2d3b4250a9d4266167ee478f1b4b1631a12b10e9daf. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - BRAVAIS - CLASSIFICATION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot  
experiments/external\_sources/materials/snapshots/nist-bravais-classification.html;  
sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; scope: diffraction  
classification across all fourteen Bravais lattices.

Comparison and custody records:

- **sft-mat-cryst-reciprocal-001-authority-correspondence:** predicted complete-scatterer-carrier\_\_path-phase-return-relation\_\_translation-dual-support\_\_peak-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-scatterer-carrier\_\_path-phase-return-relation\_\_translation-dual-support\_\_peak-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered reciprocal support and diffraction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:7fe0ba8c722df50c77b1ae34c1574c76db1044d3115cd0de6c04b557de63cfac; derivation seal sha256:2cf710a5f2a9502019e2d3816d744823c545bd807f7de28a0f80368e5d6d9856; engine receipt sha256:bb69b8443063baf90849c3e3fbb54fe746bf27a39cbb5892e3b7988d38e35d78 at receipts/engine/model\_admitted/SFT-MAT-CRYST-RECIPROCAL-001-bb69b8443063baf9.json; empirical validation sha256:71cca974d8629c82315c237652911b0b5577f25513e2fa24cdfec188ef578ea0; measurement receipt sha256:b4975788f2336635d491d04088bc4f366ecdc98223a688a48adabce8c40f16f8; isolation sha256:7ce6b243c09044f0a801fd8ae4c47c89f24bec45f413bb579ba34fb0b8e017e3; custody sha256:2f6fc08ab9a531f9412ae510e8bf4f74b19507eb4c3f421d074c2ff4a0a5b0e8.

## 16. Quasicrystalline aperiodic order

Claim identity: SFT-MAT-CRYST-QUASICRYSTAL-001

**Question and exact theorem.** Quasicrystalline order is long-range coherent recurrence with sharp reciprocal support and no finite translation period; fivefold order is thereby admitted only outside periodic-lattice closure.

complete-aperiodic-carrier\_\_long-range-recurrence-without-translation\_\_forbidden-periodic-order-retained\_\_sharp-nonperiodic-diffraction\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001

- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                               | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-aperiodic-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | long-range-recurrence-without-translation | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | forbidden-periodic-order-retained         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | sharp-nonperiodic-diffraction             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace  | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure         | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                             | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-aperiodic-carrier\_\_long-range-recurrence-without-translation\_\_forbidden-periodic-order-retained\_\_sharp-nonperiodic-diffraction\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : f482fe7557c1f3d9096abc4c5b6ab86ff984ddeb47b893b09a2df71291ecca34.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : dce768e3d1858fac538cdf97980ba03c7c7a351a5d3243eae790fb0d6fe1b82.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : b018cbd8640c5d2740f37cd8fb376d36267f8057724af2a3ad80b4bf6ef62ec7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : e752ea45c00eb03832377dbb9c0bae8c0df9aa99f886ff58cb8cc499898244.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 8713bd0cc130a811d5e81c9332eb037108cbdcecb35a4ea3afd962489cb870f0; certificate  
sha256 : 6315b1810b5134e92f29441df2c6d737beebb9e3277bcb4032a37e8d06f5c938; external-validation  
hash sha256 : 25e7cbc99190b490ca1de85a180d406e5f69d1060cc67b8232a4b4d1cdfc488f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : c5eb79da6aa59b45f7955fda6e77963fcbcc4468da12df970f9ce6c9df522f7c. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot  
experiments/external\_sources/materials/snapshots/nist-quasicrystal-measurement.html;  
sha256 : a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; scope: microscope  
confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- **sft-mat-cryst-quasicrystal-001-authority-correspondence:** predicted complete-aperiodic-carrier\_\_long-range-recurrence-without-translation\_\_forbidden-periodic-order-retained\_\_sharp-nonperiodic-diffraction\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-aperiodic-carrier\_\_long-range-recurrence-without-translation\_\_forbidden-periodic-order-retained\_\_sharp-nonperiodic-diffraction\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered quasicrystalline aperiodic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:7b1915442585a900282440427561c6f7b4df02ce32449a92d2ea47ff898edc5d; derivation seal sha256:6a4386e007b87fab581b9006924276970082bced9cd7259a4384243c894a9a07; engine receipt sha256:77617aba1e766abd8b64273551afa2ca856c9f67a68f869d86448ef1c5266e21 at receipts/engine/model\_admitted/SFT-MAT-CRYST-QUASICRYSTAL-001-77617aba1e766abd.json; empirical validation sha256:977cb5c6105037ae75d97e2312a249489f97a3fa3fe524912be1b7a5fc79bb29; measurement receipt sha256:0ad97e77ff4a78474e22cdfc34fa63f25835bd04395e2ce541ef8d2e2fec0bf9; isolation sha256:cf3c21b892268b1d10e85c3b0d1f1f505778ea51b5f4939b71f2288730876745; custody sha256:d89c32434f6f04e88eb371e35d05e15a90f83cf32e23b31d9f17ec49afb58967.

## 17. Acoustic lattice excitation branches

Claim identity: SFT-MAT-CRYST-PHONON-001

**Question and exact theorem.** A rank-three lattice has exactly one longitudinal and two transverse displacement orientations, forcing three acoustic excitation branches.

rank-three-displacement-carrier\_\_one-longitudinal-two-transverse\_\_collective-excitation-recurrence\_\_three-acoustic-branches\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001

- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | rank-three-displacement-carrier          | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | one-longitudinal-two-transverse          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | collective-excitation-recurrence         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | three-acoustic-branches                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: rank-three-displacement-carrier\_\_one-longitudinal-two-transverse\_\_collective-excitation-recurrence\_\_three-acoustic-branches\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.



- **three-acoustic-branches:** one longitudinal plus two transverse rank-three orientations are retained; passed `true`.

**Detailed mathematical and physical meaning.** A propagation direction consumes one rank-three orientation. The remaining two independent cross-axis orientations are transverse. The complete census is: longitudinal/propagation-axis, transverse/first-cross-axis, transverse/second-cross-axis, yielding three acoustic branches without an irrational normalization or continuous-mode premise.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:d2e24b132e0f5c972671ff849c506788f98e5f442ca5e40467459c911b8e4636.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:274bdf879f97e05579ec9587adb448fbfa54aaae500b5c950c6b1702e2192c84.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:82599c5b8d2e6423bcd6b667f6a82fc0e740bed1f5baf566351026c7353152f6.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:1a63a5afef7b1226f407ae481433648a075f0dfe892a946fd213606d40485070.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:32104da4daf1fd380faf112d1ca088d4bffc61a352e61ffa4c246635b881e4d3; certificate

sha256:e712e3cecd112890f70c47e00d2985ebb8941fb0a3933a188b1522998c457edc; external-validation

hash sha256:e6f329a8b9200d04c64b33e216bae238ce8bb9fcd662f88aea11ae209aa5a186.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:6913f74d3c5f5fbb3d58697fbb956c68bb97d0cd5240834d73dd97e6109226ff. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html;  
sha256:4db74747d716ccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html;  
sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-cryst-phonon-001-authority-correspondence:** predicted rank-three-displacement-carrier\_\_one-longitudinal-two-transverse\_\_collective-excitation-recurrence\_\_three-acoustic-branches\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived rank-three-displacement-carrier\_\_one-longitudinal-two-transverse\_\_collective-excitation-recurrence\_\_three-acoustic-branches\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered acoustic lattice excitation branches observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal\_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

#### Immutable evidence identities. Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:d76418e36f165132c2d2757608ccfd33cbf2caabbbf7849321efeb8a6ceec504c; derivation seal sha256:91b3320ea3273de7f8f7250145371464466d13e21357218c3c7688707b79a25a; engine receipt sha256:8c3472817c5da55f7509b6085db046570a7109ea880dd7e38a086f72c07b216c at receipts/engine/model\_admitted/SFT-MAT-CRYST-PHONON-001-8c3472817c5da55f.json; empirical validation sha256:79983d2c6c9ccc400c7d5b9a724ca15387801673a1881287815dbdefe12706ca; measurement receipt sha256:ae8c2aac5b764cf95d5996bbecc43548237405912e7bca76b77a1f77161bb877; isolation sha256:088998a56bf790b485497a4f0d902cf7c2cb52cba726aac1e09c34716acfbfe5; custody sha256:dac4e779449deddb57373829413ebff00af16ad5936d9cf9b3f2b72ec1a778d3.

## 18. Exact quasicrystal substitution inflation

Claim identity: SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002

**Question and exact theorem.** The two-label substitution A to AB and B to A generates an exact positive finite successor chain whose population pair never admits a positive rational fixed scale satisfying the recurrence.

two-labelled-aperiodic-word-carrier\_\_first-label-to-pair-second-label-to-first-substitution\_\_finite-word-provenance-and-nonperiodic-population-retained\_\_positive-successor-and-no-rational-fixed-scale\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-MAT-CRYST-QUASICRYSTAL-001

The repository graph audit independently confirms that this claim reaches **SFT-ROOT-THERE-IS-NO-NOTHING**.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor crystal\_quasicrystal grammar preserving two-labelled-aperiodic-word-carrier, first-label-to-pair-second-label-to-first-substitution, finite-word-provenance-and-nonperiodic-population-retained and positive-successor-and-no-rational-fixed-scale.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                                | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | two-labelled-aperiodic-word-carrier                        | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | first-label-to-pair-second-label-to-first-substitution     | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | finite-word-provenance-and-nonperiodic-population-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | positive-successor-and-no-rational-fixed-scale             | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace                   | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                                 | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                          | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                              | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: two-labelled-aperiodic-word-carrier\_\_first-label-to-pair-second-label-to-first-substitution\_\_finite-word-provenance-and-nonperiodic-population-retained\_\_positive-successor-and-no-rational-fixed-scale\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- substitution: six positive successors reproduce the declared recurrence; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a

universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:6baf2e887f2ad563e892b4b309c33d7c64d01903860f8c4c44e0b15bd9637512.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:653aca8f3f9546781c62f5e9b6bdb94ddf0b6462f460e38d59775e088eaada7a.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:6b6ba95cbbf2e29b9240cab9960e0b6d05e9c8038e9a8abbad4e99f30b2194bd.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:47aeae71239f94c8706673ec051643230a1b6874b53c5a45a27082ae544a1e54.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:8cbdc5aaab90614797e081352f548ef5022193a74c0e1082d3136687b0d68ba1; certificate  
sha256:6e2476ed2a1c3d4c619f2883987c2e27002bdbcab0035705a2fe538e3d5c3bb6; external-validation  
hash sha256:6e2476ed2a1c3d4c619f2883987c2e27002bdbcab0035705a2fe538e3d5c3bb6.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:6a0c71797dcd96a3ef4ba7ac7e435a44f46230005a964544d63b2492d860b4a0. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-QUASICRYSTAL-APERIODIC-2026-07-27** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot  
experiments/external\_sources/materials/snapshots/nist-quasicrystal-measurement.html;  
sha256:a1b7cc0a5882502bb31b9706225b926c9fffb1e8d93ea5c0043249376c4d6ff18; scope: post-seal  
claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: `True`
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

#### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed

- Every positive finite generated carrier in the registered successor crystal\_quasicrystal grammar preserving two-labelled-aperiodic-word-carrier, first-label-to-pair-second-label-to-first-substitution, finite-word-provenance-and-nonperiodic-population-retained and positive-successor-and-no-rational-fixed-scale.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:1a0b9bcf3caf449f37b9bb0b2a6064bbdea94e1cf1cdfaa7d9a9a7ac77b946da; derivation seal sha256:e238ce3615d8bddc05e3ed617bf643de8a5f67a3674b79a8eb9bfc90203e53a3; engine receipt sha256:ac6f202a08f452225c0fe8a378d95f216de0c49b147d5a3b1c11a1a20fe242f0 at receipts/engine/model\_admitted/SFT-MAT-CRYST-QUASICRYSTAL-INFLATION-002-ac6f202a08f45222.json; empirical validation sha256:3ab04ed10e478a9583e79b4a72699385f23bc90781885ed16c4700407ea2063a; measurement receipt sha256:4555a6c602a633cac8b36021f8bf6425de5b9395cf7df626871c70857622d512; isolation sha256:0c86b7ea5498aa0a2c3c5ee338b79c3ab58bb855e834d9db19f6f67433aed2e7; custody sha256:b078b06c9bbd3d8961e0a6b25b7c066fc4ab55cf744f3a20b10a7c57a0e71e32.

## 19. Acoustic, optical and thermal mode limits

Claim identity: SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002

**Question and exact theorem.** Uniform shared displacement supplies a gapless acoustic class, a retained two-constituent basis supplies an opposed optical class, and positive rank-three support contains the exact cube count of low-support modes and three displacement modes per site at complete support.

rank-three-labelled-displacement-carrier\_\_shared-and-opposed-basis-displacement-relation\_\_acoustic-optical-and-complete-mode-support-retained\_\_gapless-shared-mode-and-rank-three-cube-count\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001
- SFT-MAT-CRYST-PHONON-001
- SFT-MAT-THERM-HEAT-CAPACITY-001

The repository graph audit independently confirms that this claim reaches **SFT-ROOT-THERE-IS-NO-NOTHING**.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor crystal\_quasicrystal grammar preserving rank-three-labelled-displacement-carrier, shared-and-opposed-basis-displacement-relation, acoustic-optical-and-complete-mode-support-retained and gapless-shared-mode-and-rank-three-cube-count.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                         | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | rank-three-labelled-displacement-carrier            | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | shared-and-opposed-basis-displacement-relation      | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | acoustic-optical-and-complete-mode-support-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | gapless-shared-mode-and-rank-three-cube-count       | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace            | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                          | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                   | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                       | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: rank-three-labelled-displacement-carrier\_\_shared-and-opposed-basis-displacement-relation\_\_acoustic-optical-and-complete-mode-support-retained\_\_gapless-shared-mode-and-rank-three-cube-count\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **mode-support**: shared/opposed classes, three directions and cube counts reconstruct exactly; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:f4c3423093e584fe900a9f0ba616ca878be81ab2e31eda877a510579f98aa307.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5c544f51ddbc572dac1e1294db6634e15c7d7547762aaa2bbce56021664a9874.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt



sha256:a7c1d1e7be607560a5b99d1e133480c90affe9f3f97f82df128d29deb1ea712d.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:1513991d2d07cb7a0c1c673c20824ab45793e31e0934a37eeeb7d60b4d0286c8.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:15fbe0593f058d90ae68a29ae9542eab4bd640398e8297b12ec62a3419adc571; certificate

sha256:8ff5ab0de67996fa2a97558cbb4a86897a2caf4cb8e6439a3f0300125807403d; external-validation

hash sha256:8ff5ab0de67996fa2a97558cbb4a86897a2caf4cb8e6439a3f0300125807403d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:edb33397f9512d80b106927d5839c7601a60d4b000a10c53c0d49649fd226967. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - PHONON - THERMAL - LIMITS - 2026 - 07 - 27** - National Institute of Standards and Technology;  
<https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication960-11.pdf>; snapshot experiments/external\_source  
s/materials/snapshots/nist-phonon-thermal-limits-2026-07-27.txt;  
sha256:208d3ff48584c6548ccb00d3b1559ba63d66b1fabe027af9bb499504d5cad64b; scope: post-seal  
claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: `True`
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor crystal\_quasicrystal grammar preserving rank-three-labelled-displacement-carrier, shared-and-opposed-basis-displacement-relation, acoustic-optical-and-complete-mode-support-retained and gapless-shared-mode-and-rank-three-cube-count.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source  
manifest sha256:747cf3a2b40171ad8987eb1f6b539fc7e6019fe27ab1871647c424352a9d019e; derivation  
seal sha256:887b47334d338944c8d3b4f6fb0819ba228ecad61ea6986dcaad9912333beddd; engine receipt  
sha256:658c8a9808a71ff33f048f529b88e8189db27a9d4321aad19f9944df4249eab6 at receipts/engine/  
model\_admitted/SFT-MAT-CRYST-PHONON-THERMAL-LIMITS-002-658c8a9808a71ff3.json; empirical  
validation sha256:ae7de6e618edc99685896d9578e59cf835e95576599bd3ac5f2f7cab13c7a453;  
measurement receipt sha256:9ef211f19dc0d100f0f5a22d6e11df0097d11dc72f300711601678c045298e76;  
isolation sha256:e6e27b5f28a15fe84d3930fdf06bcb2a87956db1d52ce50d77e43b4721d1e2c8; custody  
sha256:d78acee38aef99baedf775f7acf5ab233d2f3137e25a8f63032d9d4beeedad7b.

### 13. Defects Microstructure

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

### 20. Point-defect identity

Claim identity: SFT-MAT-DEFECT-POINT-001

**Question and exact theorem.** A point defect is the least localized retained difference between an actual site record and its declared reference-lattice word.

site-and-occupant-carrier\_\_local-reference-difference\_\_defect-provenance-retained\_\_resolution-localized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | site-and-occupant-carrier  | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | local-reference-difference | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | defect-provenance-retained               | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | resolution-localized                     | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256 : d5cca37e5943297a8a5b2d3d85d2a054989f1a133065f6d35ed160baaa65cfd9`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256 : f68f961248b9bc315028a34f7cabd89e0b1f40a716840e7fae8f3a0bf451eac3`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
`sha256 : 3f18a95e86b4d63ffc8ae28d1018e1b039db7a8de3cd6e79b9e1cfe87a40d635`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:7dd783b6de1acf2c3cf3ee8c73e69fd6b520ec7fb967d22b238900a5d0b23b8c.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:3059cc726018f803ac57d5d892c51d517508bfa3be460c81d502fdc7d2c68244; certificate sha256:c759d08df8a75a473cd5e9dec1433b27cd200bb7d4cca6abb72cabb0f4d37eae; external-validation hash sha256:3c2b96d2f15fc90e4531999b446db3da556e38c8b51ef1660d68d2855aef2cf3.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:7988e4e687fecb304714c434cb7545ce9d70fa8f823acf23c085c562186f7272. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external\_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eafa55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-point-001-authority-correspondence; predicted site-and-occupant-carrier\_\_local-reference-difference\_\_defect-provenance-retained\_\_resolution-localized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived site-and-occupant-carrier\_\_local-reference-difference\_\_defect-provenance-retained\_\_resolution-localized\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered point-defect identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:ce5dba442d917fdfaad1110e0264bcecad74c6a4c9be9ef58ada814cbe2d7cf1; derivation seal sha256:25283d53bed00346043921382e190f5b91210459c356f8b8bffeaa89ea2aa456; engine receipt sha256:b1b3d3707d89ec075a1c23ad71477dc4af6341260e33d7e712aea0d4c35d4b22 at receipts/engine/model\_admitted/SFT-MAT-DEFECT-POINT-001-b1b3d3707d89ec07.json; empirical validation sha256:a94db58787562293e8bb51a26af1d3f6966955e3d91a4255ca338c8c4eb7daea; measurement receipt sha256:38aecf7fa7d9859f15d89dc3a4baf8b8d0b60043dcc2052a2a76b90835f55a43; isolation sha256:a019338b83f6ca0ef9f51f77476ad7d44a4673ef585264cc7173c3bddcde9759; custody

sha256:d2acebdeb9581146a364ab315d10012ef51ac622920719654b9919208b89890e.

## 21. Vacancy as structural absence

Claim identity: SFT-MAT-DEFECT-VACANCY-001

**Question and exact theorem.** A vacancy is a retained reference site carrying structural occupant absence, never deletion of the site or a numerical-zero material amount.

reference-site-carrier\_\_occupant-absence-held-label\_\_site-identity-preserved\_\_empty-one-not-numerical-zero\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                 | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | reference-site-carrier      | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | occupant-absence-held-label | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | site-identity-preserved     | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | empty-one-not-numerical-zero             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: reference-site-carrier\_\_occupant-absence-held-label\_\_site-identity-preserved\_\_empty-one-not-numerical-zero\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:34bcd6eefbd08e1ae00fee7c919cbaa8d558a45544826b10a57c2c7bd698f9de.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5a8c40e74cfe96c496d6ff7e40fa33f620678a3a3a5afa2f3ba76056e2d6c7e7.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:0e53cc842ec91c069e757ab5bb956cb974cf58b4cbfe3f5b671dbb3b85de8551.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:2a6374a8e168c792adec3263140527f62b65afebcd1c2a85a14fe573562262ee.



**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:505a9fdbc929e4fe4912e574520857cc16b9843a3599ba905f951ef28e415e1a; certificate sha256:072c8a2d131081cd814ed1c6d543f5a27b00b28d1743049df66be684c1a19b1e; external-validation hash sha256:1dc1c03612c8e602721414875702b896f302df78952ae98f40bee3f3bb53a80f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:d66f5170de57eaea5c0a8c4e930eb3e728aaeed6a13417345f946147d5fe3d83. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external\_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eae5a55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-vacancy-001-authority-correspondence: predicted reference-site-carrier\_\_occupant-absence-held-label\_\_site-identity-preserved\_\_empty-one-not-numerical-zero\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived reference-site-carrier\_\_occupant-absence-held-label\_\_site-identity-preserved\_\_empty-one-not-numerical-zero\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered vacancy as structural absence observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:8b6b1466e80a30239147f070f141ead32c10c87ce24d155d975af5d548b78089; derivation seal sha256:b4b7599659a0361406a02a2850911db2a54c6f046756a9562f280abd80656e6b; engine receipt sha256:e380e7e72cf8fb701cc5cf6bd37e711db7fa7cc9a9fef348aebb81fec9d73c53 at receipts/engine/model\_admitted/SFT-MAT-DEFECT-VACANCY-001-e380e7e72cf8fb70.json; empirical validation sha256:6b86a2e46c5ebfa7a9310081ee9972a6f67bc2d21d4684747b684a2a26521959; measurement receipt sha256:2d2091fe1ae56f8660db3b54d2cb6b8747cfcef0a10d58857c60c5a4c9cf857c; isolation sha256:ae1764141e70fda66147266e60303f406510c93e4f2b9099554fb0c5c4c0ad18; custody sha256:2a8ecf59f12ea4aa2c6785dc707dd2b1052495407984155615a5a2ba719e8082.

22. Interstitial and substitutional defects

Claim identity: SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001

**Question and exact theorem.** Interstitial and substitutional defects are distinguished respectively by an added occupied site relation and a changed occupant label at a retained reference site.

complete-local-site-carrier\_\_occupant-to-reference-comparison\_\_extra-site-versus-replaced-label\_\_defect-kind-distinguished\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                      | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | complete-local-site-carrier      | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | occupant-to-reference-comparison | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | extra-site-versus-replaced-label | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | defect-kind-distinguished                | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: complete-local-site-carrier\_\_occupant-to-reference-comparison\_\_extra-site-versus-replaced-label\_\_defect-kind-distinguished\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier**: the generated carrier label is held and nonempty; passed **true**.
- **relation**: the generated relation and organization are separately retained; passed **true**.
- **observation**: the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : f3948b872639a13d3ee135782f30c8aff9a7773c9fdf26149a4866148efaa330.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 9adeb2d2247f93e2869d911066bf4021bd6108567408c919a97b0c473380fe15.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : ba6464150d8d66f60708c21cf335227879f90b82d93bd8b62b1ef6011aaea5e1.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 3b95d8ae638b3a67b12120e44708eadb6d42a6a0db6034c93420958988deb289.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:d26a7f8e2a347a232ba359abd7b605a579728ef6581446703faf206ce7b37f9d; certificate sha256:b7a03f2805f3eeab439d48fef49cd06f8670e0d7637638a18a4edd759c7b01ee; external-validation hash sha256:337ae07bd1c202cf5affc6fa9cf3f4c6045a8d83a55056f625a351ad6fd99929.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:2b552685e6c97371ba6be2557169b0890be1cc01817fc2a39de49338915614c1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external\_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eae55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-interstitial-substitution-001-authority-correspondence: predicted complete-local-site-carrier\_\_occupant-to-reference-comparison\_\_extra-site-versus-replaced-label\_\_defect-kind-distinguished\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-local-site-carrier\_\_occupant-to-reference-comparison\_\_extra-site-versus-replaced-label\_\_defect-kind-distinguished\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered interstitial and substitutional defects observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:b8b0fce4b02e0890ef18ec6beb51e1064210749c16b5dcf939dee868c30a8a4e; derivation seal sha256:28dea2afe3f1fcdcf79309764e992de951478be45317f96c14af4e0e839b8640a; engine receipt sha256:47a87b557fa200d5e2177b408e83e080eb7d358c52c10f61c3817b60b34707c7 at receipts/engine/model\_admitted/SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001-47a87b557fa200d5.json; empirical validation sha256:4e6ee278b992402fd86a3894cf2d3ce3b6e29c3cb964073d732f8b1e6f66b673; measurement receipt sha256:e78d1384af71ffceb827e8a592a7cd061f1534e806e90b511867da52567835fd; isolation sha256:5c8ac345a9fea15a0989b063f23ceb749d6a7e46b80c9b45cc28951c28f4f23c; custody sha256:8f9a5e072757dc231f8a117bb38e7f9d6c5dc43a4051c82e0019ab109d7a988b.

### 23. Dislocation line and slip

Claim identity: SFT-MAT-DEFECT-DISLOCATION-001

**Question and exact theorem.** A dislocation is a line-supported lattice mismatch whose complete circuit retains a nonclosing held translation and supplies a lawful slip path.

line-supported-mismatch-carrier\_\_closure-failure-around-loop\_\_burgers-path-record-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                     | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | line-supported-mismatch-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | closure-failure-around-loop     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | burgers-path-record-retained    | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | slip-system-bounded                      | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: `line-supported-mismatch-carrier__closure-failure-a-round-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256:d59f3ceee9a9ea4c5fedc027dca3afe12a11b3071f83718e717f5ea4d7962b94`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7d81ee24b84c6e40d34cd4b294fb323ec65c1baf32a075bbba85a363860296e0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
`sha256:6f2b30a9d67d8c00448c0baccd1113371c830b432e882418e1938dc936c54ed0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
`sha256:870afb6bf3c0fa3e16cb9c36ffaf1fffb021c357a0cb1f7df6080efb52a9d7288`.



**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256: ad2fbb72e7f7bffff1236de535b9cc4ddf9618d802e99ba9238b109b1e619136c; certificate sha256: eba1b94470fbb42e12dfdbb14fb076926ce197f9965de57ce9e47b36f17a0746; external-validation hash sha256: 039ab49886efc164de273e4dd73105148fbcc04ed8e98851087a40d5474150fe.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: e7f7cc953536991061b1f57ee711cea8d62059945102b7d6622beb403907358a. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - SUPERCONDUCTING - GRAIN - BOUNDARIES - 2026-07-24 - National Institute of Standards and Technology; [https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature-snapshot-experiments/external\\_sources/materials/snapshots/nist-superconducting-grain-boundaries.html](https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature-snapshot-experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html); sha256: 1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- sft-mat-defect-dislocation-001-authority-correspondence: predicted line-supported-mismatch-carrier\_\_closure-failure-around-loop\_\_burgers-path-record-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived line-supported-mismatch-carrier\_\_closure-failure-around-loop\_\_burgers-path-record-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dislocation line and slip observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: 6188c63c7eca703914838536c9b19141566ce78b61fe8a0b217cd8c483e9dac3; derivation seal sha256: 956b315813b9db004e985d97806bcf3ad06e798b1c442da065d7a1737b696f2f; engine receipt sha256: ff273682b7e9bf3e07a0ebe94391f6a7f6046677fe786751d96667017dbdaa50 at receipts/engine/model\_admitted/SFT-MAT-DEFECT-DISLOCATION-001-ff273682b7e9bf3e.json; empirical validation sha256: 74c2249f8ad197da4a480dfc1c7f5a336bd21868f2b600eb0388eff6b8d0c0d9; measurement receipt sha256: 9ddba1e33caf86d459b837718673e37a49246d7fed354968f71ce5b33376e7cd; isolation sha256: 0e4803cccc3ea43aa1cf21b67ac24af49e54f17f8fa33c9d452ccbfb535666406; custody sha256: d8faf7892749b1143d64114e17b66476b9329ff8a8919918c02542daa4f249ea.

## 24. Grain and grain-boundary identity

Claim identity: SFT-MAT-MICRO-GRAIN-BOUNDARY-001

**Question and exact theorem.** A grain is a maximal connected orientation-coherent crystal region; its boundary is the retained adjacency where two such orientation classes meet.

orientation-coherent-region\_\_adjacent-orientation-mismatch\_\_interface-network-retained\_\_grain-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                   | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | orientation-coherent-region   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | adjacent-orientation-mismatch | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | interface-network-retained    | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | grain-scale-declared                     | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: `orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: c942c9634762db6520d32ff1ac8e7d7a1c63e39d2c641022dcd7138ef3598216.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 220e22dfdd09c0f9d947632d1b79d338308a1b410141a93266443951748ef628.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: dff2ad7f91bf18ad74cad7113fd23d4b572fc232ae3f3dacd4a2505c0501db95.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 4bed4e2d09f13bdd5c2a7afc570796ce789516387035f49315e09e75991a704b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256: 349f54df0d5774c84cd33aa94c0bd9445971a549f8d5040319e8c1bb49c31cda; certificate sha256: 77a0e2b7f0df63949e54011d392bbc08d9bd34adb66f336af40b02a4ae3368e7; external-validation hash sha256: 834416d05874d2011eb1ffdf3f172e77f75eb414b9ab7d09635a4ac2362a0ea45.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 17e0d614290974fd2dc557705e24b055f6725241f36b930e4e3fed566c413d49. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - TEXTURE - PHASE - FRACTION - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256: b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-micro-grain-boundary-001-authority-correspondence: predicted orientation-coherent-region\_\_adjacent-orientation-mismatch\_\_interface-network-retained\_\_grain-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived orientation-coherent-region\_\_adjacent-orientation-mismatch\_\_interface-network-retained\_\_grain-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered grain and grain-boundary identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: 98bcef7ff326b2a30a2c04f4526b617cc2e6b5a08adb067693acf16d4d1d1f67; derivation seal sha256: e514b24f767337ca53e38189e80c4ba8a8acb74613b672b8a5fcfdd042ab9144; engine receipt sha256: ac7a66efd3aedc5341c9fbbd7d72a278c5f2e9e3420f3ac6c5d121b9af1505fb at receipts/engine/model\_admitted/SFT-MAT-MICRO-GRAIN-BOUNDARY-001-ac7a66efd3aedc53.json; empirical validation sha256: db61a7d6a0f2d4f932107505d4ce4551d79e5139bf4aabb45252e82265b5f1cf; measurement receipt sha256: 28bae4d3ddb832f965a34d4a5f50d64ca9a7d2818cc04cfe3eceddd3f53bc3c7d; isolation sha256: b23111358cbbd16ddffebadea0f98f4d4710932725c2d69776d218c8afa597b0; custody sha256: b7738555c09fa628fe8a367ef171ca1c2e41c56a620fbc9f304df2afc63d428.

## 25. Material interface law

Claim identity: SFT-MAT-MICRO-INTERFACE-001

**Question and exact theorem.** A material interface is the complete boundary relation joining distinct material or phase carriers with explicit structural, content and transfer compatibility records.

two-material-boundary-carrier\_\_cross-boundary-compatibility\_\_traction-content-and-structure-ledger\_\_interface-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                           | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | two-material-boundary-carrier         | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | cross-boundary-compatibility          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | traction-content-and-structure-ledger | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | interface-scale-declared                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: two-material-boundary-carrier\_\_cross-boundary-compatibility\_\_traction-content-and-structure-ledger\_\_interface-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:c1e4dfd258e36706fa84e3cd75aaca7071913c560ebf12f38ff39bbcdf3d222e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:38e2f1f70bfd4f0bc36468142b4370421240bc7e98c6c54d8d0436482af344e6.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:c1db60d562c2617ef8de5671afd5d54569ef6985af0e15a0014cc202d4747fca.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:3439e9c2f088c1bb6c5ce841494806e4d057028abcf2413c50f6c51c1a129bdf.



**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256: 057c33aa17088d597f45fd732daa9b8051863699e686069cfa4abb4d15629506; certificate sha256: 644657ee1c81408bd1464213a5340603387c8ba06f5db9339efcd369df0fdbf0; external-validation hash sha256: f722c72b28540d303f2c1be3ed5525483c8f7771260a1132e5fc02a2517180fb.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 9f783a523f300809cccd2615a4c90f84a593fd2fbd5267829fe4312999299b5. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - POLYMER - INTERFACE - CONSORTIUM - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-interface-consortium.html; sha256: 2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635; scope: surface damage, interfacial adhesion and interface-property measurement.

Comparison and custody records:

- sft-mat-micro-interface-001-authority-correspondence: predicted two-material-boundary-carrier\_\_cross-boundary-compatibility\_\_traction-content-and-structure-ledger\_\_interface-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived two-material-boundary-carrier\_\_cross-boundary-compatibility\_\_traction-content-and-structure-ledger\_\_interface-scale-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material interface law observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: a20fe778e5b1b9c80a7b99792163778ffe132c5828b30a105a4f8dcf48e297e1; derivation seal sha256: 52261a548994417867de3243a46b64609a8e73134b022c544173fcc2d5b3f129; engine receipt sha256: a75d45c8a0bd91e7f206b23cc49ca7c97649ccc12672945d9699ea6e1c8eb1bd at receipts/engine/model\_admitted/SFT-MAT-MICRO-INTERFACE-001-a75d45c8a0bd91e7.json; empirical validation sha256: d7d15758d7c7347d521260dd8c03fdd20873caee2dccc34297ab1ee1994a911b; measurement receipt sha256: 2789b2f3f5f5d2da7dabda5141971f793ac0c3aa17426e32a28f0a1dd66fe03c; isolation sha256: 62c4e87fefaa3b164744c4ab25db1267d0676a9fa2507c9733253ee4d2c723e7; custody sha256: ac3e49780e5c320a0c9b97e62913dbc25e34efa901a369b2f41be3150e774490.

## 26. Diffusive constituent transport

Claim identity: SFT-MAT-MICRO-DIFFUSION-001

**Question and exact theorem.** Diffusion is counted adjacent-site redistribution of retained constituent labels with complete global content conservation and declared support/time recurrence.

labelled-constituent-carrier\_\_adjacent-site-transition-count\_\_global-content-conserved\_\_time-and-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                    | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | labelled-constituent-carrier   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | adjacent-site-transition-count | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | global-content-conserved       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | time-and-support-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: labelled-constituent-carrier\_\_adjacent-site-transition-count\_\_global-content-conserved\_\_time-and-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:1d11be3a1ad882f5cd29b311457e62302459d6833393f74d4e942187679fae66.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ed5233a8868de6e7a8153374b3de47540029249169de8412bcef7af92ebfb5d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:0a97789d5adcb49aef646568c5f015237315bb8fa83088af4563c2e6bd4bc54b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:48a8135d5b012c395466cb8e4986b7799d63e98542cd8ca4fa9d311f782ba773.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256: 9c59f387133a29636589558b2530ee2118991e6afe52a24776db39acffb14f43; certificate sha256: 5e1381704dc4bb016cca880cdf834a2838db3c214f1ecbd148234faf2c29a052; external-validation hash sha256: 64cf3e5bd60590e55b421e0fbf9f707a7adb5412ef80884d4445ab68dfd899a6.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 06c38d5113645138ec19c40e94ce2a9cdb3ace8703038b4f73d9197bfbf084c5. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-POINT-DEFECTS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external\_sources/materials/snapshots/nist-point-defects.html; sha256: b885250bd645ed8eae55a29d99ea4436b691f77c17c74a457acb6acf88335a; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-micro-diffusion-001-authority-correspondence: predicted labelled-constituent-carrier\_\_adjacent-site-transition-count\_\_global-content-conserved\_\_time-and-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived labelled-constituent-carrier\_\_adjacent-site-transition-count\_\_global-content-conserved\_\_time-and-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered diffusive constituent transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: 40f7cd3e3ea9de556a959e4850c76b1d2b8cb4801a32bcd9ebb56d720beeb74; derivation seal sha256: fe6aabdae82bc2c12a9539bf9af505c44f86e4fc12a9f8e911c8aea9cf8c2469; engine receipt sha256: 3e6618450ee819a245f475c45283d761896e350e8c0e1462eb2dabc10c400881 at receipts/engine/model\_admitted/SFT-MAT-MICRO-DIFFUSION-001-3e6618450ee819a2.json; empirical validation sha256: 3a20a291f5964b0d603cb10e0cb85139d358db3dba65b48bc57dc3ce8996b56e; measurement receipt sha256: d5d1645eced54a2feb7096397771785e507c1d1fe702218f2732de6c9e0ad0d4; isolation sha256: a3eab68585733f829113f1f4cc7e7fe19431f1984508c8ed8967fde128196956; custody sha256: 8da37d5bfaf2abcc2106df7c0398aa92c2f6ef38027ea47d4c74c4f90cecd65b.

## 27. Nucleation and phase growth

Claim identity: SFT-MAT-MICRO-NUCLEATION-GROWTH-001

**Question and exact theorem.** A nucleus is the least generated cluster whose interior recurrence persists against its complete boundary relation; growth appends compatible cells while retaining phase and transfer provenance.

candidate-phase-cluster-carrier\_\_boundary-versus-interior-recurrence\_\_critical-persistence-and-growth\_\_condition-and-size-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                         | Exact elimination / retention basis                                                                                                                                                               |
|--------------|------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | candidate-phase-cluster-carrier     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased | boundary-versus-interior-recurrence | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed             | critical-persistence-and-growth     | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |

| Axis        | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                      |
|-------------|----------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| observation | observation-boundary-unrecorded        | condition-and-size-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                 |
| record      | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.            |
| provenance  | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality  | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension   | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: candidate-phase-cluster-carrier\_\_boundary-versus-interior-recurrence\_\_critical-persistence-and-growth\_\_condition-and-size-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier**: the generated carrier label is held and nonempty; passed **true**.
- **relation**: the generated relation and organization are separately retained; passed **true**.
- **observation**: the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:02e1b41356ebd232f4e1ffce226a6e14589a289969615744ce95a85aad756a91.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:85311f60dd07cce5138f6bd6e045ddff85be231164ee115db8a452a521f1fee8.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:56ef43ac04bcd811cb45b4c7c6fdbcf6ccdb17e704f75735d8372bd501a556a.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:dd63ed8dc2853cf4afc78234cae08767527ac748ef137aa0f4a17520975b4ce6.



**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256: 9683ac6de970ada2d2d72b0be6c6d861cc8ddddd1b21b397d0d9b60b6defbedbc; certificate sha256: 1dbae78f92c78f583b62df211ba0173ab8f2c89a7c42f608e32c4b285d976cc8; external-validation hash sha256: c3b79b0253401ef2d732672ae16b5c5ea73b0c67e363e3f8a7a2f5247d212895.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 1c85d46f9369e6f5de9f63e0cbd6cbb7f02c8739fb3b08ad26d55f7238bbc31e. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SOLIDIFICATION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/publications/solidification-0>; snapshot experiments/external\_sources/materials/snapshots/nist-solidification.html; sha256: d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- sft-mat-micro-nucleation-growth-001-authority-correspondence: predicted candidate-phase-cluster-carrier\_\_boundary-versus-interior-recurrence\_\_critical-persistence-and-growth\_\_condition-and-size-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived candidate-phase-cluster-carrier\_\_boundary-versus-interior-recurrence\_\_critical-persistence-and-growth\_\_condition-and-size-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered nucleation and phase growth observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects\_microstructure grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: 5a3a4319df079e15b89fcc578f2f7ad41536a4d633b5fcea4746de982194b30c; derivation seal sha256: 00b4f5e631130452dec26095c7e8906936b03dd6d06a591f3a22989b1a66c1df; engine receipt sha256: a9245f3d9e9386799ada6b1cc5f501b011bfa74c4b21132e17030eae3e79bdd at receipts/engine/model\_admitted/SFT-MAT-MICRO-NUCLEATION-GROWTH-001-a9245f3d9e938679.json; empirical validation sha256: 2a188999b40d5869f6b9b7ff919265c7bcfd95f77b61d002f917d8d71b7ba49; measurement receipt sha256: c2a6cc6f62a2898a9dfc49b69b928f8aea8f9d17b2110bda7b6eb6eba6342a3d; isolation sha256: b8d3766cf94a9c7e8e9e750abb2bf1a0feef8f57dba717ddb3236c425ac6764b; custody sha256: 90ae992b4e7826a09190aad26ac3d29e2df6487b1e1529c8ef4b21b5f718c1c4.

# 14. Electronic Semiconductor

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

## 28. Electronic bands and structural gaps

Claim identity: SFT-MAT-ELEC-BAND-GAP-001

**Question and exact theorem.** Electronic bands are complete phase-compatible recurrence classes on a material network; a band gap is a declared interval with no compatible generated state path.

`complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis    | Eliminated form               | Forced form                    | Exact elimination / retention basis                                                                                                    |
|---------|-------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | complete-lattice-state-support | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | phase-compatible-path-classes            | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | allowed-and-absent-support-retained      | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | energy-observation-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e7658e1f06537e680b3ac39b7f4cc55681b1f55427f53eac74a237a7914ca6ac`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2430cf6e628f3745292c24ba39678422bf65e31c5bafae7c88e050479ec28fb5`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:ccb0fb8846d5f8b815561ff55751d3207480100b14da0039bbbdd6dd480954b2.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:bfde1e54c6b003d0a0606d7d52bfc1046a37dd8f1357f843a5fc77d332188f2a.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:1e5c2bccf261f989160ab67cea576566eb955e4ef4e8e027f5cef5f3b063634c; certificate

sha256:21963d4131f0a68880134816c335271db01e439c0b6a68b848658dcbb318d1bd; external-validation

hash sha256:c100ff6484b4b8bf70291a8f5b401b7de9a126a6cac9f6a74ad43142ec5a41ae.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:68ff9628a37dce468af12c6384628b91da884cec8991a7cc7e4e45528a7c86d1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TOPOLOGICAL - INSULATORS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot  
`experiments/external_sources/materials/snapshots/nist-topological-insulators.html`;  
sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- `sft-mat-elec-band-gap-001-authority-correspondence: predicted complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; source-derived `complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered electronic bands and structural gaps observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

**Immutable evidence identities.** Pre-source branch seal

`experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

sha256:44addc87284413f5cb47df08df006e5926357c33b694da2ae9da9862470876c6; derivation seal

sha256:175a8e789f8b27dd035c876d34d6ad6c7f59c521f797fd73ebde832f3f81d385; engine receipt

sha256:7b3d73980e43e5e27c0796cda5840f72f2139cd68462f5de28c32b6fd289f085 at

receipts/engine/model\_admitted/SFT-MAT-ELEC-BAND-GAP-001-7b3d73980e43e5e2.json; empirical validation sha256:7940d848f5f262579b0b16624dc6ba450141f83939e11ec7508f2c14b97aea6e; measurement receipt sha256:87710ab5dcb4ae5cc803063185f54704445502dbebf37fb7afffa72b7f0d29; isolation sha256:28fc22f923f5c9d892f7e2e01dccdfde00144bae2fced6a676eef828729958f4; custody sha256:5d21c6ba0ac6008740250a9ad2c4d5b32112de387241c602fd7c36e703c3128a.

29. Conductor, semiconductor and insulator distinction

Claim identity: SFT-MAT-ELEC-CONDUCTOR-CLASS-001

**Question and exact theorem.** Conductor, semiconductor and insulator classes are forced by whether occupied carrier support connects to accessible states directly, across a finite activation gap, or not within the declared conditions.

carrier-access-support\_\_occupied-to-accessible-state-relation\_\_gap-and-filling-retained\_\_condition-bounded-transport-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves carrier-access-support, occupied-to-accessible-state-relation, gap-and-filling-retained and condition-bounded-transport-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis    | Eliminated form               | Forced form            | Exact elimination / retention basis                                                                                                    |
|---------|-------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | carrier-access-support | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | occupied-to-accessible-state-relation    | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | gap-and-filling-retained                 | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | condition-bounded-transport-class        | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6b5fc1f6d32f1642235870027f3f6d09f09a59e0640b2c8e0ceed9372a3354a0`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8bc23cae5864aba8f8171dd7eabe70270b7240ec4dd08547ac1f140c0eab3df7`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt



sha256:6a3880e037cb135ac340b234e00c15070e7aa00ca0d95ab77f580baab73b002d.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:22f9b7b6ae28eef1be31d92d3cc3dea0076e729e6321efcec9f9b1998e18189f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:e51bfd79933191b3db914530ce1a3f2fe977a4e15d0a373a00bfd0620f7dca2b; certificate

sha256:40be778743ea52098d5752b1130728f7507bc2b71ea29381c51cfba7898bd47c; external-validation

hash sha256:1caf6cf61b772309acea478f82ffc1af61f3837d67f720225570c7b1955149c2.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:9c9409a2e572bb2d0836dca51f779846076dceabda444c74ff1840b1ae8b95a6. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TOPOLOGICAL - INSULATORS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot  
[experiments/external\\_sources/materials/snapshots/nist-topological-insulators.html](https://www.nist.gov/programs-projects/topological-insulators/experiments/external_sources/materials/snapshots/nist-topological-insulators.html);  
sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological invariants, insulating bulk, band gap and metallic boundaries.
- **NIST - SEMICONDUCTORS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot  
[experiments/external\\_sources/materials/snapshots/nist-semiconductors.html](https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors/experiments/external_sources/materials/snapshots/nist-semiconductors.html);  
sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- `sft-mat-elec-conductor-class-001-authority-correspondence: predicted carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered conductor, semiconductor and insulator distinction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered `electronic_semiconductor` grammar that preserves `carrier-access-support`, `occupied-to-accessible-state-relation`, `gap-and-filling-retained` and `condition-bounded-transport-class`.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3a0672b7756ff9d49ab3b20a85f2c1915685f491d53339920343d42872af33af; derivation seal sha256:82986bf06b4f76150a9f44e3485710dde93593cb487b90aff3d4518af3f5b29b; engine receipt sha256:e66114ebeeef098723db07db0f8fcf46b242222fa43dfc62cafd7fe664832ca2 at receipts/engine/model\_admitted/SFT-MAT-ELEC-CONDUCTOR-CLASS-001-e66114ebeeef0987.json; empirical validation sha256:f4a79e3f2a36290a2a22ba128e7587792a20995bcb71d7eed2ab65c2cbbf0dba; measurement receipt sha256:5cdfb489014fb0be924fde87b453d556866f08a9f42d39b4a2de8460e51b467c; isolation sha256:31ff4726366c6ee968c9e2a86e07d4f4481376fd4c23fc08e32844ea5fd39622; custody sha256:dff32d01f05e8099cb18a076661a0d64c6a7fe2d9bc75d4612b2c046c2a7b273.

### 30. Electron and hole carrier duality

Claim identity: SFT-MAT-ELEC-CARRIER-DUALITY-001

**Question and exact theorem.** A binary exclusion-bounded state yields two transport descriptions: retained occupation and its held local absence, corresponding after sealing to electron and hole carriers.

binary-occupation-carrier\_\_occupied-and-held-absence-pair\_\_charge-orientation-retained\_\_two-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | binary-occupation-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | occupied-and-held-absence-pair           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | charge-orientation-retained              | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | two-carrier-classes                      | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: binary-occupation-carrier\_\_occupied-and-held-absence-pair\_\_charge-orientation-retained\_\_two-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 6913b47e4395df4244aac3f1636c31e345f1650091defd63a5a55ad53215427c.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: a06921b3e66f349f59a1691bdf6941ce10a2bb46d011ff17d496c3e9879cb00b.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: e1f4603d1ca5bf94be4382a5ff0171fb92d97f2ed2dac597bdcdcff1969cd9e2.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 8488d36cd152a36b3fca8f7c07b9aee54038f04802540d3e561994b31f723093.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 2c00743cc5db244ed341a55c207f31476829df1719b36131ab05a2e9bdd1d2a7; certificate

sha256: d95c160677403b50e44e149e3775ceab1a29413ec3de7c3840d02d669e98325d; external-validation

hash sha256: fee3f69c239830d86d49cf5c348de9f74b0ba9301d4365196d7a093b2bcb179c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: ba6eaeeda55aaefbe09d5d56edd2f15d1f36e1cd04ad57756489bc2a67100d93. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SEMICONDUCTOR - CARRIERS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot experiments/external\_sources/materials/snapshots/nist-semiconductor-carriers.html; sha256: 05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; scope: electron and hole carrier signs, doping, transport and Hall measurement.
- **NIST - OPTICAL - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-optical-materials.html; sha256: 3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- **sft-mat-elec-carrier-duality-001-authority-correspondence**: predicted binary-occupation-carrier\_\_occupied-and-held-absence-pair\_\_charge-orientation-retained\_\_two-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived binary-occupation-carrier\_\_occupied-and-held-absence-pair\_\_charge-orientation-retained\_\_two-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered electron and hole carrier duality observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:487f354580f1b942ef88107d8a7e67acaeb612e4800a2d3db21957d7393baa69; derivation seal sha256:02e62b0bb16a65e63274071f63abc8695d4dc712151cc16d37c39dea4b57dcee; engine receipt sha256:fd598a7661fd3dad3da24311efdcba303cbcee8dad7ddcd14e32c6e69afa7a0a at receipts/engine/model\_admitted/SFT-MAT-ELEC-CARRIER-DUALITY-001-fd598a7661fd3dad.json; empirical validation sha256:32242bbc2ab119d300a72f453ece708ab5c3c3431cf5d19c4d742a0d2a85b81a; measurement receipt sha256:fdc881b8019988cb2fa9ffcaf6c97a23f6e9abe4541a24471546cd30626b139c; isolation sha256:dbdf98301457a308b3ac52745fe174bc74d6241df9a017d8fe11b0655ffa36b0; custody sha256:529bbb4a33e307039b20ec33f0fc389b025fcd8244504553e92cbc392540b299.

### 31. Exclusion-bounded state occupation

Claim identity: SFT-MAT-ELEC-OCCUPATION-001

**Question and exact theorem.** An exclusion-labelled canonical material mode admits structural absence or one retained indistinguishable occupation; a duplicate occupation is rejected.

canonical-mode-carrier\_\_empty-or-singly-held-support\_\_indistinguishable-duplication-rejected\_\_occupation-bound-one\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | canonical-mode-carrier                   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | empty-or-singly-held-support             | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | indistinguishable-duplication-rejected   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | occupation-bound-one                     | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: canonical-mode-carrier\_\_empty-or-singly-held-support\_\_indistinguishable-duplication-rejected\_\_occupation-bound-one\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt



sha256:9e095d2545b97492889f9c036a377954bc6c30b71ea4df6cf590c33a7ff73e15.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:489cedb9542a3ae1a73c6c0b0cfcae0a35d5b408bb26dcc08cbbac120e976a04.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:450062506d9f4b952b6cfc258ddee57e835465c3ba6069f3fa260372f2bd7be5.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:02cac4eb83231697c6648ffde545483083d1b1beb4f661539dde2c0d7ade8718.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:68927cecb586cf05cfe48019cdd0639a70cf2b46c72e876832380c4b24a11265; certificate

sha256:e5f507b0a7e69aa7d8ad60023d63d64a9590dee11e5f7b8667d863fb2f58209a; external-validation

hash sha256:c996a7a959521342393af3d9614300fa090a7d658dc386a86baffb1f4268760a.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:586a491bdc820fe537ca3c2684b67f21a9afdd214debf310fdc5f78d2ecc7daf. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-elec-occupation-001-authority-correspondence:** predicted canonical-mode-carrier\_\_empty-or-singly-held-support\_\_indistinguishable-duplication-rejected\_\_occupation-bound-one\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived canonical-mode-carrier\_\_empty-or-singly-held-support\_\_indistinguishable-duplication-rejected\_\_occupation-bound-one\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered exclusion-bounded state occupation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:2d0fe62702f566ed470e063bb5e5b66dd78ec57a1ecf5952711e588f64e6f0a0; derivation seal sha256:a0a7516941112ac9fc011c492b523eedf7b8aee7a793fc958fc72c9dacc71d9; engine receipt sha256:506e8ef22d4cde99c31da1531b362820047c16b74619ef23b75785d57ef7cc68 at receipts/engine/model\_admitted/SFT-MAT-ELEC-OCCUPATION-001-506e8ef22d4cde99.json; empirical validation sha256:ee35ddf7d203268b4935a82979d01e280f6475b227d378ec8b8eff821b9381df; measurement receipt sha256:084ad1d875948d1719f83b9b6a861e91642abbbaa1885ef0ce125aa3a60706a7; isolation sha256:938d56f9988c6c5ec659b1c08f398e04a05f1f4b0b848dcf4430a32356ad6967; custody sha256:02e2391d12546aa8a4d5e40c33b22264075664191749b1011b94efbf0606c846.

## 32. Doping as controlled constituent substitution

Claim identity: SFT-MAT-SEMI-DOPING-001

**Question and exact theorem.** Doping is a source-bound substitution or addition that retains host organization while introducing an exact carrier-support difference.

host-lattice-and-dopant-carrier\_\_valence-support-difference\_\_host-structure-and-provenance-retained\_\_concentration-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | host-lattice-and-dopant-carrier          | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | valence-support-difference               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | host-structure-and-provenance-retained   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | concentration-bounded                    | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: host-lattice-and-dopant-carrier\_\_valence-support-difference\_\_host-structure-and-provenance-retained\_\_concentration-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:56562612ce8c35623d0f0eccc35d803dedbbb5024e8a9b6ddd9af1ea5273c1dc.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: e0b7aa75ce3cc58f9f38cccc833e5fffb828ed6a780b76c27e72333ae94cc79d.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: cce931a0a9fc2d4bf92c7abfb6695c3c7131c95cf90e190f274acdf42801dd19.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: bb0257ed565ef5277fab5ae273fec802528089e7c14e6b69b5869605228db7ce.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 1b2754be6101b1b3994aa165f56cdc9719922bda78bb8a3e7b49541a130c5cec; certificate

sha256: bb931bfb8986710e3d9c87afc938b21108e51f8711df610b710f2d2d37447c36; external-validation

hash sha256: af6af3fceedf777a897ccb954d3a94939fda8a9a940635212ca5f0c7ff90910d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: 122bb299288127d263f0e8816b29197caf3b27fd5ab9450db9e64b49090bfc3. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256: 355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-semi-doping-001-authority-correspondence**: predicted host-lattice-and-dopant-carrier\_\_valence-support-difference\_\_host-structure-and-provenance-retained\_\_concentration-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived host-lattice-and-dopant-carrier\_\_valence-support-difference\_\_host-structure-and-provenance-retained\_\_concentration-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered doping as controlled constituent substitution observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest

sha256:7d6ba4126f6021c12701e140f9662df0dcac166c04f0411174670731bcaf36f2; derivation seal sha256:2c55ece8edc00804dddf74490222c7b077a2b1fb138d08925bacd67c16fa258d; engine receipt sha256:2dac934d9906877ee45538a866d4b4e5d01bca7db0fb4b83d1d13f6f3dfe5bc0 at receipts/engine/model\_admitted/SFT-MAT-SEMI-DOPING-001-2dac934d9906877e.json; empirical validation sha256:4b2237c7244ac6417cf900786078c5e3e67a6690e0716fa7cd6ce20c2bd1e4b9; measurement receipt sha256:76ac92ae9fbcfb18e6d6249bc79a99402af54af309c24eeb36531ec6dc8d07b8; isolation sha256:65640601d6884e72972e3ccfd89d29f60ac5ac53b9d2f2642fb10fcc37a5041e7; custody sha256:75feef131ff8f5a61484f23bc35e133e4d4dcf9aa845ebd919d5e0050a69d51b.

33. p-type and n-type organization

Claim identity: SFT-MAT-SEMI-PN-TYPE-001

**Question and exact theorem.** The two held orientations of a doped carrier imbalance force occupation-majority and absence-majority semiconductor classes.

doped-support-carrier\_\_excess-occupation-or-absence\_\_opposed-charge-orientations\_\_two-majority-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis    | Eliminated form               | Forced form           | Exact elimination / retention basis                                                                                                    |
|---------|-------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | doped-support-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | excess-occupation-or-absence             | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | opposed-charge-orientations              | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | two-majority-carrier-classes             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:56ed005fb5cdc0097997f439b277475193ecbc6b55d69b1a477b88ee060926ec`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb92a6f95ff3cdea45d2e28b701382cb2a7f52dae78b06701b10a88ec5f994ec`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt



sha256:4b980f33219ddfc1ad61861742c6de890fe9d280fdaa67a1f556462fcbeaa528.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:d8e94a97ea900fc4dfdec8667a3e360ee94746b72f6b7e344184e234fc506842.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:e452a66255f1a519151033fcce9e6faa6547ddefcecff55bca8aa84762cbfd35; certificate

sha256:30d82bd025a5aef37135c2532e86294b973a60acc75b57d4c7832a10a1aeb2e; external-validation

hash sha256:008460b8937e1e40da3576894f4ebb00fdd06ac578102266d2755dfd33919f57.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 3 claim-specific source discriminators were required; their ordered identity is

sha256:c9d08fdf23cf1d1c4a0cbd021530dcec9fe237faec27dcda06c3bc8c6ba06828. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SEMICONDUCTOR - CARRIERS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot  
experiments/external\_sources/materials/snapshots/nist-semiconductor-carriers.html;  
sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; scope: electron and hole carrier signs, doping, transport and Hall measurement.

Comparison and custody records:

- sft-mat-semi-pn-type-001-authority-correspondence: predicted doped-support-carrier\_\_excess-occupation-or-absence\_\_opposed-charge-orientations\_\_two-majority-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived doped-support-carrier\_\_excess-occupation-or-absence\_\_opposed-charge-orientations\_\_two-majority-carrier-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered p-type and n-type organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest

sha256:c315630088c00248b7a14519816565651cba129871a017b8b196d44835b21999; derivation seal

sha256:508e9b18d4e217dbe7bb9880439e5ad62146420eedddbd5f4d38ab155c89bbc1; engine receipt

sha256:09af6c90f13d03c26aa83c7e45157858c3cc873c105d60d24b5d8ecf5416bb56 at

receipts/engine/model\_admitted/SFT-MAT-SEMI-PN-TYPE-001-09af6c90f13d03c2.json; empirical

validation sha256:de8958fcad4ba8ece21a19486d0c82b04ba99b43f1aefa231a80e2a69b25851d;  
measurement receipt sha256:4401c9d604bc206b01f0b0240170a20ab4dbcdbac070de79b10ebfa528e1ffaa;  
isolation sha256:d097f756290910c99b0b5274c7508ea5752523009ebbbae60bc424a248147a40; custody  
sha256:f248e50cffb18f18a9831313e6d9d11fb1281018d9cf6fe04bfe88b3687c4896.

34. p-n junction and depletion organization

Claim identity: SFT-MAT-SEMI-JUNCTION-001

**Question and exact theorem.** Joining opposed doped regions forces interfacial carrier transfer until a retained depletion/space-charge organization balances transport and field response at declared conditions.

opposed-doped-region-carrier\_\_cross-interface-carrier-transfer\_\_space-charge-and  
-field-retained\_\_equilibrium-and-bias-bounded\_\_complete-state-transition-boundar  
y-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra  
-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                          | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | opposed-doped-region-carr<br>ier     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | cross-interface-carrier-t<br>ransfer | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | space-charge-and-field-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | equilibrium-and-bias-bounded             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 04fea8715f50e7048d9416d376c53e3450f90c1c8994797f356f59fc28bb26fd.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: b60cab4caeac9244a50a12bfbcd9a9f6971a485a5c9dca8d0dc056c9e1461973.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 341cc639ba0af26441cfc73b6163439bb85143317cd5e75e78911ae88dc2e7b5.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: bed3d1635d1a890677f6efc509da67b859b05cda4243428f4bfa2cb10c603d12.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: e6eb0781d6783a76498a288507518312a16cd86e3b3062787095f1b02f171edd; certificate

sha256: e9b13a5eb19c146ded679178d0be32f2a73e257475860e846b49a09f9cb13d16; external-validation hash sha256: d4b7e5fdc300feba712eba719d137d821e3bf69c7e235835a0281a7832a17eea.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: 28acbaad472972858e5363c8410375f0fe98fecce35a163c6af00e8e96a968c8. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - JOSEPHSON - STANDARD - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot  
experiments/external\_sources/materials/snapshots/nist-josephson.html;  
sha256: 371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; scope:  
superconductive tunnel junctions and quantized Josephson voltage.
- NIST - SEMICONDUCTORS - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot  
experiments/external\_sources/materials/snapshots/nist-semiconductors.html;  
sha256: ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:  
semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-semi-junction-001-authority-correspondence: predicted opposed-doped-region-carrier\_\_cross-interface-carrier-transfer\_\_space-charge-and-field-retained\_\_equilibrium-and-bias-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-d-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived opposed-doped-region-carrier\_\_cross-interface-carrier-transfer\_\_space-charge-and-field-retained\_\_equilibrium-and-bias-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered p-n junction and depletion organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:4d54dd75b86ef466bb3ae5cbc87a0ca7f701301e43197d0ab078c61c49750fc3; derivation seal sha256:555154742ac70136d63a345d0f10ab9c0c161554e1e8014f4b9ba7d52847822c; engine receipt sha256:779d6b98fcddc660d82e4349746ac115bf9463663074b27bac1a52fba754d2e0 at receipts/engine/model\_admitted/SFT-MAT-SEMI-JUNCTION-001-779d6b98fcddc660.json; empirical validation sha256:bcae9c1e255550e984a5e79458201c89f70b0258eba44b7b1903d1bba3cf78d3; measurement receipt sha256:35ceeda6bfff041209f8db359a7dfcba8f80dafb726fdce296417f61bf22051e3; isolation sha256:76cfb9f4f250904e62748743c6525a455e0f0750a8adcbabd145a21d2c142a99; custody sha256:533feb47e28eae5a49bc786c08228949054a2dfef1623d130133575a3ec9d5a6.

## 35. Semiconductor transport relation

Claim identity: SFT-MAT-SEMI-TRANSPORT-001

**Question and exact theorem.** Semiconductor transport is the complete counted motion of occupied and held-absence carriers through accessible support under declared field, gradient and scattering conditions.

labelled-mobile-carrier-support\_\_field-and-gradient-transition-count\_\_scattering-provenance-retained\_\_mobility-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | labelled-mobile-carrier-support          | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | field-and-gradient-transition-count      | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | scattering-provenance-retained           | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | mobility-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: labelled-mobile-carrier-support\_\_field-and-gradient-transition-count\_\_scattering-provenance-retained\_\_mobility-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:0e7b64dccdbdae3f781a411c669794f0bb3a3e5bdc46971765b794957d212a9.



- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:013b382715fc6ec176e0dc3408ede980c818daf766daf96c476885b998c34ddd.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:126ab210a36b80039e752758c90fa83b171da3f4688ecbfab8fb9478a547d6f8.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:bbf482bcded2d0df85a2fcb371710d3e58ddec033cd86e9729e94016097fd96b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:aedaad9f2cdf6e6887baf53d955eae7a0fb8b34bef6cbbb1ef936381ff1b3def2; certificate

sha256:1f9ea281dc6bffeca2b178a18d709033fcb47d9f2666f9691f5c20615e0f8693; external-validation

hash sha256:07cc521721d714dd3db3311e20236df3c3daf59ff0064523b530398b1e8f4db1.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:416db3b9005e4976355fa7996612862d673d7aced19bdae0fe27a041a619ee80. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- **sft-mat-semi-transport-001-authority-correspondence**: predicted labelled-mobile-carrier-support\_\_field-and-gradient-transition-count\_\_scattering-provenance-retained\_\_mobility-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived labelled-mobile-carrier-support\_\_field-and-gradient-transition-count\_\_scattering-provenance-retained\_\_mobility-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered semiconductor transport relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3af983283357631283d3947def3fd55314859ce664d93c96ae2d9a4aa1d2e69c; derivation seal sha256:fcd13ccc157a4adf9d8bd511d17f75266e4d4514d3c5a8008bb2c4a9fe2e2a2c; engine receipt sha256:7a9e1a097ec9c3c48321750ba1f42b728a9bec278024122acdb49c71d1bfeceb at receipts/engine/model\_admitted/SFT-MAT-SEMI-TRANSPORT-001-7a9e1a097ec9c3c4.json; empirical validation sha256:6b0761d0b8abf30d604d8cd0c0d8fc0bca015f1b61338f43fd2910a8b178aa52; measurement receipt sha256:8b454ccd9aee42a77be377617c292db6ea57bd02cbebae033513adf4e802cf3f; isolation sha256:9a8ad3fe9172c8975e080fdd5a07d52725f60710ce518c29d8a6ea02fa14784b; custody sha256:db2e9f0628622dc8e4a0e9ae541b76556fe17d30ad12204a0b24021289e87f25.

### 36. Interband optical transition

Claim identity: SFT-MAT-SEMI-OPTICAL-001

**Question and exact theorem.** An interband optical response exists only when a probe supplies a complete compatible transition between retained occupied and accessible band states under the material selection rules.

initial-final-band-carrier\_\_probe-energy-compatible-transition\_\_occupation-gap-and-momentum-retained\_\_selection-and-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | initial-final-band-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | probe-energy-compatible-transition       | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | occupation-gap-and-momentum-retained     | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | selection-and-resolution-bounded         | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: initial-final-band-carrier\_\_probe-energy-compatible-transition\_\_occupation-gap-and-momentum-retained\_\_selection-and-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:c936b0d5bb1ae8aec13da9742921982d562b55f6d4774bd1aab6e334a1e5b753.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8701918a766daef17bc285510ac5721438c7e56272c88217cd085a9d3e145154.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: bc252118ced3be1bd50755e418df34a442529528051ab45a471d57b3cfcfb8f6.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: ceec9c1581794a42484f4f193268ea44aecb938181070c46a7f3f65721a2f46a.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: a3afed5ac8947640ccdafaf35625a2212acb73b39ad06c991752279f3cad2019; certificate

sha256: 8076ebc7ac57b55ce725e87f69dcce17a303f9329a50b04a951c98f157d8053e; external-validation

hash sha256: f68ffc662b3b7c7128a78f5abc04b5a0e0eaa09941e02795cea9ced09ac95ef0.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: 278dc1caac24eb6c922c3a6bf6e833a99a93040d869c9a79be2ae4104dbf0bf3. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - OPTICAL - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-optical-materials.html; sha256: 3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.
- **NIST - SEMICONDUCTORS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot experiments/external\_sources/materials/snapshots/nist-semiconductors.html; sha256: ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- **sft-mat-semi-optical-001-authority-correspondence**: predicted initial-final-band-carrier\_\_probe-energy-compatible-transition\_\_occupation-gap-and-momentum-retained\_\_selection-and-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived initial-final-band-carrier\_\_probe-energy-compatible-transition\_\_occupation-gap-and-momentum-retained\_\_selection-and-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered interband optical transition observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic\_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:8d8279fa3008b12b0b1051877f0af738d00569ad9ba188399d1eba5df11f24f8; derivation seal sha256:8e4a5b2c675b5e98ec5e642838d2228f57b6aee5ee1ac95b8b21082fbcf31b8a; engine receipt sha256:809590cfd0ac95b5b110d63963478dbeff718642ced0621bb7bd593ce0f09e93 at receipts/engine/model\_admitted/SFT-MAT-SEMI-OPTICAL-001-809590cfd0ac95b5.json; empirical validation sha256:d223c4d8de7571cd7987e65eab435a979ca11071a431f928801ba7ff40fc18f7; measurement receipt sha256:bbdfdb43fb4478403d0f7fbdebc1396367ef0eb93db04319e389aad5fb1104d3; isolation sha256:beff5fea3faf3ab08ccdde2b1a5053453c9a471eee182464a6cd2ce6a342af1; custody sha256:2a01435955d2824cb67bf66500b68c1a34f4933eeb7ecee81a2e04ca4db6e43.

### 37. Exact p-n junction rectification

Claim identity: SFT-MAT-SEMI-RECTIFICATION-002

**Question and exact theorem.** The two held bias orientations of a retained p-n junction force distinct forward barrier-opening and reverse barrier-strengthening transition classes without signed proof quantities.

junction-barrier-and-held-bias-carrier\_\_orientation-conditioned-barrier-transition\_\_forward-opening-and-reverse-strengthening-retained\_\_complete-two-orientation-transport-census\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-MAT-SEMI-JUNCTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor electronic\_semiconductor grammar preserving junction-barrier-and-held-bias-carrier, orientation-conditioned-barrier-transition, forward-opening-and-reverse-strengthening-retained and complete-two-orientation-transport-census.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                        | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | junction-barrier-and-held-bias-carrier             | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | orientation-conditioned-barrier-transition         | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | forward-opening-and-reverse-strengthening-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | complete-two-orientation-transport-census          | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace           | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                         | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                  | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                      | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: junction-barrier-and-held-bias-carrier\_\_orientation-conditioned-barrier-transition\_\_forward-opening-and-reverse-strengthening-retained\_\_complete-two-orientation-transport-census\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- bias-census: forward and reverse held orientations yield distinct barrier states; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

#### Adverse controls.

- false\_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256: 72091144fd5e5e2f17fb0af1236f878771e05c91a88f2d97aa1b5e2187a75b40.
- tampered\_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 9929efcd682e523925a4517b24689bc98c434b00c3121fd02619458c61f8c0bc.
- tampered\_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 1fd0a27d51a77ca70c0055f7e0b90401b16bb9eddda5a4c552a9733f42ea9c2c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt



sha256:1ddd52f23e78c87eb0542e05a59b000a8285827ad99039256e09441808f242fb.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:274dadcd795cbad20b0196e6035f10b5d8be04f5669237100a5e40812bacb1d9; certificate sha256:336a6381c77c7668726d3aff4dc72e9d7f6b8f8edaf623d2703d3f950dd3290e; external-validation hash sha256:336a6381c77c7668726d3aff4dc72e9d7f6b8f8edaf623d2703d3f950dd3290e.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:5f25f10dcea8438b7df990add66ee5fbf6485ee5cb207a08aa417df77a3419b2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-PN-RECTIFICATION-2026-07-27 - National Institute of Standards and Technology; <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir4414.pdf>; snapshot experiments/external\_sources/materials/snapshots/nist-pn-rectification-2026-07-27.txt; sha256:ca301b8fc47ba509909982ca2b077c238b92f3ff4978ed7377d086071c519ba4; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

**Explicit exclusions.**

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor electronic\_semiconductor grammar preserving junction-barrier-and-held-bias-carrier, orientation-conditioned-barrier-transition, forward-opening-and-reverse-strengthening-retained and complete-two-orientation-transport-census.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:3808d1a1c973341ac5988f70736ad6acf2671f1d6de4e44ca6a6c76f39e9a696; derivation seal sha256:4235421a299754c0130b0539c84de12ccc13f420cb83c72cd00d73bc69c6ee34; engine receipt sha256:e71ca636880f795b1e68037e7ad6f62ce1388b4a8a8d6c6a05399847beaa836c at receipts/engine/model\_admitted/SFT-MAT-SEMI-RECTIFICATION-002-e71ca636880f795b.json; empirical validation sha256:c9e1c1cda237826ef15921efe0dd41a8bc39c8cd60cd3357c3670a45d14f70f7; measurement receipt sha256:06c4e475020998315f98335528e79fcb642bc1176030bca0f7f729022ac026b9; isolation sha256:65b5f7ae534d603f11f917cc1061ba772035490b6a36f4e51b868a596df904b4; custody sha256:e970053159f8be987db9dd34005e22f72951151146d54ed1a9a82ab1bf89168a.

## 15. Superconducting Superfluid Topological

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

38. Coherent paired-carrier state

Claim identity: SFT-MAT-SC-PAIR-001

**Question and exact theorem.** The least exclusion-compatible collective carrier joins two opposed fermionic labels into one integer-composite recurrence class.

two-fermion-joint-carrier\_\_opposed-label-pairing\_\_integer-composite-recurrence\_\_pair-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form               | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | two-fermion-joint-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | opposed-label-pairing     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | integer-composite-recurrence             | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | pair-support-bounded                     | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:588d928b1e15481455a5bc3aed20e5ecc939285f701ef568ab8854c37a456594`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1fce1ab61ac5cdfd56f21d9a82a0f122803c48d6893b8182a0ec054e44125969`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:06212265fed2d4576c34bb29a7fa1db2397ec20e3ecdda764ca853158bb137eb`.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:d85c54fbbab7de7503b8d8e13076bebd80a9d80d2bf9d3463f2ba5d851c6bf6e.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:ccdabd49d56b90757f60cc9f3d1de2c6b1e70499d23109147029465b0c1a2078; certificate sha256:4425dcd748752cd4959cb4846753a73d83912db2c333b36d3295d3212ba1f9a1; external-validation hash sha256:cd91799f0836165d952f55222fdc867f9baf72874a1e3c76abb6f573efb6766f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:6c15a4430a3b63ae8b2d9b46dc34a353ae01acb710cd8481e153960a6e86cc26. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-sc-pair-001-authority-correspondence: predicted two-fermion-joint-carrier\_\_opposed-label-pairing\_\_integer-composite-recurrence\_\_pair-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived two-fermion-joint-carrier\_\_opposed-label-pairing\_\_integer-composite-recurrence\_\_pair-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered coherent paired-carrier state observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:d93b4d5e2b622347f1676960ee60f23945f6a7ced001e0126d43a7c8b8ef3a8e; derivation seal sha256:a1338a1e0276c18373474215b722f972da5d845bbbd492ccb324d205c46553b5; engine receipt sha256:f65c0f0c9bd7124faa85bdb47c873a6abc42e8e8708b6d1e1a59820b33115855 at receipts/engine/model\_admitted/SFT-MAT-SC-PAIR-001-f65c0f0c9bd7124f.json; empirical validation sha256:713c8da110495dfd47aaea6dd319378273a24160916d2c4318d89f2f2fd1d278; measurement receipt sha256:61a58faeb48a968867e3b01590511b20540c5a14b184e41518714cf58b250eba; isolation sha256:7785d4374ef3eff679275b146a79ab9f5de3cea69341d630072727b1c8d59a9f; custody

sha256:9c164976a0cf9ee4d1d0855105a2d8a6351a716ee3bee38ae12ec035fb2c8d5c.

39. Dissipation-closed superconducting transport

Claim identity: SFT-MAT-SC-ZERO-RESISTANCE-001

**Question and exact theorem.** Superconducting zero-resistance transport is phase-locked pair recurrence on complete connected support with no retained label-compatible dissipative transition.

phase-locked-pair-network\_\_collective-translation-recurrence\_\_momentum-randomizing-paths-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                       | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | phase-locked-pair-network         | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | collective-translation-recurrence | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | momentum-randomizing-paths-absent        | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | critical-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: phase-locked-pair-network\_\_collective-translation-recurrence\_\_momentum-randomizing-paths-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:8d318b6cd754552f1635d2d3376fb19a7943dac948ddf64cdb2c469dd2d57981.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2d3c2643f67651e4ca0ce67d52da102053664b47fd07c1aec55f2625f3b5820d.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:4d3bbf5b42e26656f92e162c71c1b9b62f1636bacdd84ef0eade4dc87e68aa53.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:51e5ed61aacce99183b4b16d20ea87706585d4c167fda8309eca0ac820dd396.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:b1b9adcd5b385285baebd3bcc32081e7bcb07bc9a7423d1d0d8f4b24623b8b64; certificate  
sha256:e767c0f4c1ff932262442a7d1c6b5ae435d89b7adfc2ede2eb66da249e8a5872; external-validation  
hash sha256:bc66886b883b3b04e280b2c157934a163799a8e3abcd37cc54ea9170516dc3b5.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:c5d02ac8fd4e460114d7267f20dc67b2ba9ff4b6cb76be84227c69b692b6557a0. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - SUPERCONDUCTING - FLUX - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superconducting-flux.html;  
sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero  
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-zero-resistance-001-authority-correspondence: predicted phase-locked-pair-network\_\_collective-translation-recurrence\_\_momentum-randomizing-paths-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived phase-locked-pair-network\_\_collective-translation-recurrence\_\_momentum-randomizing-paths-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dissipation-closed superconducting transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256:b9e796760dea97e1e9f71baab1a6c8bdfd297371b6fbdd5aa16ee8c36edf8c366; derivation seal  
sha256:edbbea4729b78617df2c8f1ecdb55716528ac140a38827276580103223e2b971; engine receipt  
sha256:12eb3dc5c5b6718a083650f79097da45e15e1c0f9a7d6d962fb3afdf672576ea at  
receipts/engine/model\_admitted/SFT-MAT-SC-ZERO-RESISTANCE-001-12eb3dc5c5b6718a.json;  
empirical validation sha256:4463105306d4f42745346031c4a24c8910c65ae30aaf2ea125d0108575905655;

measurement receipt sha256:1bebb8a089facd8820e247e0c2b56def398143bf54d99b01e82b17c475edc90b; isolation sha256:3a31a682ebb7914cb6dfe9a572940f32f1a21e3af617a0e474f367180900fffee; custody sha256:4b18283de437e6a909184edac099a512c370d092730bd30c5ef62b3927e90677.

40. Magnetic flux exclusion in the superconducting phase

Claim identity: SFT-MAT-SC-MEISSNER-001

**Question and exact theorem.** A phase-locked charged carrier network generates a boundary response whose complete bulk recurrence closes an incompatible applied magnetic distinction, yielding flux exclusion apart from the declared penetration boundary.

coherent-boundary-current-carrier\_\_applied-field-response-relation\_\_bulk-phase-lock-retained\_\_penetration-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis    | Eliminated form               | Forced form                       | Exact elimination / retention basis                                                                                                    |
|---------|-------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | coherent-boundary-current-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | applied-field-response-relation          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | bulk-phase-lock-retained                 | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | penetration-boundary-declared            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:41fdd77f83e8d59c2bfcf7bfbfd501d37db3c95f5bc0b83d16b44837f40d88f75`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e4bd038bb4b94d51173bdfb9230bddd24ccc1144277314f85858688c05c4e10d`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:b0b3d381e964f5faaa7119aeaac1005b9b4d579b20faed23038adc1dd012e4cc.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:e1e636a30e4b8e61b27e684806bc21fbf45696298651b7107e96b7f089e0a892.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:928c10c30b33c7559a3dbbc0c913420ed2ab81b7bc1d15273d91edeb97eee1d0; certificate

sha256:06e8deb4ea6d9b5aa05a4dfec72f2b735bfa417cc18126dfef1493310c209e47; external-validation

hash sha256:b50c818dc8d34d3a26d99222c94ed78c6d5e46fe14e683dadfd25d9519ca71ce.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:0c547443d909099a887a74f3853eb1e7fc46b03829baa87ede9b6fae8b103170. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SUPERCONDUCTING - FLUX - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superconducting-flux.html;  
sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero  
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-meissner-001-authority-correspondence: predicted coherent-boundary-current-carrier\_\_applied-field-response-relation\_\_bulk-phase-lock-retained\_\_penetration-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived coherent-boundary-current-carrier\_\_applied-field-response-relation\_\_bulk-phase-lock-retained\_\_penetration-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered magnetic flux exclusion in the superconducting phase observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest

sha256:b59e8e71bf983f7a08fd389c4b3c03789f7d0f5a8bd9b4c5f8864fbfb35608e8; derivation seal

sha256:f9146fca79784576aa3a1589231dca173d7efe82cd3885b978b60ee3e13ef2bf; engine receipt

sha256:2cd04f09d3f7371d20ce1b7209cc1dc496ea19835d019733d1fda439b0195fce at

receipts/engine/model\_admitted/SFT-MAT-SC-MEISSNER-001-2cd04f09d3f7371d.json; empirical

validation sha256:ed1c150af3c3925bcb4915bb464cdcb311c930e5f83b00b8d6175f225bb0f044;  
measurement receipt sha256:3f7d9d6861cb68227b9a4061d8ab60bd253c3e2a7a84b9134c79a13fa58ec14e;  
isolation sha256:f141e62bf73a7f12e615ada65bbff9248644c0f2540ddc848f69ebca25a2b0b4; custody  
sha256:713d272d7326c231a0ba2e21288be842ea24652fdb5f251e11611b0f3113d7ae.

41. Superconducting flux quantization

Claim identity: SFT-MAT-SC-FLUX-QUANTIZATION-001

**Question and exact theorem.** Single-valued return of a paired-carrier phase around a closed path forces whole winding classes and therefore discrete flux support.

closed-phase-loop-carrier\_\_whole-return-winding-relation\_\_paired-charge-label-retained\_\_integer-flux-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                   | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | closed-phase-loop-carrier     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | whole-return-winding-relation | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | paired-charge-label-retained             | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | integer-flux-classes                     | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: f0e9a5316c0cbe435499c31ffe40b86071cc48d07b4534c765299a5a35bf0f9d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: d7f3655cbeb3b9850a02dc2d27cfd0a9c033adbdc9caa63cba999c19e29c974.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: c69e26fe4417441d2552b06d56b73199016f8853baa5f0e2a5f34bcac21d8bcd.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 64af51fc38ea344b01e2131b58ecba3a1728ed24158136831a4a38b3507f95cb.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: ba1d06e0c591660c25b9d494d140c2eb9c5b5ad7a60667e026d6c7c3f845b47d; certificate  
sha256: 7cf210102f6487dc10ff5b4ceb32d3bf816945f961e947c3b153c669c4d4c632; external-validation  
hash sha256: 76c78e4fa8a0436139f72942375cfe051a5b9bb6dd6496e8e17cbb720971c2f6.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: a59499d54f66da57c505ea1a9006d13183cf8c79f97b6694d0562a7789a16489. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - SUPERCONDUCTING - FLUX - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superconducting-flux.html;  
sha256: 47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero  
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-flux-quantization-001-authority-correspondence: predicted closed-phase-loop-carrier\_\_whole-return-winding-relation\_\_paired-charge-label-retained\_\_integer-flux-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived closed-phase-loop-carrier\_\_whole-return-winding-relation\_\_paired-charge-label-retained\_\_integer-flux-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered superconducting flux quantization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256: c8b02fef0c388c57e46670d576003802b43b86270c1afaff3f410a87423f6911; derivation seal  
sha256: 85f5457eddebd585da75c677d2b205089e95a4558c7cc6f9af8783fe82ea8e0e; engine receipt  
sha256: 5222d9301d5077db8f7fe1adccb876b76af5caeec23508a322fc8393ac62e22b at  
receipts/engine/model\_admitted/SFT-MAT-SC-FLUX-QUANTIZATION-001-5222d9301d5077db.json;  
empirical validation sha256: 0b4b3abdbac20b0e33a7ee4a688a80a5a4090133d5ad8bb7d756508ea41e47aa;  
measurement receipt sha256: ee23932a5a16f9a89e0b3dfd7839be86e66580e95a8c9a7621fef766dd025324;

isolation sha256:b80a2d163e7842712cce6b99df6f2c0337bcf60b91f002180b285ac4d1ab8ed6; custody sha256:9d61fe50609999ec7322ca722ebf3962578a79550c05102d4c8a2a06e82e31dc.

42. Weak-link coherent transfer

Claim identity: SFT-MAT-SC-JOSEPHSON-001

**Question and exact theorem.** A weak link between phase-locked pair supports carries a coherent transfer recurrence determined by their retained phase distinction without dissolving pair identity.

two-condensate-and-link-carrier\_\_phase-difference-transfer-recurrence\_\_pair-identity-retained-across-link\_\_junction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                          | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | two-condensate-and-link-carrier      | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | phase-difference-transfer-recurrence | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | pair-identity-retained-across-link       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | junction-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: two-condensate-and-link-carrier\_\_phase-difference-transfer-recurrence\_\_pair-identity-retained-across-link\_\_junction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:399845f807dcd160de6e469aad258bfa25f83f72dc4570e597b8b67651d0aab2.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:c0e22c5b1eff9f4e8d5756445e63ce5b47873c214a9e6225659f30250ae45e44.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:7b8e05bab5f8a0f80750a7c96e67151bc76e63513889500c048513a184222c3.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : f9a41df391a6e3a93cb1c58b912920fb74a9f3d3cfe4d8231ca7cded8f9f5816.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 4d646761ef86e203d4dc92fc307815d351f2f955293517f5549e5dcc1e949b39; certificate  
sha256 : 28ea20d0b7536a61f4df9e90490a956ebd4d4d57e33dbf816d7a7011b0260a1b; external-validation  
hash sha256 : 74274c0b5926ea435009d31d46f20c137898814c025f01bac6ccec42b1905e3c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : ee4e8e99314f54e4c03074f8570f0e1abe2959be6265b7bcf80807a92d298924. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - JOSEPHSON - STANDARD - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot  
experiments/external\_sources/materials/snapshots/nist-josephson.html;  
sha256 : 371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; scope:  
superconductive tunnel junctions and quantized Josephson voltage.

Comparison and custody records:

- sft-mat-sc-josephson-001-authority-correspondence: predicted two-condensate-and-link-carrier\_\_phase-difference-transfer-recurrence\_\_pair-identity-retained-across-link\_\_junction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived two-condensate-and-link-carrier\_\_phase-difference-transfer-recurrence\_\_pair-identity-retained-across-link\_\_junction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered weak-link coherent transfer observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256 : de9469a89d2867722b9eae6fa3760649daaa5553d70c469eb7cf83e15e2525a; derivation seal  
sha256 : 3bd854b2b132fd641e4e5f5955b5ffce6518e0c7ac28bcdd65660d450b12d489; engine receipt  
sha256 : 7081c6da685356122c147a5935e3b767d81a0c0c6ce6d5e0a7b75f6c0b4da6d0 at  
receipts/engine/model\_admitted/SFT-MAT-SC-JOSEPHSON-001-7081c6da68535612.json; empirical  
validation sha256 : d871b3b487789842475feacb4d5f08c14ca3faa00faab4518a71d1f3cf2ceb03;

measurement receipt sha256:2faf6ca3e067d3c131f05245b0149df1d19104e157da47b7accab873cd704cda; isolation sha256:7bc6387d84c77a4c1527b40b42bd1d5b808abd5603f45727497e2621e7d90a08; custody sha256:48ac2f8f2ee7cacdc2cdd908a0e4e30408ceced8311b194c717c481488f74827.

43. Dissipation-closed superfluid transport

Claim identity: SFT-MAT-SF-SUPERFLUID-001

**Question and exact theorem.** Superfluid flow is shared-phase recurrence of a neutral collective carrier when the complete accessible transition census contains no allowed dissipative path below its first retained excitation boundary.

neutral-coherent-fluid-carrier\_\_shared-phase-flow-recurrence\_\_loss-paths-below-gap-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                    | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | neutral-coherent-fluid-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | shared-phase-flow-recurrence   | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | loss-paths-below-gap-absent              | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | critical-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256:53aeb9bd83d18de3cd8d06065a5ef322fc212de21e769e8ca85c254f3a235c1e`.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0dd29512db07cf4c534254d6059cf520fc5edc37285f2abc39c5a4c037549c9b`.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
`sha256:f6e09bc043b7b7342f350da3404ad33d5781dccd6d6a9ee2adb0e1d3c39d5b21`.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 502b285ce7563d3d21f1c79789e6ae682a6022b3e36db0a107366d25c3c7fa2b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : f56b139525ec8db1263aebf85b98f0d1417c8c2ef70d4f4ecef2abb94bc22a8e; certificate  
sha256 : 737b7188dd1647b964cd6655764a2a2db0aea236180922a145ffbdfb1ffa9ad; external-validation  
hash sha256 : dbdd999b36e350be4f7fc5b44e9ef260cea53c782eccf9d0cb8e356c5eb0cc93.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 00b0b7c16330fdbcbb90a5ea51473a63afe04ee29fbb606a9c345ba91717e8eee. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SUPERFLUID - CIRCULATION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superfluid-circulation.html;  
sha256 : 49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; scope: superfluid  
flow, vortices and discrete quantized circulation.

Comparison and custody records:

- **sft-mat-sf-superfluid-001-authority-correspondence:** predicted neutral-coherent-fluid-carrier\_\_shared-phase-flow-recurrence\_\_loss-paths-below-gap-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived neutral-coherent-fluid-carrier\_\_shared-phase-flow-recurrence\_\_loss-paths-below-gap-absent\_\_critical-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered dissipation-closed superfluid transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256 : cb2ec54d212e954c015f00c0dac3b0b91ad8433dddd8dfeb5e842f81a20a3158; derivation seal  
sha256 : c4fa97a72987df4fb8e8db6b3e67e94202e5bd15d51c09701501a72e26228d5f; engine receipt  
sha256 : f97e7d21474da7f1f117d9dfd57a2a714f0844f88ece7bf4d90daf91e8e79b3b at  
receipts/engine/model\_admitted/SFT-MAT-SF-SUPERFLUID-001-f97e7d21474da7f1.json; empirical  
validation sha256 : e10012064a7012eaad97c30443e19ed8db341b8c01a039ad34cec73910fc9ebd;  
measurement receipt sha256 : f5e2e939e5d8bbbed493cbca552c915b723dbef99a7a0602b8a960eac7d1a8b2d;

isolation sha256:5c0d5f2cea3b7d6c7242bf7c842f4a0713086806e9d89c4909cff5b976545779; custody sha256:65f0b4b610b2e75226f3cae0b0a7eb10f9d9093837f3fc6cb940dde6bbfb33e8.

44. Quantized superfluid circulation

Claim identity: SFT-MAT-SF-CIRCULATION-001

**Question and exact theorem.** Single-valued return of a superfluid phase around a closed path forces whole winding and discrete circulation classes with a retained vortex-core boundary.

closed-superflow-loop-carrier\_\_whole-phase-return-winding\_\_vortex-core-boundary-retained\_\_integer-circulation-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                   | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | closed-superflow-loop-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | whole-phase-return-winding    | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | vortex-core-boundary-retained            | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | integer-circulation-classes              | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: f02eec7391d7631590d2bdf547034d8deb9704bd1216a21b9f245de6f8279715.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 47e3e962a411f836a67dcfcfb25ec672026bec43ec887acfbfb46cdd0e5d2fd37.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 4e454943e4f2ef6f47e060c907ae345a5b14d1aa029de2e0f12f423693052ff7.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 65529a60c0bbbd22624bcc4ef2744b5a19a59898aaea44440259859ce3e8d166.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256 : 8309220cfb4f71ac91bb6c78e1b8e93d5fd7d982a212003a218b11b52185129b; certificate  
sha256 : c2182c823e85f1379edf6bbcae692fb0a0597df0bf835c2b4a567d343189c797; external-validation  
hash sha256 : c4741e9aec860d4696db44b536e7ed688067581da702492261813d3450aa487d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256 : 3b4dda8db9ff0cf748bd3069c4aa37b649f64bf399b70947c75841856eaf0257. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - SUPERFLUID - CIRCULATION - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superfluid-circulation.html;  
sha256 : 49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; scope: superfluid  
flow, vortices and discrete quantized circulation.

Comparison and custody records:

- sft-mat-sf-circulation-001-authority-correspondence: predicted closed-superflow-loop-carrier\_\_whole-phase-return-winding\_\_vortex-core-boundary-retained\_\_integer-circulation-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived closed-superflow-loop-carrier\_\_whole-phase-return-winding\_\_vortex-core-boundary-retained\_\_integer-circulation-classes\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered quantized superfluid circulation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256 : f16ac71b88734483350d41ca16b2b40556ee06e0987c89a426fb1985a4b5907a; derivation seal  
sha256 : b372c91d91c4134850d6c8fb14dca1272e0daaeaacce8ca48d23bd0f66a3498b; engine receipt  
sha256 : d7416c15ecaa36ba63cd90357f14a677eba102cc7a50158fc5fda860d820aebd at  
receipts/engine/model\_admitted/SFT-MAT-SF-CIRCULATION-001-d7416c15ecaa36ba.json; empirical  
validation sha256 : d0a3ab6803a908c6d5278dadb5d3099c6ef00e2f4b1c721818568eaa604bcb2d;  
measurement receipt sha256 : 01272ae44e4f4e7d9ba40480b6afd4158ecdb7a6bdd572fceb256005fe603f75;

isolation sha256:a6bf12743e23eb7e27f83abfac2fab464763d8dcc4f16491218c03255d3b3e3a; custody sha256:b9112c2aaa8190490bcecf35ccc1ef0b1a0a82050c5612ebe0472af434.

45. Topological material invariant

Claim identity: SFT-MAT-TOPO-INVARIANT-001

**Question and exact theorem.** A material invariant is topological when its complete global path class is unchanged by every generated local deformation that retains connectivity and boundary records.

complete-global-path-carrier\_\_connectivity-preserving-equivalence\_\_local-deformations-quotiented\_\_integer-path-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                         | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | complete-global-path-carrier        | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | connectivity-preserving-equivalence | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | local-deformations-quotiented            | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | integer-path-class                       | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:13efa656a19aecf32a12b6813904ce4d2fcde0814a104238b0f2940d99e2c78e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e181bd02e17925831b4a6b22c0d3ae5c3578c1a6ce53367eeabced88df33d8a6.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:8905e525f2d1831fae398bb4d42ebe70cd9d0eb4243660c738c48c791372c83e.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 2b0c618640e78c9e6a61d0cd591b39a31111d3a4bcb515fa4230310307ad5b0f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: 52be2a7f7455b7444b99c8ee1fb48683ea99fdce6dd35ca5f94546db602cac5a; certificate  
sha256: 324729b6e2e4e99286bacf7e230b6054ad0584c770944fde8bac4f7d7e3ca46f; external-validation  
hash sha256: d5da1e69cff9663a4438aa6a46b03c0d80ee306d6917c10d13e544d6c476d28d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: e25423817b22188b5ad5603572fe72556d6d623de47ea8d79cac1a22ac17e8e8. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - TOPOLOGICAL - INSULATORS - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot  
experiments/external\_sources/materials/snapshots/nist-topological-insulators.html;  
sha256: 4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological  
invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-topo-invariant-001-authority-correspondence: predicted complete-global-path-carrier\_\_connectivity-preserving-equivalence\_\_local-deformations-quotiented\_\_integer-path-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-global-path-carrier\_\_connectivity-preserving-equivalence\_\_local-deformations-quotiented\_\_integer-path-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered topological material invariant observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256: 39b15b0d045f1b1d1b7ff8ad2ee76ac387e0f1a555cdf04c6cfe0a7e5c93e16a; derivation seal  
sha256: 355e1bda262e641217018e5bfd12142fb515f53642ce68873f49b667d47e9dbb; engine receipt  
sha256: 0aa06ea813c11acea4d063c83c11e606519be491423b0806a196e2876fe9391d at  
receipts/engine/model\_admitted/SFT-MAT-TOPO-INVARIANT-001-0aa06ea813c11ace.json; empirical  
validation sha256: e2546e5a5c2cffbad153836b434eee2a7d90ffd41f42024f1d71c7fbfa5e9c84;  
measurement receipt sha256: c2ed694d7588b3efc388b89c3d8225bce6a6b223050cfa18c3666f6752c8d69b;

isolation sha256:d877347a8eb8de659a662a2d14c4f0f00357e99386626f2e72ebd2da2b6a733e; custody sha256:1ec350459cdf675e1b52403b1065a89816d482f69231144349ef769376bfac96.

46. Bulk-boundary protected transport

Claim identity: SFT-MAT-TOPO-BULK-BOUNDARY-001

**Question and exact theorem.** A nontrivial bulk path class forces a boundary recurrence whose removal would require closing the retained bulk distinction; compatible local backscattering paths are therefore absent within the declared gap.

bulk-and-boundary-joint-carrier\_\_global-class-boundary-obstruction\_\_local-backscatter-paths-excluded\_\_gap-and-boundary-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves bulk-and-boundary-joint-carrier, global-class-boundary-obstruction, local-backscatter-paths-excluded and gap-and-boundary-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                       | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | bulk-and-boundary-joint-carrier   | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | global-class-boundary-obstruction | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | local-backscatter-paths-excluded         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | gap-and-boundary-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bulk-and-boundary-joint-carrier\_\_global-class-boundary-obstruction\_\_local-backscatter-paths-excluded\_\_gap-and-boundary-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: f204c5e7146eb04130ad4807231ad0637e358eeb604715a0c22917bb8e16da4a.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8ed623d8b4faf8ed6f5d8eabc89a41c340ddf240b0b45ecc9e15bd39089bc68f.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: ad1aad827f8955b0b0041cf25546c6841d11c1b396331884019eeb73aab28e00.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 4d467a33e09e676142c88ff0e26ce9d516bfa0201a789ecef92a812414280b8f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256 : c1ad057b49304b4368ba4fad93e4f0ee05abf12ee690c16fd5c5d7c3a8eda48b; certificate  
sha256 : b0899dd406db35b3d8d9d3b00a5254964967f26d618d1a345bb4f7a4f3537c19; external-validation  
hash sha256 : 5b8e12fedd650e5769d332dbaf6afda85a21d1f6ab5c33aede7b21fedad07d32.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 1d4a59568502972f6927fa8cb2944be5ea47b884c616a7a56c4d71c0d8fede16. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TOPOLOGICAL - INSULATORS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/topological-insulators>; snapshot  
experiments/external\_sources/materials/snapshots/nist-topological-insulators.html;  
sha256 : 4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological  
invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-topo-bulk-boundary-001-authority-correspondence: predicted bulk-and-boundary-joint-carrier\_\_global-class-boundary-obstruction\_\_local-backscatter-paths-excluded\_\_gap-and-boundary-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bulk-and-boundary-joint-carrier\_\_global-class-boundary-obstruction\_\_local-backscatter-paths-excluded\_\_gap-and-boundary-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered bulk-boundary protected transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting\_superfluid\_topological grammar that preserves bulk-and-boundary-joint-carrier, global-class-boundary-obstruction, local-backscatter-paths-excluded and gap-and-boundary-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256 : 624e50eb6c37a7cb6aa302bb729524f97d404589da3d4dacfb4b391c2f0fd9ee; derivation seal  
sha256 : 786171e535a2997aaaa15642857733ac66028d16eab119c2349d6501ea056489; engine receipt  
sha256 : abb28732eccdc748940dce9e3629db011dc41a205ef3b968b06d5cf1c0fb4f1b at  
receipts/engine/model\_admitted/SFT-MAT-TOPO-BULK-BOUNDARY-001-abb28732eccdc748.json;  
empirical validation sha256 : 74b655977a85543c9d7d4722f055436f75559358e69d248ace5caaa8253b1b88;

measurement receipt sha256:6fe03baefebf6c5f8b5194955b9d1132490c993a55fc531f5407d89ae26235b7;  
isolation sha256:6b2e7114ed5081ebd7faf0314e4d1e0aa595766da8b28fd6ea5bbb6f8dc75bc0; custody  
sha256:03b7784db09f06251f28e2779209ce357bee47b308dce3fabf2cb5625b6e2dc5.

## 47. Superconducting isotope-response record

Claim identity: SFT-MAT-SC-ISOTOPE-RESPONSE-002

**Question and exact theorem.** A superconducting isotope comparison is admissible only when isotope identity, constituent-mass distinction and transition response are retained together while the observation remains incapable of selecting the pairing law.

paired-material-and-isotope-identity-carrier\_\_isotope-substitution-to-transition-response-relation\_\_pairing-law-independent-provenance-retained\_\_specimen-method-condition-bounded-response\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-SC-PAIR-001
- SFT-MAT-PHASE-TRANSITION-001
- SFT-CHEM-ELEM-ISOTOPE-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving paired-material-and-isotope-identity-carrier, isotope-substitution-to-transition-response-relation, pairing-law-independent-provenance-retained and specimen-method-condition-bounded-response.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                          | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | paired-material-and-isotope-identity-carrier         | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | isotope-substitution-to-transition-response-relation | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | pairing-law-independent-provenance-retained          | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | specimen-method-condition-bounded-response           | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace             | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                           | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                    | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                        | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: `paired-material-and-isotope-identity-carrier__isotope-substitution-to-transition-response-relation__pairing-law-independent-provenance-retained__specimen-method-condition-bounded-response__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `record-boundary`: isotope identity and transition response are retained without a universal exponent; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 94fd92ce0920aed5e96f9ac44e642ab502467bb08a12d724a44c2c2d21c9398f.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: f21319ea8ef1ac3766fecbd963eacd861eb50473dcf9179c57e4935140c92658.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: e2db6f23721473fd237164511f5c59498218b933478ddaf811445464216cdd7f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: c16ace5c2e9c0f270d5d3ae27ef67924b682328cbcdc69b58d803fe0d0c60029.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: e4da47dc86627db27b968f024b231d219853ae5f4e3b645c14a7bc0e6093ea84; certificate  
sha256: bd352525e4d60943a208319d14de9b2a49f8fd6ac05f10ea747d2f0e801405e3; external-validation



hash sha256:bd352525e4d60943a208319d14de9b2a49f8fd6ac05f10ea747d2f0e801405e3.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:f7fe74a4077c16022bd5816fa81794f75fb1b7a9be1aedb285474284704e694a. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27 - National Institute of Standards and Technology; [https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147\\_A1b.pdf](https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147_A1b.pdf); snapshot experiments/external\_sources/materials/snapshots/nist-superconducting-isotope-2026-07-27.txt; sha256:48fa89f9f63d467e9259e766f81fd8f854924ad479ec0006d14075cfe48d3ae3; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

**Explicit exclusions.**

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving paired-material-and-isotope-identity-carrier, isotope-substitution-to-transition-response-relation, pairing-law-independent-provenance-retained and specimen-method-condition-bounded-response.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:9cc83826c3a6be6add3771eb5e8946b13bfdedb21a8e4c7858277c74ac0ca0a9; derivation seal sha256:71ac33cc8762d33fd9f32a0930ae2c7e9fc20b15fbfec30946d731a6ffbc5efc; engine receipt sha256:a55394460feb15801fb15a215f8e6c6ab5b9fb62523db33307c4f721c15dc1f6 at receipts/engine/model\_admitted/SFT-MAT-SC-ISOTOPE-RESPONSE-002-a55394460feb1580.json; empirical validation sha256:f2d5a758829778c576c552b919b86b71464aaa09f2e926e02e9dff32d034e0bf; measurement receipt sha256:3d1c99fa2505455dd15bedca5b4a2d02c7fffafebd31cfa330f7dc7686ecfeeb; isolation sha256:505e2002a7536d3a6689bc18f93811c6179c22bc8f52754b75580209c58c542b; custody sha256:cfad539f900cf896749cefd972a21fffcfd0bbe1e91cefdba8fb4d1137a4a51.

## 48. Integer and primary fractional Hall classes

Claim identity: SFT-MAT-HALL-QUANTIZATION-002

**Question and exact theorem.** A gapped oriented two-generator material carrier retains whole winding classes and a separately bounded primary hierarchy of reduced positive odd-denominator parts, invariant under gap-preserving deformation.

gapped-oriented-two-generator-material-carrier\_\_whole-winding-and-reduced-odd-part-relation\_\_gap-preserving-transport-classes-retained\_\_integer-and-primary-odd-denominator-filling-census\_\_complete-state-transition-boundary-trace\_\_root-bound

-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-TOPO-INVARIANT-001
- SFT-MAT-ELEC-CARRIER-DUALITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving gapped-oriented-two-generator-material-carrier, whole-winding-and-reduced-odd-part-relation, gap-preserving-transport-classes-retained and integer-and-primary-odd-denominator-filling-census.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                        | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | gapped-oriented-two-generator-material-carrier     | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | whole-winding-and-reduced-odd-part-relation        | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | gap-preserving-transport-classes-retained          | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | integer-and-primary-odd-denominator-filling-census | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace           | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                         | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                  | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |

| Axis      | Eliminated form                      | Forced form   | Exact elimination / retention basis                                                                 |
|-----------|--------------------------------------|---------------|-----------------------------------------------------------------------------------------------------|
| extension | free-fit-exception<br>-or-extra-rule | no-extra-rule | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present. |

**Unique survivor, base and successor.** Sole survivor: gapped-oriented-two-generator-material-carrier\_\_whole-winding-and-reduced-odd-part-relation\_\_gap-preserving-transport-classes-retained\_\_integer-and-primary-odd-denominator-filling-census\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- hall-census: the bounded primary hierarchy contains only reduced positive odd-denominator parts; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- false\_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: efdabcd3c4418e00dcb59d89337e8c44bcacf27ba5c4436d0aa642fc45cc492f9.
- tampered\_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: fb1c4724e66f9325dc667f98edbfe8f7eea78a89066058c5b775b2bfb88ec2b9.
- tampered\_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: a4e744f2ecee5e97eb0f6e47dd983fb87b6c837ab7f935d7f03b51c0268749cc.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 8f0c7e1c3bb60b2866b5329f881997595405b99cf2520e4cf080fe63800f0982.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 3dcaf426d44e402de2b381525079f7b2f183435006a131c8efc1ee2b0877862f; certificate  
sha256: fa11d3b9672c0cb258bf232d11fd91bf5d31df65359fc501fa8af42599dbf2e0; external-validation  
hash sha256: fa11d3b9672c0cb258bf232d11fd91bf5d31df65359fc501fa8af42599dbf2e0.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 3 claim-specific source discriminators were required; their ordered identity is  
sha256: 2551ebd4dc35246430f129face2ce828648425f4c82491bf347ce4c37d907108. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-HALL-QUANTIZATION-2026-07-27 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external\_sources/materials/snapshots/nist-hall-quantization-2026-07-27.html;  
sha256: 4536e34ab401e72216f8b9010bafcf9f1cacdbbfc47ca8fc998b781fa9125994c; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

#### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving gapped-oriented-two-generator-material-carrier, whole-winding-and-reduced-odd-part-relation, gap-preserving-transport-classes-retained and integer-and-primary-odd-denominator-filling-census.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:d9290b7cb8c1eba4c906fcec12feb75d9af06433ecf31acef8ed28f8815231988; derivation seal sha256:d20afc47519f5439e90081f10d46ee922c9ade6f7c477841dc669c59e5677379; engine receipt sha256:8bf14f74a78ae92fd3ff5878df726d92cbac72870b8af7f44601acf2b01397f5 at receipts/engine/model\_admitted/SFT-MAT-HALL-QUANTIZATION-002-8bf14f74a78ae92f.json; empirical validation sha256:f53d1d556220384209dc27cc0306054c51b5de98895eb8d0695e3c3cadece320; measurement receipt sha256:3a469de7be7ae142f5d62b887d5328d625080a048be09342e457bbcc5d86bb66; isolation sha256:928f854b15080f487374967b75ab4e6ba0890c1d04168758285303f4e70f0cfc; custody sha256:be256a30cf9a0ea3ad6a654e2f75833deaddc556892def83888c95ae99d8f0fb.

## 49. Exact topological boundary-mode count

Claim identity: SFT-MAT-TOP0-EDGE-COUNT-002

**Question and exact theorem.** For two distinct adjacent whole bulk winding classes, the exact protected boundary recurrence count is their positive whole gap with orientation retained as a label rather than a signed number.

adjacent-bulk-winding-pair-carrier\_\_positive-whole-winding-gap-relation\_\_bulk-orientation-and-boundary-recurrence-retained\_\_exact-protected-edge-class-count\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001
- SFT-MAT-HALL-QUANTIZATION-002
- SFT-MAT-TOPO-BULK-BOUNDARY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving adjacent-bulk-winding-pair-carrier, positive-whole-winding-gap-relation, bulk-orientation-and-boundary-recurrence-retained and exact-protected-edge-class-count.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                       | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | adjacent-bulk-winding-pair-carrier                | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | positive-whole-winding-gap-relation               | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | bulk-orientation-and-boundary-recurrence-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | exact-protected-edge-class-count                  | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace          | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                        | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                 | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                     | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: adjacent-bulk-winding-pair-carrier\_\_positive-whole-winding-gap-relation\_\_bulk-orientation-and-boundary-recurrence-retained\_\_exact-protected-edge-class-count\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

**Operational witnesses.**

- edge-gap: bulk classes five and two force three boundary classes; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : b0f8a1570544fe9e553b882d2599e7eb2c174700b4ba3a54af58951a5eab4531.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 0435e9e62c8543856b653f65753315b2720a09a2a8c63d8a4b17e43d328c5c6b.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 6a85996d29e292434ba2ad82021ff6c843a64e1cf1ccf64ada1efc2b7d39f816.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 4e2210aa723d5207ae3e372cf3b8277d4fe7023863ccab3eef2b4ce9e3132a2d.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 68e58fe1d0215233990f1fac100ed7b05b73412a0fd8e54d63e8c235d3ae8a11; certificate  
sha256 : bab99db1cc19f41183013609ec5b8ff2a0384c102f8296c8625320820eacee25; external-validation  
hash sha256 : bab99db1cc19f41183013609ec5b8ff2a0384c102f8296c8625320820eacee25.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 6faca2d50bb9d04fa3b6f0418840d7433420dd842e94e195b0e330114bc786a2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - HALL - QUANTIZATION - 2026 - 07 - 27** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external\_sources/materials/snapshots/nist-hall-quantization-2026-07-27.html;  
sha256 : 4536e34ab401e72216f8b9010bafc9f1cacdbbfc47ca8fc998b781fa9125994c; scope: post-seal claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: True
- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

#### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule



- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor superconducting\_superfluid\_topological grammar preserving adjacent-bulk-winding-pair-carrier, positive-whole-winding-gap-relation, bulk-orientation-and-boundary-recurrence-retained and exact-protected-edge-class-count.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:4f4001fc31772c1638559bd8221ac0bce34d58e0c13b761118a5d68f32af7b7f; derivation seal sha256:1b1b1e75f7751982016da82cf278cc89f9c5ff8e8eba04fbfd9302a46a5f0005; engine receipt sha256:5b742a33b9d5b5822b2ff2c8db6375cbe0b9f3e605643f187fee05f43a921c57 at receipts/engine/model\_admitted/SFT-MAT-TOP0-EDGE-COUNT-002-5b742a33b9d5b582.json; empirical validation sha256:5cac72ae62bf75b24d851e95ace663da5c23e2af0aab8998a300fa047eea0012; measurement receipt sha256:81460434a3dacd79b30d5e6ed2a7e9e1f717b3ec5ebef047eff1afb352d2f3cf; isolation sha256:e63e4d1570fddfad718942053f80e4f9bfecdc47fc34fd58412f66b0bb9f9ebc2; custody sha256:9dfcc9d49aacd64cf256ffa9f50b353709b943c016f4fac3881755be01c05e31.

## 16. Mechanical

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

### 50. Stress and strain records

Claim identity: SFT-MAT-MECH-STRESS-STRAIN-001

**Question and exact theorem.** Stress is oriented load transfer per generated boundary support; strain is the exact retained change of internal separation relative to the same reference material state.

material-region-and-boundary-carrier\_\_load-transfer-and-shape-change\_\_direction-support-and-reference-retained\_\_method-and-scale-bounded\_\_complete-state-transit ion-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches **SFT - ROOT - THERE - IS - NO - NOTHING**.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-region-and-boundary-carrier     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | load-transfer-and-shape-change           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | direction-support-and-reference-retained | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | method-and-scale-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: **material-region-and-boundary-carrier\_\_load-transfer-and-shape-change\_\_direction-support-and-reference-retained\_\_method-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule**.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: **depth\_independent**. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier**: the generated carrier label is held and nonempty; passed **true**.
- **relation**: the generated relation and organization are separately retained; passed **true**.
- **observation**: the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a

universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:adb5d05a21d454b6bf5a2017f8be3f371d55fbd9a0b7ad25b947caa373ad534f.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:109a21a2e46dcd23f5466f23be5156d57b500eff7391011d89ecc112be5e6872.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:6391dbcd50225bace528b7afb330ee156b47e635d52cdd610f75541e72b83b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:04e5725ac444cc83fe68338efe05502691abd15b8808ea95150d1c8bb7212fe6.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:60cf9be4f23f11d4bef081dbc622511a6b00ba9a95d7c50d982b096248f1af5e; certificate  
sha256:166b798626def8349847d2bbf1f68dca63d749c0a498661c151eb032cc0da216; external-validation  
hash sha256:64538778c885dec7c2c0b9e159dc79e019cbcc17cace487879a2ca3a86e5b28f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256:58cbbf4e31983da152d2e9df9813867504aea0339fd3dd88255134648592ea34. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FATIGUE - FRACTURE - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot  
experiments/external\_sources/materials/snapshots/nist-fatigue-fracture.html;  
sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:  
processing-structure-property-performance, defects, fatigue and fracture.
- **NIST - SUPERCONDUCTING - GRAIN - BOUNDARIES - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature>;  
snapshot experiments/external\_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2;  
scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- **sft-mat-mech-stress-strain-001-authority-correspondence:** predicted material-region-and-boundary-carrier\_\_load-transfer-and-shape-change\_\_direction-support-and-reference-retained\_\_method-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-region-and-boundary-carrier\_\_load-transfer-and-shape-change\_\_direction-support-and-reference-retained\_\_method-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered stress and strain records observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:72ec1a5c263bd16dcc14f88e80cde6ddbce93c49574494ea6d506ef427d69a6b; derivation seal sha256:d3b9e2aa529d399c2335b68a60fb2429b910b89d6a5c9463d99af63692002ffa; engine receipt sha256:85482dc44ea7d4480cdd7474370679a5ef92c2b374ade6acd5604e5da740e766 at receipts/engine/model\_admitted/SFT-MAT-MECH-STRESS-STRAIN-001-85482dc44ea7d448.json; empirical validation sha256:b255306584d1626058d63a03e6c05d19c2039983e3939726f8e7e9d1feae2f65; measurement receipt sha256:905fe24fd20a32060c8e829ca181725b21b382c21b9580ce56ac405699ed9f02; isolation sha256:4eb94e72da4ebae40c69edca01de9e83b273d60729611f3ac44fa71099fd27e2; custody sha256:f680002c5dbcf1e4fa1ab346fedf28362a1c9de3b8c2c7e067d99fe7112d62b.

**51. Elastic recovery**

Claim identity: SFT-MAT-MECH-ELASTICITY-001

**Question and exact theorem.** Elastic response is a material transition path whose load release returns every retained constituent and adjacency relation to the original canonical state.

bond-network-state-carrier\_\_reversible-displacement-path\_\_original-adjacency-classes-restored\_\_elastic-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001

- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bond-network-state-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | reversible-displacement-path             | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | original-adjacency-class-restored        | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | elastic-boundary-declared                | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bond-network-state-carrier\_\_reversible-displacement-path\_\_original-adjacency-class-restored\_\_elastic-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : d965c15a37557dcfc3605cc273e1a1880c9e41083dbcf9d3a708f979636fd61.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : cd6eb0e5456c7e6e8a8e5f7fd98dfc5b0a7eae9a32d7ecce593b09f29c908bae.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : ac6d6673b36ac8725d7bfe8f9d12e9924b9024d0edd255e90883210b8efb8fe9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 47aaed4f4a0ea0724465aceb9bf4aba1ef44ce469cad8d3d06fa3e0f47f21236.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : b0a0f914de0a4b46b3d7efd5c82bdd05cd890bec50d642f36863a2e98494e644; certificate  
sha256 : 9d1e0777b17da1c0589983ef1da7f0c3ec0ba76e129d86df0a1b1a58f37c3c93; external-validation  
hash sha256 : a0374dd3a632fe0a266ad5bf3f615aac82e7a7b413a508f0b7c44bcd0e433ad.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : 1f5231592213d1cb22ac6eee1db2e48d66131790c95c6401a6712d5e12f4bc25. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SEMICONDUCTORS - 2026 - 07 - 24 -** National Institute of Standards and Technology;  
<https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot  
experiments/external\_sources/materials/snapshots/nist-semiconductors.html;  
sha256 : ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope:  
semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- **sft-mat-mech-elasticity-001-authority-correspondence:** predicted bond-network-state-carrier\_\_reversible-displacement-path\_\_original-adjacency-class-restored\_\_elastic-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bond-network-state-carrier\_\_reversible-displacement-path\_\_original-adjacency-class-restored\_\_elastic-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered elastic recovery observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity



- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:b346b8fee12d428022c4ae51969a24edb3f509f9de381374c5735ea1eb0d92a5; derivation seal sha256:260626b207e5763c1058d0fc4ec4281cb5ce31d71314b505f4027525115ab722; engine receipt sha256:b490d2bc4771b5f91d7458764359748fcb9fd73ecb665b5f66637be92c9d9b17 at receipts/engine/model\_admitted/SFT-MAT-MECH-ELASTICITY-001-b490d2bc4771b5f9.json; empirical validation sha256:91e76ec99100ab2b4bcb9de89117654abe30c261b652a123188a14eccb7d5e2e; measurement receipt sha256:0fc70106063eb948741b12ccaf1d72685502d7185847683b38e43c272ba1b467; isolation sha256:5f44ba6f53e33a525c41ec5a037db2da522c7f9ab958f32f1cc61771aa19cae6; custody sha256:4746d3544ccf6e014f821317b7525878c79c6de8fa91be7cb75968f039c58fc0.

## 52. Plastic deformation

Claim identity: SFT-MAT-MECH-PLASTICITY-001

**Question and exact theorem.** Plastic response retains material content and phase identity while a defect-mediated transition changes the canonical adjacency organization after load release.

bond-and-defect-network-carrier\_\_irreversible-same-material-reorganization\_\_content-retained-adjacency-changed\_\_yield-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                               | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bond-and-defect-network-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | irreversible-same-material-reorganization | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | content-retained-adjacency-changed        | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | yield-boundary-declared                   | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace  | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure         | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                             | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bond-and-defect-network-carrier\_\_irreversible-same-material-reorganization\_\_content-retained-adjacency-changed\_\_yield-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:9150c7c34bf6ef1d0009e84afa667c913cb1142a022c9425f3fc6a140fffd3b9d.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a7d6901fc35b585477d6b688a690b3c162e2e621559cad8b4d96464a4ec0170f.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:844e5f851097100443c4f3d0dfa284fbae50e59dff142d5d5c9b8fdddf629d3.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:316c01f0fe9eb2fb72cde97a9c3e7af4199b41650b03f8a77be1d286fd0b766f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:f15e6957edff21a3f20f016b86d2c1eec8d49f37c03d565312ea934715ed36d1; certificate  
sha256:db298f3a0e0fc3ab181511dd1bea67a8856c29679fb939576d2814ae6dc44876; external-validation  
hash sha256:20e178859f18007683e55801f7a6cb2240a269f32c4aa11dd20fa58e04218056.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:c36087458657bcd08909aee6600aa4bcabf26dce1f9f96112864f0e05ec356d9. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TEXTURE - PHASE - FRACTION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot  
experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html;  
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase  
fractions, texture, deformation and uncertainty.

Comparison and custody records:

- **sft-mat-mech-plasticity-001-authority-correspondence**: predicted bond-and-defect-network-carrier\_\_irreversible-same-material-reorganization\_\_content-retained-adjacency-changed\_\_yield-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bond-and-defect-network-carrier\_\_irreversible-same-material-reorganization\_\_content-retained-adjacency-changed\_\_yield-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered plastic deformation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3006a582b76abd466eb5e052f89c2465497a3c270e20fe6db6dbb339179a8ac2; derivation seal sha256:12ebaca61ddae97d7c9eb477672cc0de736f0ab3c7f9af57c6d2e80f45d61158; engine receipt sha256:77ce9abdbdc078fd565e3a48d5a5b2f65b128d91f104012cc32b4c4a02eb7d2f1 at receipts/engine/model\_admitted/SFT-MAT-MECH-PLASTICITY-001-77ce9abdbdc078fd5.json; empirical validation sha256:b3532d1895dd1296d276e1735e26099cf264926c90c65d74200d5e625debdfe7; measurement receipt sha256:391ccc840a880b49ecdd2940ed67b3596c473564b0d566fe1b63c106bffed26a; isolation sha256:d873591d0a8d9bb028422f4b911183a711f9c215eb9344d67d2f95798c1a8f7e; custody sha256:efd5e346f7ffd5349aa7bb765688da67e26c84fda13093741e87732a585a073b.

### 53. Dislocation-mediated slip

Claim identity: SFT-MAT-MECH-SLIP-001

**Question and exact theorem.** Slip is a counted whole-lattice translation propagated by a retained dislocation along a compatible plane/direction without requiring simultaneous rupture of the complete plane.

dislocation-and-slip-plane-carrier\_\_whole-translation-advance\_\_burgers-label-and-content-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | dislocation-and-slip-plane-carrier       | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | whole-translation-advance                | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | burgers-label-and-content-retained       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | slip-system-bounded                      | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 89538877c541c0e3b80b081a953e40d057a01f631b47a0e8282b46c7a892f43f.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 0151dd250ba88c10559598d8778c71c3c11e36e9edcd1034f8ee2fc45cbfb0c5.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : f077732e53b69bf50948fe5266a30e63f5f80a7293c8bc4ae7d2a46e0d248e76.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 16203aa2aa450696e0421d92c1b3cc0a003c3fd86ca1413a987ca923e90c1af7.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : ce30006112583d5a105cd9e64a8deec6bdef9c9a58e2ccafa5b65005bd351415; certificate  
sha256 : c9a1f12156a69f5368449a3cc781bbff9923903bd31e5d7f38c8d32b9beec75; external-validation  
hash sha256 : 1ca21fd13495c4d6ec9d83b742a4d8006a197f875c48dccdd1140cb435248696.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : a5c1c011368a43bfcccc04cf8c412ee846c22c570ed3a97aa228d2280274c1f1. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - QUASICRYSTAL - MEASUREMENT - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot  
experiments/external\_sources/materials/snapshots/nist-quasicrystal-measurement.html;  
sha256 : a1b7cc0a5882502bb31b9706225b926c9fffb1e8d93ea5c0043249376c4d6ff18; scope: microscope  
confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- **sft-mat-mech-slip-001-authority-correspondence:** predicted dislocation-and-slip-plane-carrier\_\_whole-translation-advance\_\_burgers-label-and-content-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived dislocation-and-slip-plane-carrier\_\_whole-translation-advance\_\_burgers-label-and-content-retained\_\_slip-system-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered dislocation-mediated slip observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed



- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1e7c4abd9525c4f5aa9107b87bff628f69fee285b2d2a20cf4c4b5ddb071f60a; derivation seal sha256:5c9bc421d35c9d3b9ed2d15aacce9dda69982b549984f8c43b1d769f9d16501e; engine receipt sha256:ca7d3c3104a0dfdf5dde9127e49e575edfbba784c1c2d58859ab2f22267eb1e3 at receipts/engine/model\_admitted/SFT-MAT-MECH-SLIP-001-ca7d3c3104a0dfdf.json; empirical validation sha256:262f49da288b06c0eadc6b88cf734f56ef39fe5c5f9ca0093b53805735c49a9b; measurement receipt sha256:612c261a80046a405268cd495f42234bf4b68f6b2b4e82c49826f0939b1e8b52; isolation sha256:7274e98371ac622e94f1b2247c5c951a7650f0e5270ff34fc798957594909bbd; custody sha256:d3ef1d91e8259ecd6710a62a7ebf7ee7c3741be9f8d9e6b5668b3591fa03c339.

## 54. Elastic modulus as response ratio

Claim identity: SFT-MAT-MECH-MODULUS-001

**Question and exact theorem.** An elastic modulus is the exact stress/strain response relation for one declared material direction, condition, reference state and reversible response window.

elastic-state-pair-carrier\_\_stress-to-strain-exact-relation\_\_direction-condition-and-range-retained\_\_linear-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | elastic-state-pair-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | stress-to-strain-exact-relation          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | direction-condition-and-range-retained   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | linear-window-bounded                    | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:7d6248f498d15a2edbf0aea112d99e5a2f1daf19a804222723337b55c86c4ace.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:06222eed10c6b068ebb6c92d7456fde86013ee12119f30ebb92fe895b8f0fffc.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:2bbd140e1af9ff6e2ba9427d90da1fbe2b21c9c8a379fecc1902d3f131edf130.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:9d740d421a442846bdb6849544d4930fc9837d3a2f4ef6f6a838b16d1b9541f1.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:18bf6748d138471a11dcd6ba96c75fe25842d895dd65d2f2ced937d8eca68784; certificate

sha256:36744a94fc531a579c5f99ec93300f4892e6f6e9dd73cd52fe56f5d70fba9de6; external-validation

hash sha256:111c32e35dd4f098864a80c781a12cc6cf9e944f89cbac2640eb53a44f1a4854.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:85c03f12333c98d0238b67dfce07d56b00dbd561755af223f44fbc04919ff7e9. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-SEMICONDUCTORS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot experiments/external\_sources/materials/snapshots/nist-semiconductors.html; sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- **sft-mat-mech-modulus-001-authority-correspondence**: predicted elastic-state-pair-carrier\_\_stress-to-strain-exact-relation\_\_direction-condition-and-range-retained\_\_linear-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived elastic-state-pair-carrier\_\_stress-to-strain-exact-relation\_\_direction-condition-and-range-retained\_\_linear-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered elastic modulus as response ratio observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1153be6043acfefa94d2197d88eb7ba687bea8d2e7db45b379d2b884ec2fe603; derivation seal sha256:e9f4cc9b2d4c4888f1326180b6143f911b195fa9076791fc8bb4e403fb286e78; engine receipt sha256:fd39c697c448c56d56a6d63557ec19a7b249f2ca0ff72bae1f895ee7fc77fd88 at receipts/engine/model\_admitted/SFT-MAT-MECH-MODULUS-001-fd39c697c448c56d.json; empirical validation sha256:841d85ef828079b201e69f2e3cb0bfc8c17992ebc430516174765f5a3641725a; measurement receipt sha256:e195b8c1f1a29cab72fdd2ba5198b76a7c3eaf63c8afce760acfd8ab09ba87bf; isolation sha256:9281062f09a9ad8c389d3cc4cd86a231f35c1720f0bb32a9799f953a5285a3da; custody sha256:0269158f503aae4190601227607fce846f38422cd5356c8b20326c355ff2d890.

## 55. Strength and hardness boundaries

Claim identity: SFT-MAT-MECH-STRENGTH-HARDNESS-001

**Question and exact theorem.** Strength and hardness are method-bounded thresholds at which a declared loading path first produces retained plastic, indentation or failure organization.

material-and-test-carrier\_\_load-to-persistent-change-threshold\_\_method-geometry-and-scale-retained\_\_threshold-not-universal-constant\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-and-test-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | load-to-persistent-change-threshold      | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | method-geometry-and-scale-retained       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | threshold-not-universal-constant         | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: material-and-test-carrier\_\_load-to-persistent-change-threshold\_\_method-geometry-and-scale-retained\_\_threshold-not-universal-constant\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 862336e016548ff4c2c82fd3ac77ca0323be9c77556e971a6aebbacd0c76fb5.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: bc14a47bca29bd87794372fa028e2771a41e0b2d2e2b3f1ec19666f2e362e0c7.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 4be80645c59da9b639daaaf753ec6ce5c2df9dec4c06899ac073ceb7572efc38.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: c2c68a86de634604b832653e2398fde3c925a952212ee0b01e72d81390035d1d.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: bd81bef0333505d69e1bb6b71f2ea590586d3910f93d18e8c9770aa2d979ea68; certificate

sha256: 6e74e10acadf20f87e2b884dbb28242db27c0b03c2917387319e000cb73953a8; external-validation

hash sha256: 05602a8fb8c3d7a23827d52a2847b910135f7aa50c9622b0198f2e704299548c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256: 306bd49533d46cb19fcf0627564636eda8ee8cc1a15790255742306b9d9a8c67. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - POLYMER - COMPOSITES - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-composites.html; sha256: 0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- **sft-mat-mech-strength-hardness-001-authority-correspondence**: predicted material-and-test-carrier\_\_load-to-persistent-change-threshold\_\_method-geometry-and-scale-retained\_\_threshold-not-universal-constant\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-and-test-carrier\_\_load-to-persistent-change-threshold\_\_method-geometry-and-scale-retained\_\_threshold-not-universal-constant\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered strength and hardness boundaries observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest



sha256:8c586386b9a6b3354402a0c4f30ff03c2f27d866530dcb39d08757be0caad2e2; derivation seal  
 sha256:74b8b045099cf7851df7063a11da3c2a0e7870c7148bf04fb703f638f213e800; engine receipt  
 sha256:2db59637af7814e2711eba819d183bafc63ff7b259573bed972188dd7aaf2a08 at receipts/engine/model\_admitted/SFT-MAT-MECH-STRENGTH-HARDNESS-001-2db59637af7814e2.json; empirical validation  
 sha256:0e7d03af5c7af07d3e7714193bd5d51bb707adabef48ef5c6d82ed7100c6176d; measurement receipt  
 sha256:43ad4c5ed82a361d710e614767541948b7402e476eac6a6b773b321d8ee0ec09; isolation  
 sha256:cb858f2adbe1786193d85075804549f15206839b715fad6d7a87296ed263bf6c; custody  
 sha256:7605813ca3176eda76d4d8ce6a2512d1a68bb9f25cf017efef4d138523c4197f.

## 56. Fracture and toughness

Claim identity: SFT-MAT-MECH-FRACTURE-001

**Question and exact theorem.** Fracture is connected bond-path separation creating new retained boundary support; toughness is the declared resistance relation for crack initiation or advance with a complete work ledger.

crack-and-bond-network-carrier\_\_advance-versus-resistance-relation\_\_new-boundary  
 -and-energy-ledger-retained\_\_geometry-and-rate-bounded\_\_complete-state-transitio  
 n-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_  
 \_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | crack-and-bond-network-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | advance-versus-resistance-relation       | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | new-boundary-and-energy-ledger-retained  | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | geometry-and-rate-bounded                | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: crack-and-bond-network-carrier\_\_advance-versus-resistance-relation\_\_new-boundary-and-energy-ledger-retained\_\_geometry-and-rate-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : a5733f9eb6f9903bd7d45570055f974ecaedde3f68a6a62ed63ebde4607095d2.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 6a5d0a7bb0a8e2f66788209faf7630ad1686264d0c376bc6aa1c3c9c934ce1e4.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: f27d462d208ffce1fd4f85a280d48e599aefab736bdf059196e25dc0714e5118.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: 6509c61f547b936474888b2417bcd4a5c72cfd3a0ae057432717257c1443a7e0.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: d6bc4e4588e8d5185f39670215ad2f0e681c18f43d64af3415da20f97af3e995; certificate

sha256: f13fcb4efc42df89e9e318ffd78f831ed7652201114029d301e13d7ace8e69e6; external-validation

hash sha256: a99696668e62b4f2d3dce17f8752af6a911915de7dffa7eb432db42293f12515.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256: 01b75c79f54df1e6372093d29ee1c45470f56ff393dbed86e238d8a2c2e91a74. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FATIGUE - FRACTURE - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot experiments/external\_sources/materials/snapshots/nist-fatigue-fracture.html; sha256: 449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope: processing-structure-property-performance, defects, fatigue and fracture.
- **NIST - POLYMER - COMPOSITES - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-composites.html; sha256: 0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- **sft-mat-mech-fracture-001-authority-correspondence**: predicted crack-and-bond-network-carrier\_\_advance-versus-resistance-relation\_\_new-boundary-and-energy-ledger-retained\_\_geometry-and-rate-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived crack-and-bond-network-carrier\_\_advance-versus-resistance-relation\_\_new-boundary-and-energy-ledger-retained\_\_geometry-and-rate-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered fracture and toughness observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:bbc88ba37928e3ac9514e4b72395806d1d0a2cc60f604090a4fabd73f7b3bdbc; derivation seal sha256:8b5df3b57c3581998968ea6f3337fb3d1ade12a540eaae72c4dc0b7e3bf1a88b; engine receipt sha256:66cd7e1bb5e969fb2cf9920411459685ec4b19cce6ddc54415e5bf1e449497cf at receipts/engine/model\_admitted/SFT-MAT-MECH-FRACTURE-001-66cd7e1bb5e969fb.json; empirical validation sha256:31ff5bbedcc5575ff326542e7720d80181ab36505299b0828dd244e41ab283b0; measurement receipt sha256:f5fcd156b6b0806c04a181af2e91fb90172cfe31ca081e8d9fb7b3f2e123eabb; isolation sha256:90c60d61e5bac21634efcd59ce8c04fb5e939f5118bbfad4c168a6032a6678e1; custody sha256:1f63d814bc73955153dabccf74d08cbe10b6657f10c76bb37a2872dff423ae46.

## 57. Fatigue and creep

Claim identity: SFT-MAT-MECH-FATIGUE-CREEP-001

**Question and exact theorem.** Fatigue and creep are distinct history-dependent material processes: recurrence-counted damage under cyclic load and retained deformation under sustained load and time/temperature conditions.

history-labelled-material-carrier\_\_cyclic-or-sustained-transition-accumulation\_\_damage-defect-and-time-provenance\_\_load-temperature-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

**Boundary:** Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                                 | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | history-labelled-material-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | cyclic-or-sustained-transition-accumulation | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | damage-defect-and-time-provenance           | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | load-temperature-time-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace    | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                  | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure           | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                               | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: history-labelled-material-carrier\_\_cyclic-or-sustained-transition-accumulation\_\_damage-defect-and-time-provenance\_\_load-temperature-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

**Base:** One complete material carrier retains its generated relation, organization, observation boundary and proof record.

**Successor:** Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

**Closure scope:** depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:138e417ec271e2d16689255fb807b76d4e11a863a53f75a4be29e4664f40edaa.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:f4a38be94309bdd54a09b8c098de0bf328e971aaf99abb5afda17dce0565aaa3.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:5964c78e400b2cccbe8f604adf24d54cb4afaadca4cf3524164fcc0b10c9811e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:96533e94cd35ecc95cede0615590e5f0915afcc9ae1e73b589bc9ad5feb6ff99.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:50718c3f7afe9f2f21ff7e49da32cb9694d31dceb34873026cf5ab117835fb04; certificate  
sha256:b9dc76d661417daa718f8f2c9614ab248424d3786dd7fb08cfc03b9d7680ae6; external-validation  
hash sha256:22152c9fcd29cc509c7d11b289060e20c3b52f613b2a4721bd6678fd3f58976a.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256:30bae96d9d87e88765e2e13ea6f741387b1141dca008910c29f8318ec151b536. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FATIGUE - FRACTURE - 2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot  
experiments/external\_sources/materials/snapshots/nist-fatigue-fracture.html;  
sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:  
processing-structure-property-performance, defects, fatigue and fracture.
- **NIST - POLYMER - COMPOSITES - 2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/polymer-composites>; snapshot  
experiments/external\_sources/materials/snapshots/nist-polymer-composites.html;  
sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer  
matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- **sft-mat-mech-fatigue-creep-001-authority-correspondence:** predicted history-labelled-material-carrier\_\_cyclic-or-sustained-transition-accumulation\_\_damage-defect-and-time-provenance\_\_load-temperature-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived history-labelled-material-carrier\_\_cyclic-or-sustained-transition-accumulation\_\_damage-defect-and-time-provenance\_\_load-temperature-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule;  
exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered fatigue and creep observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor



- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:6259261c2f433280ac5b338213ad30a639c165bf120e029689cd7cc6b0f61e4b; derivation seal sha256:161e752c1fbcc5111fa1dca42857a94f4d8918ae581e3e23f81e30df3059a745; engine receipt sha256:0533e9af7333170230d829ce66828f6bc1982defb9c8fb311cf01e53196f13e8 at receipts/engine/model\_admitted/SFT-MAT-MECH-FATIGUE-CREEP-001-0533e9af73331702.json; empirical validation sha256:b2728846e5e1df985715934a704a0d3e072542786645908f030c00d21547ca7a; measurement receipt sha256:645d29b7f2bb33fcad245033ea69bd2534dba99f4df7b9a74d6b153c77f1df80; isolation sha256:fc65a26867d6ef55044a2f9e33afc6460d9a90381f191f27c46fe0a05893a6f1; custody sha256:a2f03701274180403fafd17c220b8ad21dad330cab01ff5cb46bd025c103928a.

## 17. Thermal Magnetic Optical

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

### 58. Heat capacity and excitation storage

Claim identity: SFT-MAT-THERM-HEAT-CAPACITY-001

**Question and exact theorem.** Material heat capacity is the exact relation between supplied thermal carrier change and the resulting accessible excitation-support change at declared conditions.

material-excitation-support\_\_energy-to-state-count-change\_\_condition-and-phase-retained\_\_temperature-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001

- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-excitation-support              | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | energy-to-state-count-change             | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | condition-and-phase-retained             | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | temperature-window-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: material-excitation-support\_\_energy-to-state-count-change\_\_condition-and-phase-retained\_\_temperature-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

**Operational witnesses.**

- carrier: the generated carrier label is held and nonempty; passed true.

- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 9b1c92f07930722df9def1146b48afb9053a331fe13649f8dbc811937f62bda5.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f5a5463e58ef4db286448b3ccf1df2c4a0f2f2c5934e286ad3c5c109878ac038.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : a6b3bf01252235db636981ba4ca9b966662f720f8b2f46ea7bfc07d9beb3bdc9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 09b10d18c04b2ebd6f0139afdc0e36b99291ceb2381d3b1485aa6dbff83644ee.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256 : c85a958591bc27b8bea1709497f13281fc169c363ae08e5f0d3c2d231bc34257; certificate

sha256 : 3b8c30953cea0c023f330da888aa7ee92a4162a93e01cd00fa210aa29c8a15ad; external-validation

hash sha256 : 609c83bbdd41d06080381f919f9aac4e58eb32bf48a496e68e78b41ed213737c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256 : 962425c0c7e89286623b46cb21a73a7bfd40a4e3549f825ec5937dd421beca5d. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256 : 4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- **sft-mat-therm-heat-capacity-001-authority-correspondence:** predicted material-excitation-support\_\_energy-to-state-count-change\_\_condition-and-phase-retained\_\_temperature-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-excitation-support\_\_energy-to-state-count-change\_\_condition-and-phase-retained\_\_temperature-window-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered heat capacity and excitation storage observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:e42ae574ea210e58e7db4db2b4dea63b4c2a952a2c3003d1494486b831a51cce; derivation seal sha256:7abc09c00bae8245182e239280d27c4e0fcd5752a0f60ddf6adc1531bfebc332; engine receipt sha256:9a919059d9d03fcdc366ad97dd95eed602d6df50b8db4d49e3acb1d492dcf1d8 at receipts/engine/model\_admitted/SFT-MAT-THERM-HEAT-CAPACITY-001-9a919059d9d03fcd.json; empirical validation sha256:6e1bf64a4d135d263d2f6e1a0a7ce530df19e721915617a6ed4e4cc56060a319; measurement receipt sha256:5c348ebe8e6ee46f9ac074932fb49e4af0501d044da5826744228e4d7d8afc9e; isolation sha256:e3b1e98bbbb1fa295e5694748d0a260d5e9516397fdad151680d860c0d743008; custody sha256:cb2428cec3ca8bfac3b1cd7e5da284e0d4f80482444e02eb6f6b01292f22deac.

## 59. Thermal conduction

Claim identity: SFT-MAT-THERM-CONDUCTION-001

**Question and exact theorem.** Thermal conduction is counted adjacent transfer of retained energy labels through connected material support with complete source, boundary and scattering provenance.

energy-labelled-cell-network\_\_adjacent-energy-transfer-count\_\_total-energy-ledge  
r-retained\_\_gradient-time-support-bounded\_\_complete-state-transition-boundary-tr  
ace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rul  
e

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001

- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | energy-labelled-cell-network             | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | adjacent-energy-transfer-count           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | total-energy-ledger-retained             | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | gradient-time-support-bounded            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: energy-labelled-cell-network\_\_adjacent-energy-transfer-count\_\_total-energy-ledger-retained\_\_gradient-time-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:19069236c3c4b3a7dffaab7b12bbef6c542b7fa84585b15712bd64dabb87e580.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e830330106bc942c9651b68e4685a201d4266513d442e71e66fd2321a9463b3e.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:14fe4fe4b473610bb862eb6beff2dfd07c036ad39ee6a12848a26d4e24a8af58.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:7a817c1177ec039fa173c7e2f4b2458c3bc9a478948cec1e1e7b45539ae54486.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:e6979253f0608812554abf41f81ef0099f77c5cc4c214da12304a1b55437d891; certificate  
sha256:8e9fd289da0b25f75129f0b35c2ace40504a81b64a66962a0b25cfa6e7386d8e; external-validation  
hash sha256:31ba40a4f94cd387d02921b23d30d4c580558c96bdbfe0ef2e03fce22fcc583a.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:f0aa7c18ee1acf7349c5163388a2ca04b1c881e3486769bc42c642df190e9803. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html;  
sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal  
conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- **sft-mat-therm-conduction-001-authority-correspondence:** predicted energy-labelled-cell-network\_\_adjacent-energy-transfer-count\_\_total-energy-ledger-retained\_\_gradient-time-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived energy-labelled-cell-network\_\_adjacent-energy-transfer-count\_\_total-energy-ledger-retained\_\_gradient-time-support-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered thermal conduction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity



- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:e897e584b3dd80bc4168883855f14430c86cd1a0d62bda4ad2814b4c6ebf8554; derivation seal sha256:1fdc4a1db8b717fa6573da8c1c79cf3d21145d4cc07736eb12ed7fd13e1e50c5; engine receipt sha256:1bc2b97d196c9750d41d454a0aa47fa7b36d142e108da02aeba0e7fed8134e33 at receipts/engine/model\_admitted/SFT-MAT-THERM-CONDUCTION-001-1bc2b97d196c9750.json; empirical validation sha256:e0c3e7778a79d6e1256434b13cb64fcc8c903bf2cadef8c62fc60890f47bd1d1; measurement receipt sha256:afa71e5c2000da0945d1c1cc12bcaa435c411f13ac286cfadcfb847b0cc71a2a; isolation sha256:e0ae991f27d27091ef5c4f2d234f76b510a90c4d02cd87451b0cc9a7af342af7; custody sha256:be4f7ae87af0abe604d3a65bbcb6ace9caf68fc7941833d5e2d5b4d44144779d.

## 60. Thermal expansion

Claim identity: SFT-MAT-THERM-EXPANSION-001

**Question and exact theorem.** Thermal expansion is the condition-bounded change of recurrent constituent separation with excitation while material identity and complete adjacency provenance remain retained.

bond-separation-network\_\_excitation-conditioned-separation-change\_\_material-identity-retained\_\_direction-and-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches **SFT - ROOT - THERE - IS - NO - NOTHING**.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bond-separation-network                  | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | excitation-conditioned-separation-change | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | material-identity-retained               | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | direction-and-temperature-bounded        | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bond-separation-network\_\_excitation-conditioned-separation-change\_\_material-identity-retained\_\_direction-and-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a

universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:0e53cee45fef353af8c1fc3a048db16697e59a4d1945eea47b3e9a9ae4ae9c0c.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:28f81c0bd83923a6393f7cf96abef9aa601ace13fbe9c2e13d8a95e762b6775e.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:35b43f42468a1c1e771a8aa75e61ee96f3241e1249bfb8bfde861a74df429ab0.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:016edca3decdf384605400b1fcfa0e140add94b17145adf5269e163906f9311f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:2430b2dfd7b2daa237185f98bc1c08842018b04eedfc432b4aca14a108bc9c25; certificate  
sha256:12cf1d124a03f9181174b282315f745c3854f4129d0540c209fd60966356acc3; external-validation  
hash sha256:f43b131aa11d8a1e83be60efe50ca5486c5336a52fb45526c6d32755c4bf3a72.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:51a71476779069d408c1bdc3413b8c879c8a17f2e40b35bbd6caca235d49ee36. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html;  
sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale  
microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- **sft-mat-therm-expansion-001-authority-correspondence:** predicted bond-separation-network\_\_excitation-conditioned-separation-change\_\_material-identity-retained\_\_direction-and-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bond-separation-network\_\_excitation-conditioned-separation-change\_\_material-identity-retained\_\_direction-and-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered thermal expansion observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:d08e166df831402c873d497a2b6ebe99c4557b02f7a6686f6bbcf42d229df9c8; derivation seal sha256:0214f5934c9a967a9aa0ac4ddcde7068fc7560b1c26f9ea683d9c5f23a5cc25a; engine receipt sha256:e43221596aae9f61e140436942ee5dbb1625f0bfa5a520f34f217d780558cbdb at receipts/engine/model\_admitted/SFT-MAT-THERM-EXPANSION-001-e43221596aae9f61.json; empirical validation sha256:be6cdeb54ff17afaa198f0414e8ea3404907e36e1b79a041e23f13f399cec260; measurement receipt sha256:b871d56ca42a2531ba7e55209ef6306482b6aeb06c6825d06320164217ab5345; isolation sha256:a299c268357a8d8afccf08b9285f38ec3440b0a33f6dc2497ec2a36ad2b1d44f; custody sha256:3f767b087bfbcc3375d204a69b2fba54d2c14a7282e893c1d63f95f35b76e021.

## 61. Ferromagnetic order

Claim identity: SFT-MAT-MAG-FERROMAGNETISM-001

**Question and exact theorem.** Ferromagnetism is connected recurrence of aligned local held-moment orientations producing a retained macroscopic orientation class.

local-moment-network\_\_aligned-shared-orientation-recurrence\_\_connected-net-moment-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | local-moment-network                     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | aligned-shared-orientation-recurrence    | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | connected-net-moment-retained            | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | ordering-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: local-moment-network\_\_aligned-shared-orientation-recurrence\_\_connected-net-moment-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise**: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:18c6e7d85f2e9146578d2645a8213fdbf355c75e79e05273dcfc9a89bdc6c8f7.
- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:09dd1edb95f74fb641872c23fef4e458b8b03d180b58c83bcf96284e7c2e84d9.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:22995d5074cfb5ecda427c7205df471e65beb5216cf1f63a3073a4d46746ab5b.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:290a1b4e2cb6f280907d51338d128896ce8ed6e11c6cd49d332f1bbbccb0340c.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:d0db4c43c76befd8e81f178bb7d60c32918904c030aec387054fd7cac0d279d3; certificate  
sha256:b6ccb098a72046791e3b63c766f6baafd7006b7b10f8a30543b374f3f2a1b383; external-validation  
hash sha256:dd32b24dd4b45270d8a255cfbbcbd691505465416487c457ea4a0d8c47875be8.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:8f81d89cf2d73a6809d674ceca2c208912e0f9998eaa5252837c8d3ce4fd2a3e. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea3808dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-mag-ferromagnetism-001-authority-correspondence**: predicted local-moment-network\_\_aligned-shared-orientation-recurrence\_\_connected-net-moment-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived local-moment-network\_\_aligned-shared-orientation-recurrence\_\_connected-net-moment-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered ferromagnetic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule



- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:874d335c33a53a6df0256d9f5e149acc452d0743a4f57132fbbad96958d83b63; derivation seal sha256:ef659703b1b78755c811fadd47b2d4d01767e0be0db9aa4ed6357fb2e6738878; engine receipt sha256:8702b789cbad4cb166e5003ed111ac105b9908c3b921c7694876a7542a47c9ab at receipts/engine/model\_admitted/SFT-MAT-MAG-FERROMAGNETISM-001-8702b789cbad4cb1.json; empirical validation sha256:81ddf6d5cd8f35488a8fd9a766f2aa59bd53d2cfce1aef09c382fea4d32521be; measurement receipt sha256:711bb821fc4020f451a199fa3d1e50db79292aec3b2096e80b7f84bc7be5ba0c; isolation sha256:706c62f32b92289aebef28bb5c38f54efd56a53938d566f43efca10e89f4062; custody sha256:e06f5ee0a0adc7afe327d4983e262f49654c4374c76cc69517f35ffb41a20a5f.

## 62. Antiferromagnetic order

Claim identity: SFT-MAT-MAG-ANTIFERROMAGNETISM-001

**Question and exact theorem.** Antiferromagnetism is stable opposed recurrence on distinguishable sublattices that retains local order while the complete macroscopic moment distinction closes.

bipartite-moment-network\_\_opposed-sublattice-recurrence\_\_local-order-global-closure-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bipartite-moment-network                 | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | opposed-sublattice-recurrence            | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | local-order-global-closure-retained      | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | ordering-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bipartite-moment-network\_\_opposed-sublattice-recurrence\_\_local-order-global-closure-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 3a97cd3e05edbee53ee54b4478bab16cf055ce518e721868e07f6c11502ba8cd.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : d91fbf826b29dfc56d95eaf525c93c5a097a496f66fd54bdc07270e4d97c1709.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 21fc3784752313c95d471cc7f4c37eba8dc481651af93a44b97ae34e2574b7f9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 00d0d75dd210b3b851ee3af7c6b1b5e9382df37025806f700f9e3c4a914ab1b2.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 554a049abd839fa53066a0d078721f9d32d1e2da2ab50d6bad079166de236149; certificate  
sha256 : 5bcf84137ee43fbe7eaa1e6c5cb69622b45205f73a6e2c53f144f0c95bcab8ce; external-validation  
hash sha256 : a46609ffce4544872f8f33cad521d02fd03fb08c46966d098291c136698c7eca.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 3 claim-specific source discriminators were required; their ordered identity is  
sha256 : e52715234c5d0275b83377bae9ec6c31e571fc23fe49d47cd26aac633767439c. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - ANTIFERROMAGNETIC - COUPLING - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding>; snapshot  
experiments/external\_sources/materials/snapshots/nist-antiferromagnetic-coupling.html;  
sha256 : b8ce5105622c02b84bffe643693c05c199df57beae79bd9c5e29f26c4360e85e; scope:  
polarized-neutron measurement of magnetic structure and antiferromagnetic coupling.

Comparison and custody records:

- **sft-mat-mag-antiferromagnetism-001-authority-correspondence:** predicted bipartite-moment-network\_\_opposed-sublattice-recurrence\_\_local-order-global-closure-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bipartite-moment-network\_\_opposed-sublattice-recurrence\_\_local-order-global-closure-retained\_\_ordering-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered antiferromagnetic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3df466cadcac208382ade4c4777ac52b418e0c3e01286abe21cd23a00899994d; derivation seal sha256:66b2d9f5998a96ae9e7a4b0f607b4b3701336de1c18415dd2cfa78eaff6de448; engine receipt sha256:4e024776e6572a09e72fd47738e804b731ccffff0685efb0993023aecafc15c7 at receipts/engine/model\_admitted/SFT-MAT-MAG-ANTIFERROMAGNETISM-001-4e024776e6572a09.json; empirical validation sha256:9c06b2a0f59c2933efdaa85f3a3f7107cdfb23da035952b54ebbb1571c31b065; measurement receipt sha256:1144769f267e47cca17f0dc4ddd226220357394daaece0f0da693ac5622311f1; isolation sha256:324097c4049c4005726db64e2a93937542c96b9be63d8c95bea908389c63d4e8; custody sha256:c52d305b11f8dbf80db49ca64d8ce900ce41725947b43d8721e896ad58ea6163.

### 63. Dielectric polarization

Claim identity: SFT-MAT-DIEL-POLARIZATION-001

**Question and exact theorem.** Dielectric polarization is a retained field-conditioned redistribution or orientation of bound held-charge labels with complete conservation and relaxation records.

bound-charge-orientation-carrier\_\_applied-field-displacement-relation\_\_charge-conservation-and-relaxation-retained\_\_frequency-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and

frequency-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                                 | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | bound-charge-orientation-carrier            | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | applied-field-displacement-relation         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | charge-conservation-and-relaxation-retained | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | frequency-condition-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace    | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                  | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure           | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                               | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: bound-charge-orientation-carrier\_\_applied-field-displacement-relation\_\_charge-conservation-and-relaxation-retained\_\_frequency-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : ab4c971eddd4a0592c52799f300cbb68ccdd19e0d17929704c030e932e3147ef.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f811d9b97206cb817d188a47fc41c6ee4857a456316be6a1d8a144544071bcd3.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : e5b70b12ddb119fd8aab8971c8adb87e5830aadd0fa9b644bfc648aa356cbdf8.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 419fd084389f321aa44b601dd00e318fb0647694ec700f5298800692c0e802c1.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 00a9ba701d96966b56fd31c7b4d206d442ec6d652fd7453700c985196d1c92b6; certificate  
sha256 : 59d7757274ab8ecbe48f4c00bc9b612f3d32ff9fa583f3cd2803d0fed7eb7482; external-validation  
hash sha256 : 53450986c1b2ebc93c03ac06621f5d9b2598b3048bfeec6730e0dda3f02dcadf.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256 : a96625233799c6676585da9b9389b00e18e4f90a9ec7b3b86d06d2b1c40ced6b. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html;  
sha256 : 355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-diel-polarization-001-authority-correspondence:** predicted bound-charge-orientation-carrier\_\_applied-field-displacement-relation\_\_charge-conservation-and-relaxation-retained\_\_frequency-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived bound-charge-orientation-carrier\_\_applied-field-displacement-relation\_\_charge-conservation-and-relaxation-retained\_\_frequency-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match **True**
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered dielectric polarization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed



- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and frequency-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1498859c348dca8e370798bb01a8832aea7e4e8d376b158d788bd6156378dc0e; derivation seal sha256:0a118061761c83630046d49ac13b24f8877554e4f646101aa11e5dba5b8f1420; engine receipt sha256:f432cb7b60b503a89a22333e6942725b659572fd12da81a606267858f3ac637d at receipts/engine/model\_admitted/SFT-MAT-DIEL-POLARIZATION-001-f432cb7b60b503a8.json; empirical validation sha256:606adb7ecb391af5f1f37d0122a6ca840f6a4b532ea334cd0fb1c4205cadc2aa; measurement receipt sha256:e912dcb76211979f7aca50f6f3348c37518499fc73f6cd86ec3c8aad480477c; isolation sha256:3abe9bbad740c8ab4999df5dbec74fe762361aee816a6c135941212a29e44b2c; custody sha256:8344d6447b2b2cb16be34b7690434539f57962edbde04bd804c4b811d17e8539.

## 64. Optical propagation and refractive response

Claim identity: SFT-MAT-OPT-REFRACTIVE-001

**Question and exact theorem.** Material refractive response is the source-bound phase/delay relation produced by coherent probe coupling to accessible material excitations while propagation, loss and dispersion records remain separate.

probe-and-material-excitation-carrier\_\_coherent-delay-and-transfer-relation\_\_frequency-phase-and-loss-retained\_\_spectral-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

**Boundary:** Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | probe-and-material-excitation-carrier    | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | coherent-delay-and-transfer-relation     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | frequency-phase-and-loss-retained        | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | spectral-condition-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: probe-and-material-excitation-carrier\_\_coherent-delay-and-transfer-relation\_\_frequency-phase-and-loss-retained\_\_spectral-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

**Base:** One complete material carrier retains its generated relation, organization, observation boundary and proof record.

**Successor:** Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

**Closure scope:** depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:9c630b0915ecb27ecd4a8e9e3a7467474e98d990d0618c7a85c74363f19b6bc7.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9ae669c1738af532adaf47a7e7ad4f3d7043304c6043f5d358a4a2940af80005.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:73d971daac95bf3c58df6f424c3181e678ef094a14cfca717ca19c769d266212.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:b5e103cfd88598d9b7655978391197ab5224bcc9d005792573d8a058c49d84c4.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:65dc955c5bc0b935e97be9bc2733c36b012a32a1a1e56f9c7c9d67b631ab63ef; certificate  
sha256:191167b5a78855912ddc9e723c619c26dce388f41e3e0573f98e36c081b53714; external-validation  
hash sha256:7338f1b3d4505446488721394e707d1a2d078e07f0c0fd0dcf8a817f8ba2905d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:d8b7289d06e63b982d20adbe30509ea26d9351d73e84dbb6f49daa777954288e. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - OPTICAL - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-optical-materials.html;  
sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of  
refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- **sft-mat-opt-refractive-001-authority-correspondence:** predicted probe-and-material-excitation-carrier\_\_coherent-delay-and-transfer-relation\_\_frequency-phase-and-loss-retained\_\_spectral-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived probe-and-material-excitation-carrier\_\_coherent-delay-and-transfer-relation\_\_frequency-phase-and-loss-retained\_\_spectral-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered optical propagation and refractive response observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:6338ee48ea6cf26874c7b32eb792944b85ed28d91cbcf78412f4f07738f8a2f3; derivation seal sha256:1b93339c2b96508f0faecd14fa3af55d3369e93d5a2cf3410c3b8cffeef3243a; engine receipt sha256:82335b4d53a29fa158d4484e54572994b56b5dda246c8654034cbdc01f485d99 at receipts/engine/model\_admitted/SFT-MAT-OPT-REFRACTIVE-001-82335b4d53a29fa1.json; empirical validation sha256:a09d090deef51116d29b164006bf17e5a792eac1e4af65e9ca63528a3ab350e1; measurement receipt sha256:f2305f1463ef3636441aa52e0a33cc956f137b10e879102d6f956a3ca8c464ec; isolation sha256:029f56e5c4af77cb4977858abb7ad0a734f216f21e388c340a7c094da910c654; custody sha256:ddcc328651656989f385b50503615b950a835d1186cc1970f2953b7db45a12e7.

## 65. Material phase transition

Claim identity: SFT-MAT-PHASE-TRANSITION-001

**Question and exact theorem.** A material phase transition is a provenance-retaining change between stable macro-recurrence and symmetry classes with explicit conditions, path and hysteresis boundary.

initial-final-phase-carrier\_\_stable-macro-fibre-change\_\_symmetry-content-and-path-ledger\_\_condition-and-hysteresis-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | initial-final-phase-carrier              | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | stable-macro-fibre-change                | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | symmetry-content-and-path-ledger         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | condition-and-hysteresis-bounded         | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: initial-final-phase-carrier\_\_stable-macro-fibre-change\_\_symmetry-content-and-path-ledger\_\_condition-and-hysteresis-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 699bb0538a254745ef67cc361ac8f4844e8c1fa2538837ac5938e52053255541.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 3192cf970f56809f280e80c2c9820e5498273e7e42e1999057f39d8970014427.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : e183813b92b1fe15c3e6eccee915107ea2ab934872dcf80d50f64eea683505b9.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 406f3613f24df373d9f299a2659d9ad0e2bcf07a282d0ba2cebccb1230a0620b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : a5971eb324e6f0a7959b7a16ebaee120a132038c909c3c221b6c9a779eef1c7c; certificate  
sha256 : c7ca23959f208aede92422515466358ab2b11bb5ecbefd73d808513ca2d5dc7a; external-validation  
hash sha256 : cafe572aaadaa8b0658384852fee24d1d662945aa2b0d6c7931694fe40e7de38.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256 : e20ae266b0ec1dd474ba078d282cf54d525ff318993fa9787c5b9fd426c2c8a6. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - MULTISCALE - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html;  
sha256 : 2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale  
microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- **NIST - SUPERCONDUCTING - FLUX - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot  
experiments/external\_sources/materials/snapshots/nist-superconducting-flux.html;  
sha256 : 47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero  
resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- **sft-mat-phase-transition-001-authority-correspondence:** predicted initial-final-phase-carrier\_\_stable-macro-fibre-change\_\_symmetry-content-and-path-ledger\_\_condition-and-hysteresis-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived initial-final-phase-carrier\_\_stable-macro-fibre-change\_\_symmetry-content-and-path-ledger\_\_condition-and-hysteresis-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material phase transition observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar



- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal\_magnetic\_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:0297cd0782de90adb0d271e4db03d88413a6636799f16e3fbdcfc3b5d84617c4; derivation seal sha256:c219d7cd41090ebb065ba94181b6f6cdea11e507feac6b44e0424b3af3e26e8b; engine receipt sha256:c40ad4320e96682f79e7b338df47728fd53e8df1455276c8a6673294e60ea745 at receipts/engine/model\_admitted/SFT-MAT-PHASE-TRANSITION-001-c40ad4320e96682f.json; empirical validation sha256:5cefdebc7a29e14018d3791b7023149b6671cfa8c81121bee0a4e4667f743cd9; measurement receipt sha256:f97aa41aceec4f79873517d7940d2804059220eee142435cde997ab2b1fe951; isolation sha256:fca5b15e153357ae3fcc8f912e9b78e0da3ec9e98cfef1944b5528738c27536c; custody sha256:fbfbbaa82bf8ce93b146cb1a97a3e85e00bc21ab4c4f20e2516ee0ccdf94433a.

## 66. Ferrimagnetic sublattice order

Claim identity: SFT-MAT-MAG-FERRIMAGNETISM-002

**Question and exact theorem.** Opposed distinguishable magnetic sublattices with unequal positive counted moment support force a nonempty partial bulk moment class distinct from aligned and equal-opposed order.

two-distinguishable-magnetic-sublattice-carrier\_\_opposed-unequal-positive-moment-relation\_\_local-opposition-and-nonempty-bulk-gap-retained\_\_partial-net-moment-order-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-MAT-MAG-FERROMAGNETISM-001

• SFT-MAT-MAG-ANTIFERROMAGNETISM-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor thermal\_magnetic\_optical grammar preserving two-distinguishable-magnetic-sublattice-carrier, opposed-unequal-positive-moment-relation, local-opposition-and-nonempty-bulk-gap-retained and partial-net-moment-order-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                     | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | two-distinguishable-magnetic-sublattice-carrier | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | opposed-unequal-positive-moment-relation        | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | local-opposition-and-nonempty-bulk-gap-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | partial-net-moment-order-class                  | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace        | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                      | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure               | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                   | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: two-distinguishable-magnetic-sublattice-carrier\_\_opposed-unequal-positive-moment-relation\_\_local-opposition-and-nonempty-bulk-gap-retained\_\_partial-net-moment-order-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

**Operational witnesses.**

- unequal-opposition: five opposed to three leaves exactly two in a held orientation; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

**Adverse controls.**

- false\_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256 : 7a1f40900a717d2d47abefffd7fa3f938bd727cf128299586801e6773007f9fd.
- tampered\_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : f969091576a41a221b052aa295f4b3fd36204a3833515368554ec03bffdcb86f.

- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:10a38363fea4202d9bea4baacaa1e3fae35f20c66884ab7e5ea925f361662f5f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:56c4c2eaa33d2a9aa6276bc9282f40ae508181277a751a908423ef0d6319be2f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:62401af2379325f04b5f3a09888974f68d2f7a8fabd1d5c319f04b0774216657; certificate

sha256:c6e26e75582fe70ec9aae72fb463dce2692d02cceff11d704ccfd7263bb8723cb; external-validation

hash sha256:c6e26e75582fe70ec9aae72fb463dce2692d02cceff11d704ccfd7263bb8723cb.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:591cbb5129d22ba5ed720fac188ac2eb94e59e4b1a97b279c6f0e541ecea06e2. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FERRIMAGNETIC - SUBLATTICES - 2026 - 07 - 27** - National Institute of Standards and Technology;  
<https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earth-transition-metal-fe1>;  
snapshot  
[experiments/external\\_sources/materials/snapshots/nist-ferrimagnetism-2026-07-27.html](https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earth-transition-metal-fe1);  
sha256:685872920e5663b2ec71636b953d0e7ddec3d1cb679fb4f17bb94b6d6d811795; scope: post-seal  
claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: `True`
- all registered rows preserved
- tampered changed label rejected

**Falsification boundary:** The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor thermal\_magnetic\_optical grammar preserving two-distinguishable-magnetic-sublattice-carrier, opposed-unequal-positive-moment-relation, local-opposition-and-nonempty-bulk-gap-retained and partial-net-moment-order-class.

**Immutable evidence identities.** Pre-source branch seal

[experiments/sealed\\_predictions/materials\\_v1\\_v2\\_successor\\_complete\\_pre\\_source.json](#); source

manifest sha256:181be2cb81d02e04eff8c616cafaa4f2344670bc2097a787aa7f2e0d52281817; derivation

seal sha256:cf87fe52b9ef52f74ba1c74afee83891689fbfa06d7655dc65e6287aa5658602; engine receipt

sha256:42f94c9f6bfbf33931ff8f8dc7603dde8134e2b6afb67e2a0b22d96942f0a29a7 at

[receipts/engine/model\\_admitted/SFT-MAT-MAG-FERRIMAGNETISM-002-42f94c9f6bfbf339.json](#);

empirical validation sha256:43fc76865edc7dcf64c02c3f89d6a339fb9b19b282f6ec6e0f9ba982122cf8eb;

measurement receipt sha256:0726f5ccf3befd94cc3579c24513bc880f1b92956c909944b8b101170f6d6d05;

isolation sha256:fc95b2f9f5452cf386454313a923348361f4da6a3d6efe5293c26ded3d395984; custody sha256:933c669680d48554da6586e68932bbc12e149d53dbaf06e28bbbdcca913901ef.

## 18. Material Classes Bulk

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

### 67. Metallic material organization

Claim identity: SFT-MAT-CLASS-METAL-001

**Question and exact theorem.** A metallic material is a connected atomic organization with delocalized carrier access across multiple constituent cells and no declared transport gap at its operating conditions.

connected-atomic-network\_\_delocalized-carrier-access\_\_translation-and-bonding-retained\_\_condition-bounded-metal-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | connected-atomic-network                 | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | delocalized-carrier-access               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | translation-and-bonding-retained         | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | condition-bounded-metal-class            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256: 75257e1c7d9eaa9bff969b6d2330e1265d2a0f9bba87c9aece8e402510ab3988`.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:1b001bf09b7d2091d3d1b8bbeb596db998ff93fd4e7203badeebe99ab867c5b4.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c7b74d8a2285a0aab500e235bd3898e00a6c65155c152646c59586906e97f402.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:cc6ff66be0902ac0ba61c87beee88d6283725c77049ca60aecc1a9358568fb3b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:bb4ef02ce0fb08afdd526d8b4a3fc086373575480832e1d9cfb5b462bb5f6bfff; certificate

sha256:2458dd1b4f8560378cf749959b1476d3cc8b0e8a8e003582c2fbb109136fbb6e; external-validation

hash sha256:61ea3121a344907c6519355dde4ec7510dcf6362f3a0fda5897da2741b33cb77.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:87fd1a3ef3fb7ef8871ddb7ecd0d074dda38732b6f59dc8a0acfa4dafb814bd1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-SOLIDIFICATION-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/publications/solidification-0>; snapshot experiments/external\_sources/materials/snapshots/nist-solidification.html; sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- **sft-mat-class-metal-001-authority-correspondence**: predicted connected-atomic-network\_\_delocalized-carrier-access\_\_translation-and-bonding-retained\_\_condition-bounded-metal-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived connected-atomic-network\_\_delocalized-carrier-access\_\_translation-and-bonding-retained\_\_condition-bounded-metal-class\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered metallic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest



sha256:5f02bd1b0f32f5693c1791781165d3bcb38fc8c49ff0ec01254dc518ced91df6; derivation seal  
 sha256:af475c9901883694ec27d4a4693e3ca1eb56373d75c42e6c0b073c66e977bcd; engine receipt  
 sha256:ff88492d77ed22ed6106d1f981e4d70142431b1bfe9aeaa7cc5a6f26966951e2 at  
 receipts/engine/model\_admitted/SFT-MAT-CLASS-METAL-001-ff88492d77ed22ed.json; empirical  
 validation sha256:c5422a5c61e616c84db792a559545f816ea6a3d6dfee8456af9a560e2d6c8588;  
 measurement receipt sha256:84f09dc3af6ef8df70c69226be6eec901246c4e203c597a51b981db60890fc1f;  
 isolation sha256:3ec6b3346deb1532d82db336b639a7be66c91b7b1c4b4146bcf4d40108b37540; custody  
 sha256:eb8356b4e42e5e256d654f54b378fdeed0f9fb5059c09c23e394e2e3d4102391.

## 68. Ceramic material organization

Claim identity: SFT-MAT-CLASS-CERAMIC-001

**Question and exact theorem.** A ceramic is an inorganic nonmetallic material whose bulk carrier is a localized ionic/covalent network with retained phase, porosity and grain/interface organization.

inorganic-network-carrier\_\_localized-ionic-covalent-joining\_\_phase-and-grain-str  
 ucture-retained\_\_processing-condition-bounded\_\_complete-state-transition-boundar  
 y-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra  
 -rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | inorganic-network-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | localized-ionic-covalent-joining         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | phase-and-grain-structure-retained       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | processing-condition-bounded             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
`sha256:76deb008a987f16ea7cf2d23813da1ce5b61f9665f182459ad93958acd125655`.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 62d814937e216c438b3031aa1287c27a1c099fc17fb38a779a32a4c0abda3867.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256: 52134d6b9c5c728526c88b12b66102baf449f59f6b8deab50eda64a5d8430d18.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256: b1df93ebe9661b7a1504c2c444af7ff8ce24fd932d309719a4a077d0effcf610.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 48db99ea691a8154b6bd4b946ecd20cd7d61481745e9e73e611241b91eb35d41; certificate

sha256: 9872a8af3e2b80be8d45b13f6f0d4175c23baf4f58eacdedd1afdee08f3fa44e; external-validation

hash sha256: fdb0f31db398bd99134453418f3830811ffc0ef4924dd1e4b3e2d9d5d24cb566.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 01eb6862c745c23944528b05975766210e00b897c0ef858c5cccb000421b24cf. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-CERAMIC-ADDITIVE-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot experiments/external\_sources/materials/snapshots/nist-ceramic-additive.html; sha256: 540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; scope: ceramic feedstock, defects, sintering, densification, phases and microstructure.

Comparison and custody records:

- **sft-mat-class-ceramic-001-authority-correspondence**: predicted inorganic-network-carrier\_\_localized-ionic-covalent-joining\_\_phase-and-grain-structure-retained\_\_processing-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived inorganic-network-carrier\_\_localized-ionic-covalent-joining\_\_phase-and-grain-structure-retained\_\_processing-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered ceramic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:36cccb1dd2e5e1628e3512f5fa6df179b27bb82e1499c77dcc1baca99f0d6238; derivation seal sha256:369e843e6bc60b72015a21594789a04f78935ae133eed27afa02dad43c23c0ea; engine receipt sha256:c8d2a975402210330273a38078b9cb98953aa151f947aef809e2a6877aa82749 at receipts/engine/model\_admitted/SFT-MAT-CLASS-CERAMIC-001-c8d2a97540221033.json; empirical validation sha256:05b56e875a7c653dbdeb13e30878e702946a5d6c85e3e47da657abf31287437e; measurement receipt sha256:d5391cef95ca7f7c7ab19f64f4b6eab3ddcd8ade6de2050843457b42c1f1bb85; isolation sha256:34ae11265cd7a19fde0a5f163f473131a9d88c056516a17c01e0a67aa9493a6f; custody sha256:c65e9e11483fa4575b83fe65d209fea3569ede6cbad5ccf6048ed0aed39d9905.

## 69. Polymeric material organization

Claim identity: SFT-MAT-CLASS-POLYMER-001

**Question and exact theorem.** A polymeric material is a population of covalently recurrent chain carriers with explicit repeat identity, sequence, length/branching distribution and interchain organization.

repeated-unit-chain-carrier\_\_covalent-backbone-recurrence\_\_sequence-length-branching-and-entanglement\_\_distribution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                                | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | repeated-unit-chain-carrier                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | covalent-backbone-recurrence               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | sequence-length-branching-and-entanglement | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | distribution-bounded                       | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace   | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                 | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure          | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                              | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: repeated-unit-chain-carrier\_\_covalent-backbone-recurrence\_\_sequence-length-branching-and-entanglement\_\_distribution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:19eea086be5305ab4d9ffbe6bbf3f5a9286aec315bf29b7d4403a02f9aeb76c3.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:7e11f2ef9579ffc07192430aeb6c40fa225c2ac06d7e114724ecc9219834280a.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:11aaf3801d8105f70582cb88872a4c36298334d504540dafcbb43a38041c1ed7.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:881a7d9a75b566364a003b319abaeca54bdf2cd3263ffbd9ab1aeb7813f0df7.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:c1061dd2f3ce337b40a2e45f157414d832614f3037b6f908a09deb0ef7f1e04d4; certificate

sha256:e2dc53b388ebdf459e8929547810cf4c3d778209fea737eb1bf27af6795d1b08; external-validation hash sha256:7c8e5ef81f03d00e04ef56779653dd0d5cfffed973bca994f80b6ae1c9ed47974.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:b4d03886dc24480e527ac4522ef8eb3213f4940617f016588852655c95ef6d8a. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-POLYMER-PROCESSING-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-processing.html; sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18; scope: polymer processing, crystallization, rheology, kinetics and recycling.

Comparison and custody records:

- **sft-mat-class-polymer-001-authority-correspondence:** predicted repeated-unit-chain-carrier\_\_covalent-backbone-recurrence\_\_sequence-length-branching-and-entanglement\_\_distribution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived repeated-unit-chain-carrier\_\_covalent-backbone-recurrence\_\_sequence-length-branching-and-entanglement\_\_distribution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered polymeric material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.



**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:6a1b61cf7b27fd57917e294a49338c0d42f55298366417e4c1c0e6ee8680e27e; derivation seal sha256:f3bf4d70c0a28bed53287d8770575102bf8e2585e68ae7310e414b9dc218ef3c; engine receipt sha256:a99eafa3c8ace458d61210eebad617b1826e865d7025839889d21e69e28f6ff6 at receipts/engine/model\_admitted/SFT-MAT-CLASS-POLYMER-001-a99eafa3c8ace458.json; empirical validation sha256:5fd119728138b9711ab1e4dc0020f406fcd5489ae5d9dda98813de8936b3983e; measurement receipt sha256:cd90b19a87eb545697cc05bd0370d1b1ac3f616e0baa870bf5a41fe44481e356; isolation sha256:11825e09dd0ac943dd33d6d22bdbbd9ff7ba42a8c0c68a92d5c72c8e5ccd07e6; custody sha256:ffa4c921c9ac95448b5471c3dfe8126bbce6d40721a51c4ba3744e339a384af7.

## 70. Composite material organization

Claim identity: SFT-MAT-CLASS-COMPOSITE-001

**Question and exact theorem.** A composite is a joint material whose distinguishable constituent carriers persist while interfaces and architecture compose their load, transport or barrier responses.

multiple-constituent-carriers\_\_interface-mediated-load-transfer\_\_constituent-identities-retained\_\_architecture-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | multiple-constituent-carriers            | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | interface-mediated-load-transfer         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | constituent-identities-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | architecture-and-scale-bounded           | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:c6c46419af214daaf146350e487c95de5bc30d082ae54dad12b5579ad28b1633.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:eccccf75b15e102f438af9f7c385a5cc558ed848ca7c8313848451f843f4856a9.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ce4a29905dae4aa34196f4664e2f919f02c6b3eab2d8cd11401b744b38a79e1b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e02674ffffcbbf54bd9a50f41cef4217f5351d525fb98fc7931dae2a22496d021.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:8d4c7ec3138919dd7154707c09abc73b7d0fcecfeb5de1ef5330929a19b6ef56; certificate

sha256:e0737335b366ab6f6be9f854b63f51a38c01a6727f89f1e92a8958ab93e87def; external-validation

hash sha256:ab91a5e87b1f7ead61352f386d150be5fec82f2d722d41a666c2f3e18d595300.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:99c767633d82ee90ffa49c7f507f599fa6fd42c297c565e2ce1dc9e4a339472c. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-POLYMER-COMPOSITES-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- **sft-mat-class-composite-001-authority-correspondence:** predicted multiple-constituent-carriers\_\_interface-mediated-load-transfer\_\_constituent-identities-retained\_\_architecture-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived multiple-constituent-carriers\_\_interface-mediated-load-transfer\_\_constituent-identities-retained\_\_architecture-and-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered composite material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:0c4d6765e00200da4b3797781c97c267cd941b131622a573de77c323c27824ab; derivation seal sha256:c8d209ef1e580b1bae19f6f4a3c8c816ecac58ba3747d66810463b448cf61240; engine receipt sha256:e068e79f1b5490cd352d10a8669b04741ce9fedbf15e89115fbfa514487b6983 at receipts/engine/model\_admitted/SFT-MAT-CLASS-COMPOSITE-001-e068e79f1b5490cd.json; empirical validation sha256:a8e4192ff987d1ec55606bf440fe850195cf076253bef45c4287a44b7dd1ceec; measurement receipt sha256:04a5264efc4594b50729f4788e24b5b702585fd07b9127b6840eb7df565b6d49; isolation sha256:1451cd65a42e6b1336107ce0653b3d3a2c9742c625bcc8d8af0e498b61c5cfc0; custody sha256:b55a8ef809423e1b68ca5143fc757b398f522cd4b4ec94fdad9342c02fa767b5.

## 71. Glass and amorphous organization

Claim identity: SFT-MAT-CLASS-GLASS-001

**Question and exact theorem.** A glass is a connected material with retained local structural recurrence but no crystalline translation class, whose macrostate depends on preparation and observation-time relaxation.

connected-nonperiodic-network\_\_local-order-without-translation\_\_history-and-relaxation-retained\_\_observation-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | connected-nonperiodic-network            | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | local-order-without-translation          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | history-and-relaxation-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | observation-time-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:c8117ff6f4fb77e532becbb51883f31b96b64fc2032238b47ee0761bfd73ce3c.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:48ccd96f2cef517058dcc908ea88079c276754135e112e1ce69ebf05da1070a2.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:dc7537443e2a29e6ec64d48fdd218123e16753cbd276ee7cbcd8adee543f54a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e4c5829e88d7ee7cf3bd6e46e6305425f6c0aae67dfa329036fbe4f8ab143be4.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:7e5fd03d9ff4c3789544066770b9bce294e351f8cc31f47b82515957acbebd5e; certificate

sha256:633a1eca778acd3c503f297e2512b44ae93ce6dde346cdcb88dde7ef694d8f51; external-validation

hash sha256:bcd2bad747a845796be107bb29cfd82be8b8a5f595131d1dadaaf578261e267d.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 3 claim-specific source discriminators were required; their ordered identity is

sha256:bca2ffd249200181edd6ffefb138fc5f42e3029ab07afc7b74f34735ef7970d2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-GLASS-TRANSITION-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics>; snapshot experiments/external\_sources/materials/snapshots/nist-glass-transition.html; sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777; scope: thermodynamic and kinetic glass-transition measurement in glass-forming materials.

Comparison and custody records:

- **sft-mat-class-glass-001-authority-correspondence:** predicted connected-nonperiodic-network\_\_local-order-without-translation\_\_history-and-relaxation-retained\_\_observation-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived connected-nonperiodic-network\_\_local-order-without-translation\_\_history-and-relaxation-retained\_\_observation-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered glass and amorphous organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.



**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:ead873ba4e541f4086b9cf10d55335b648a49936c705d212d34a67042618ba67; derivation seal sha256:20b63fcf87d6da5eff04eb20dda0fe29cf66a0f8502357ec603f104e95066877; engine receipt sha256:48121416b6e96a7ca43dfe1cd3cd81cb6122dca865c2d5757249708911c2d8bb at receipts/engine/model\_admitted/SFT-MAT-CLASS-GLASS-001-48121416b6e96a7c.json; empirical validation sha256:82fa9de65ba0566f78de7820f4837b4fa446ff6d8564270c79fd9abe6b2d82b6; measurement receipt sha256:963696e40f2dd738d98875acbeb98ee72b964edfd075b4ea88f9ca04de50fa1a; isolation sha256:2fb7bfab1a75e763ff1da7ffbc65554ff91371b787f01d199b07624b7c7f2dac; custody sha256:5b5891f1bf93f5e9ce4ef68c1d14aafbd3e29547eebf393d707bd525b16a298d.

## 72. Porous material organization

Claim identity: SFT-MAT-CLASS-POROUS-001

**Question and exact theorem.** A porous material is a joint solid/structural-absence support whose pore connectivity, interface and accessibility are retained at a declared resolution.

solid-and-empty-support-carrier\_\_connected-phase-and-pore-network\_\_interface-topology-retained\_\_resolution-and-access-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | solid-and-empty-support-carrier          | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | connected-phase-and-pore-network         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | interface-topology-retained              | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | resolution-and-access-bounded            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:86474f877d49101e7e3ace7b1ea74f2d9a093c01fe7680f3b1271e4489adc258.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:aa09a3db065810ef015d5e9c1c6c88deee9b27a1f27fe6dbf4b0cf92e5801cc6.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:de7c2b045d20ac78f59d8ec13cc4ff932cc6c3c6f7cd7c236358dbf0836e8d40.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:79a409094422197b89e8c3e50fdf69c57e508575d8db62334202a4d2f429b0c0.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:9df78b11cbc6e6c95b2d9a9e0a9f99e1c79ff7e0c979633cebef497f96e33a74; certificate

sha256:a3e148d5b2218edb7365c024ec306f7d09058931911d8cc7a88ee26c66aa1af0; external-validation hash sha256:7d669197ff1fddc9a4452d46a59fde18bc7e56e1d41e894b145b83fd6603ac8c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:13583e6afba8522a3fd7dd9d3c8fd22796763215d55db5dfbd11a552b2b30528. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- **sft-mat-class-porous-001-authority-correspondence:** predicted solid-and-empty-support-carrier\_\_connected-phase-and-pore-network\_\_interface-topology-retained\_\_resolution-and-access-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived solid-and-empty-support-carrier\_\_connected-phase-and-pore-network\_\_interface-topology-retained\_\_resolution-and-access-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered porous material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:69f8356c8af5397580d2a11cf32c7e72fd277e6e8bed3270680e722abc3cefbe; derivation seal sha256:c54ad9ff0af549841d8deaf72c13a3262cecd1f9c6babf627c99aa57579d918; engine receipt sha256:38df22075765cbfdde541b7c7c5909a5657a9a5d88c17badc55cf9e1691d896f at receipts/engine/model\_admitted/SFT-MAT-CLASS-POROUS-001-38df22075765cbfd.json; empirical validation sha256:450e94e6f317469bb6b703f57926d750ce630510f52b2625b57fb1b86b746636; measurement receipt sha256:7ba33b9a0f10949d4026a021623572caf081489251cb9fd806df2a9ca191e427; isolation sha256:21dd1f085996b65e7f1dd705fab7ba3a6e648ad01ebaa1cfb389d1ecdf6e2366; custody sha256:6b400b1a8dfc8c669425cc3c4d17624d8621bf1ca1828ad820b17ab91ea2d375.

## 73. Molecular-to-bulk correspondence

Claim identity: SFT-MAT-BULK-CORRESPONDENCE-001

**Question and exact theorem.** A bulk material law is admitted only when the complete constituent/interaction network composes to a scale-stable observation while fluctuations, interfaces and preparation history remain explicitly bounded.

complete-constituent-ensemble\_\_local-relation-to-network-composition\_\_fluctuation-interface-and-history-retained\_\_representative-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                                | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | complete-constituent-ensemble              | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | local-relation-to-network-composition      | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | fluctuation-interface-and-history-retained | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | representative-scale-boundary              | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace   | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                 | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure          | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                              | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: complete-constituent-ensemble\_\_local-relation-to-network-composition\_\_fluctuation-interface-and-history-retained\_\_representative-scale-boundary\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:8aaa2be2cb1aaf75093de967bfa9bb883f60e3bfb38d10c1501e9e67043ca4e0.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:af1ae25c467d916ea85f9b6610c5ba82923fef5eece84c75ea61a4f185806e16.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1aa726163603796dc22e356d791678eb295c11adc115abd3c4435a34c6960e8e.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:64c386dbbe1c3aa910b83a1516e7b297cbebdb40d4ca072c8985bbf628c260b1.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:2730e1897d8309a7942b0a6250869b93142c515f1222b012fab05c66195db660; certificate

sha256:dc8f66c5f350939ed51ffd3201e28c851664e6533f6eefa0bb451050bad8ee9b; external-validation hash sha256:d0944a5281e0ad757098205af86c358371b8c15f418d733c9961798ef1fafa56.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:95c6dd46934ae777e83fddc8889a1df0d2b9d0e0f586a7c703736741414c412f. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-POLYMER-COMPOSITES-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.
- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-bulk-correspondence-001-authority-correspondence: predicted complete-constituent-ensemble\_\_local-relation-to-network-k-composition\_\_fluctuation-interface-and-history-retained\_\_representative-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived complete-constituent-ensemble\_\_local-relation-to-network-composition\_\_fluctuation-interface-and-history-retained\_\_representative-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered molecular-to-bulk correspondence observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule



- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:b7c45e293722535b1fee18b96470c7d74e777249add1635ab885cfb643012bce; derivation seal sha256:18006e7469ea4d53ae3e2432398c8677dce5f5cd10dd2be3fa2658c714da22ad; engine receipt sha256:447912e5ddb8853c1ee41231f0d106962ca3940a118a3fd420384a77f06509b4 at receipts/engine/model\_admitted/SFT-MAT-BULK-CORRESPONDENCE-001-447912e5ddb8853c.json; empirical validation sha256:37e602662e5ef301a345ca2dcaa89ced48957308154af015e5d5a8e23fb71221; measurement receipt sha256:3844092558e29c9ff4835b0c2c4df9bc85fdb6b9418f73032b67837012d39920; isolation sha256:3953777fe17d106465d44bfe5c9158304296a968e2186bf3fbe0ba5891ce252d; custody sha256:feadd5dd045f6ba5e2e5f2c2688ab882456e1efeb3630d53f2e0ca0ab2066b0b.

## 74. Material anisotropy

Claim identity: SFT-MAT-BULK-ANISOTROPY-001

**Question and exact theorem.** Anisotropy is inequality of response relations across retained material directions after specimen, condition and method identities are held fixed.

direction-labelled-structure-carrier\_\_orientation-dependent-response\_\_same-material-identity-retained\_\_coordinate-and-symmetry-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches **SFT - ROOT - THERE - IS - NO - NOTHING**.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | direction-labelled-structure-carrier     | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | orientation-dependent-response           | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | same-material-identity-retained          | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | coordinate-and-symmetry-bounded          | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a

universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:6cbc7c78de53a91c49c276e589ee1970b58c34859a374b9e536116f635df6977.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:32e5a3fe10672f05fcc0c334670205b39f340f5e57431dd46f3b50d3742f3fa3.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:175cd6c6266b529caa8f9d2671fa7ae2b95dd96739c241f9fbdef8a23d8e613a.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:d1c1e14e228849b8d8b6e38554430a79cd3247aadccdb0a8de858714adf77819.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:b984029abd34332b72aa4acbe09bba4fa14c8c7e328e04abe4a2d287d0d8ec73; certificate  
sha256:d3277213f7cc1a63985c32223dc1176a3a60269bd1d12589f6b5becef88e0d77; external-validation  
hash sha256:f31046c84f6ca5ab88a49c834753bfd3a8c8fb142ffb4cb2b1148a7cd13517c1.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:5415b2701a13d019225ae29acb353bfe745f8505d009fe5b4176d0a86c279069. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TEXTURE - PHASE - FRACTION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot  
experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html;  
sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase  
fractions, texture, deformation and uncertainty.

Comparison and custody records:

- **sft-mat-bulk-anisotropy-001-authority-correspondence:** predicted direction-labelled-structure-carrier\_\_orientation-dependent-response\_\_same-material-identity-retained\_\_coordinate-and-symmetry-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived direction-labelled-structure-carrier\_\_orientation-dependent-response\_\_same-material-identity-retained\_\_coordinate-and-symmetry-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material anisotropy observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:72f09f5534c17c27099b1d5257c3786dbac85ef798492155e0d32d5a6a10e3e3; derivation seal sha256:fbdd70413189e72b586102c3e9366e8bcb48fc378e7e134073135a4e0529a77b; engine receipt sha256:c3c22fd6b9e4de8ba6af34698e7d7dba28cf604ee5d62cef540b8459446182b6 at receipts/engine/model\_admitted/SFT-MAT-BULK-ANISOTROPY-001-c3c22fd6b9e4de8b.json; empirical validation sha256:551c9e104169aae809fc0394df484bdc23a543b85a8346628b6209e3723d3b3; measurement receipt sha256:1a455d7914018cf69f07f94afb5e7a4abf16bf467f77e237738d2008f9cff2c2; isolation sha256:e984687ace695a4b2491c37d9f3cfe69d4dbb7cbb9d5f60cb8b992d004c67869; custody sha256:61e237fb7d23a7c17168cc1fb16bc841b1020e9a1d5e5df1860c43e029becd5a.

## 75. Size and surface effects

Claim identity: SFT-MAT-BULK-SIZE-SURFACE-001

**Question and exact theorem.** A material size effect occurs when the exact boundary/interior support relation changes the accessible state, transfer or mechanical census; it is never a free scale correction.

finite-bulk-and-boundary-carrier\_\_boundary-to-interior-support-ratio\_\_surface-state-and-curvature-retained\_\_size-shape-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001

- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | finite-bulk-and-boundary-carrier         | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | boundary-to-interior-support-ratio       | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | surface-state-and-curvature-retained     | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | size-shape-bounded                       | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: finite-bulk-and-boundary-carrier\_\_boundary-to-interior-support-ratio\_\_surface-state-and-curvature-retained\_\_size-shape-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a

universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : a71adddee9b5e490d396e46c2d4ce8e7d6df9fd103a368f317d5404c794f5962.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 1cb16132da27181a2ea269de5aeccca7fd5c6fb9308e530c9eb4762bdd7c6cd9.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 16e259f649e3a6a6d3f8aaf73de1ae125511d669c7b238cab1230047fa53e70b.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 0abbadc72e2bb18b61d548347a7fc809891b92f314caf46fd012efe82f96adb7.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 8b01df9a6447d154451b82d24d6fbe0a6cd3d71941e037c9c9c056cc28fdf109; certificate  
sha256 : 71bf3b0b6332824a4a93e13e69b343a7f22245cac791167a29f3c097e5d0f4c3; external-validation  
hash sha256 : 5b57f4f341beaf218e7fdd9becd318ff208a58060bf09221cbfdda95b8aea683.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256 : 96c7a8b7a44f93e71147fa23ce577cc4434028fcf92b7d9f39cab20f952bc8b5. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SURFACE - DAMAGE - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot  
[experiments/external\\_sources/materials/snapshots/nist-surface-damage.html](https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project/experiments/external_sources/materials/snapshots/nist-surface-damage.html);  
sha256 : 7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; scope: surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- **NIST - NANO - PROTOCOLS - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/nano-measurement-protocols>; snapshot  
[experiments/external\\_sources/materials/snapshots/nist-nano-protocols.html](https://www.nist.gov/mml/nano-measurement-protocols/experiments/external_sources/materials/snapshots/nist-nano-protocols.html);  
sha256 : 366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- **sft-mat-bulk-size-surface-001-authority-correspondence:** predicted finite-bulk-and-boundary-carrier\_\_boundary-to-interior-support-ratio\_\_surface-state-and-curvature-retained\_\_size-shape-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived finite-bulk-and-boundary-carrier\_\_boundary-to-interior-support-ratio\_\_surface-state-and-curvature-retained\_\_size-shape-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered size and surface effects observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.



**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material\_classes\_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:95bea428b8eefb5e8f23690a77f8b094becf3ba585242b61cb740f778c8cc4f8; derivation seal sha256:5f43cb2e35c18fe49c584059308b7af0afc7c6a72bad42a13c824446b7e64489; engine receipt sha256:d42d7f0a8dfb8cd74cb9b7857420beef13a926f5ec874c61e613976f557dac14 at receipts/engine/model\_admitted/SFT-MAT-BULK-SIZE-SURFACE-001-d42d7f0a8dfb8cd7.json; empirical validation sha256:46a2c11f07c97d5bca30736ccaac4a8677cea56d58998eaf25795921cd21e80c; measurement receipt sha256:22c019851ed0dcc75e91525362b462fb42429f0006f626a2460bac869fe59b66; isolation sha256:6a00639920ccb0d0be88becd640b378050d2bd32f0cd668d7a8022d2e2aab7fb; custody sha256:96a1bc3e068eea0c455024d13b01d9ecf1d2b794de18c5c4783991da5e4fc51c.

**76. Hydrogen-bonded water bulk-response ledger**

Claim identity: SFT-MAT-BULK-WATER-RESPONSE-002

**Question and exact theorem.** A water bulk-response claim must retain the molecular hydrogen-bond network, specimen and phase identity, boiling, solid/liquid density and heat-capacity observations with their conditions, methods and uncertainty boundaries.

molecular-network-and-bulk-specimen-carrier\_\_intermolecular-organization-to-bulk-response-relation\_\_phase-density-boiling-and-heat-capacity-records-retained\_\_id entity-condition-method-and-uncertainty-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-INTERMOLECULAR-BINDING-011
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-PHASE-TRANSITION-001
- SFT-MAT-THERM-HEAT-CAPACITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the eight registered binary preservation decisions.

Boundary: Every positive finite generated carrier in the registered successor material\_classes\_bulk grammar preserving molecular-network-and-bulk-specimen-carrier, intermolecular-organization-to-bulk-response-relation, phase-density-boiling-and-heat-capacity-records-retained and identity-condition-method-and-uncertainty-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                    | Forced form                                              | Exact elimination / retention basis                                                                                 |
|--------------|------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only      | molecular-network-and-bulk-specimen-carrier              | The material carrier cannot be reconstructed. The complete generated carrier is retained.                           |
| relation     | relation-imported-fitted-or-erased | intermolecular-organization-to-bulk-response-relation    | An imported, fitted or erased relation is not forced. Only the generated exact relation is retained.                |
| organization | organization-collapsed             | phase-density-boiling-and-heat-capacity-records-retained | Required material distinctions are merged. Every required organization distinction remains held.                    |
| observation  | observation-boundary-unrecorded    | identity-condition-method-and-uncertainty-bounded        | The observation class is unidentified. The declared observation boundary is retained.                               |
| record       | result-without-transition-trace    | complete-state-transition-boundary-trace                 | An answer without its transition trace is not auditable. The complete trace is retained.                            |
| provenance   | authority-or-target-selected-law   | root-bound-forward-forcing                               | An authority or target cannot select the law. Every decision traces to the root through admitted dependencies.      |
| generality   | finite-answer-lookup               | positive-finite-successor-closure                        | A finite lookup has no general certificate. The One base and every positive finite successor preserve the relation. |
| extension    | free-fit-exception-or-extra-rule   | no-extra-rule                                            | A free choice can manufacture the desired result. No axiom, parameter, fit or exception is present.                 |

**Unique survivor, base and successor.** Sole survivor: molecular-network-and-bulk-specimen-carrier\_\_intermolecular-organization-to-bulk-response-relation\_\_phase-density-boiling-and-heat-capacity-records-retained\_\_identity-condition-method-and-uncertainty-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: The first complete carrier retains its relation, organization, observation and proof record.

Successor: Appending one lawful positive successor preserves every earlier distinction and adds no rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

**Operational witnesses.**

- **bulk - ledger:** all nine required molecular-to-bulk evidence fields are retained; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

**Adverse controls.**

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 18b0d69913ad5bc692e76806db29181e55a3393a181f9f7d7525f066e86c78dd.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 03ddd75b6d96b73054222c608d9a9de3960543fa91c25ecb965e03e9905724c5.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 93be1ca1dd2bfd31a348aa94e09bcac089cb5a27baf8cc4d06188fd1fd1b7b1.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 8470568cf156f40d4f230a507c433c53b7ef0c2608ca12e252ce1c3b5f3f34a5.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 740847c6a0eb2ab7caafe3433bdc61a3264c9e1ac5f79bd7664bcbc4bb557e55; certificate

sha256: adb5f7d3a68e71521a0d0aa531d4600ab203beedc87c96451fb9aafb6138880c; external-validation

hash sha256: adb5f7d3a68e71521a0d0aa531d4600ab203beedc87c96451fb9aafb6138880c.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256: dde23657166381ae8b42a76512717a54870f1fc924267c72b3e0618487891a25. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - WATER - THERMOPHYSICAL - 2026-07-27** - National Institute of Standards and Technology;  
<https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=FFF&Units=CAL>; snapshot  
experiments/external\_sources/materials/snapshots/nist-water-complete-2026-07-27.html;  
sha256: ba9fc5225bc8d0d89cbe15703bff2d9dc06ef64244598c420c7666931f1befbf; scope: post-seal  
claim-specific authoritative comparison.
- **NIST - LIQUID - WATER - DENSITY - 2026-07-27** - National Institute of Standards and Technology;  
<https://www.nist.gov/document/jpcrd38200921ppdf>; snapshot experiments/external\_sources/materials/snapshots/nist-liquid-water-properties-2026-07-27.txt;  
sha256: b3af256bcffd7ff87fbf2aa00f545910a531d5a1b285fa9bb9f926f7afad9e24; scope: post-seal  
claim-specific authoritative comparison.
- **NIST - ICE - DENSITY - 2026-07-27** - National Institute of Standards and Technology;  
[https://tsapps.nist.gov/publication/get\\_pdf.cfm?pub\\_id=926353](https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=926353); snapshot  
experiments/external\_sources/materials/snapshots/nist-ice-properties-2026-07-27.txt;  
sha256: 7073b61255cf73163595c9ca25646dcf2c323dd2aca942d797f9f0e4dd6c178e; scope: post-seal  
claim-specific authoritative comparison.

Comparison and custody records:

- sealed prediction equals complete source-derived observation: `True`

- all registered rows preserved
- tampered changed label rejected

Falsification boundary: The claim fails if any registered carrier, relation, organization or observation feature is absent, any required external row fails, or any tampered row is accepted.

#### Explicit exclusions.

- no prior answer, named phenomenon, source identity or measurement may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence and opposed direction are held labels
- no axiom, free parameter, fit, learned coefficient, exception or target-derived rule
- external target content remains inaccessible until the complete successor set is sealed
- Every positive finite generated carrier in the registered successor material\_classes\_bulk grammar preserving molecular-network-and-bulk-specimen-carrier, intermolecular-organization-to-bulk-response-relation, phase-density-boiling-and-heat-capacity-records-retained and identity-condition-method-and-uncertainty-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_v1\_v2\_successor\_complete\_pre\_source.json; source manifest sha256:e181e7b9d0b370a10d0fdda4e459caf66dbc40665ab7cae962f3e73b1d218d67; derivation seal sha256:f74d7a789438f6e069854e4651e70699007c810fe1a232a9e5ad63f4d1549679; engine receipt sha256:bbc61e7fbba2d3cf5ce25e2015c3aa1713e03d741067f422760e797ab6a42774 at receipts/engine/model\_admitted/SFT-MAT-BULK-WATER-RESPONSE-002-bbc61e7fbba2d3cf.json; empirical validation sha256:0ea1185d4ba1ffe80e5c49895cb65c9f6bdde00361f4c14fd0ad8124e77c6239; measurement receipt sha256:c60a627a4bb3afc89140f7d6e4e88493a2ead819f8c5ac7c8221f3b4aad224bb; isolation sha256:4b02bcb242983a2408823c7bab95f0b878716b006850a00f86f1643c68a2cfdc; custody sha256:b97e1b5ece66313d657ab0e2dc92b92e34bea313db4f1da2009c649d06b07c29.

## 19. Processing Degradation

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

### 77. Processing-structure-property path

Claim identity: SFT-MAT-PROC-PATH-001

**Question and exact theorem.** A materials process is a complete ordered path from input carrier through retained structural states to property relations; endpoint labels alone cannot erase the path.

history-labelled-material-carrier\_\_process-to-structure-to-response-map\_\_intermediate-states-retained\_\_route-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | history-labelled-material-carrier        | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | process-to-structure-to-response-map     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | intermediate-states-retained             | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | route-condition-bounded                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: history-labelled-material-carrier\_\_process-to-structure-to-response-map\_\_intermediate-states-retained\_\_route-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: de1ccf6eb7be4b98104a537ca43370f1260822868e975e19263742b742042635.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 2a2242612fcf027a838edcc8079ef2b4a0a8103741fd3d4dac0fb3faa9783e70.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 361024627d19d9bae6cc5a148da978c4f72f58f46563b8d250daba4ecff56ba3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: c570df606d3d54c7acc82474a18b688a269c52278fa3ed9e40cf3fb952620731.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: 3de27f586e0a54a133982ba8f398e3da38fac53b3510ae159a9d1aa2f07a33c0; certificate  
sha256: 03e7e4f1fc0dc184d8177651acc4a903420c58e4b0d458d84f3f307ecb17b308; external-validation  
hash sha256: 7f4acaefbfb04449e58fbb65dc23f59ec79c48a4162681365f5b4d931ef7faaa.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: 7cbdc5dc38d614e32c43f7f765797cd184d0e4187360667ac81f809ede32d647. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - FATIGUE - FRACTURE - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot  
experiments/external\_sources/materials/snapshots/nist-fatigue-fracture.html;  
sha256: 449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope:  
processing-structure-property-performance, defects, fatigue and fracture.

Comparison and custody records:

- sft-mat-proc-path-001-authority-correspondence: predicted history-labelled-material-carrier\_\_process-to-structure-to-response-m  
ap\_\_intermediate-states-retained\_\_route-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-fo  
rcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived history-labelled-material-carrier\_\_process-to-structure-t



o-response-map\_\_intermediate-states-retained\_\_route-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered processing-structure-property path observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

#### Immutable evidence identities. Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1f883e2d57a7b61755f14fc762c6739e9ce7beb81863b37d8e95f4ad26909eeb; derivation seal sha256:702060f13eedb527da17fb13321451beb4f372bf20b7eba39441e125afe297d1; engine receipt sha256:44a5d1efe1f1541917ea77e5eed4af2fdadbf277107194470e0cd69a279a2bf6 at receipts/engine/model\_admitted/SFT-MAT-PROC-PATH-001-44a5d1efe1f15419.json; empirical validation sha256:96ece9f7a71e3deb2748bec2f31385040cbee3d68883d1f5bcf6a94d67a8cd6d; measurement receipt sha256:cbc82b899d3bc7f483ec6881a345df54813c7cbdd9a4ab034da56209bb2f053d; isolation sha256:87b63a192292d43bf9591f2f3dfa1354236cc5b24b75811f1ba2e0eacb225d5d; custody sha256:c441dd3a1ce057baa9ad7680adcd01a5ba977076a692f9633121dd6ae9017865.

## 78. Phase stability and coexistence organization

Claim identity: SFT-MAT-PROC-PHASE-DIAGRAM-001

**Question and exact theorem.** A phase map partitions generated composition/condition support by the stable material recurrence class, retaining coexistence, fraction and transition boundaries without fitted extrapolation beyond its domain.

composition-condition-support\_\_stable-phase-minimality-relation\_\_coexisting-fractions-and-boundaries\_\_declared-domain-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | composition-condition-support            | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | stable-phase-minimality-relation         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | coexisting-fractions-and-boundaries      | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | declared-domain-bounded                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: composition-condition-support\_\_stable-phase-minimality-relation\_\_coexisting-fractions-and-boundaries\_\_declared-domain-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 9054bd12994d5f41bcd6f6177b6fed1a34c64e07f8bc3fa509b475f602c4ea5c2.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 40353a579820b968d584b0a61eba404cd3123713f03e571bd52c017083c2f5cc.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : b3bf6511512a363058e595b8843d98098ef829b121614b2c706fac58032f2bf3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256 : 2f7e1698d0a482289fd10c54b7e271b8255002aff2c60c9580d49d52be03b00b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256 : 5b0d3302fca9faacc388d07b983abf6ce483d21355e3e376bafddfaeb2611318b; certificate  
sha256 : f57fb9ac93816e6dfe7d21e87d4a975ff683877da72cf233739c2ef07f175a5a; external-validation  
hash sha256 : e19d6dc3ca0588280f74e04f9517cf7ac7566243ff1d1a70e74bb3323873b877.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is  
sha256 : cac50917fffb4e7db7eaf36ef805abfc1b25c482a863f8a04e987b8d380a19de2. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html;  
sha256 : 4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal  
conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST - TEXTURE - PHASE - FRACTION - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot  
experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html;  
sha256 : b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase  
fractions, texture, deformation and uncertainty.

### Comparison and custody records:

- sft-mat-proc-phase-diagram-001-authority-correspondence: predicted composition-condition-support\_\_stable-phase-minimality-relation\_\_coexisting-fractions-and-boundaries\_\_declared-domain-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived composition-condition-support\_\_stable-phase-minimality-relation\_\_coexisting-fractions-and-boundaries\_\_declared-domain-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered phase stability and coexistence organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

### Immutable evidence identities. Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:2697defe1bb570f0e0a431bf86071a61097ef77fa6a094e7bed3496b89580dd0; derivation seal sha256:e0c1577c109e318217fbd8353609550171447bf5838f1c229e48b62946ff50ac; engine receipt sha256:b2ab10b5f18cb90daa694ebe0ed304d3a153b06b007290c11a4ef97f4e157d0d at receipts/engine/model\_admitted/SFT-MAT-PROC-PHASE-DIAGRAM-001-b2ab10b5f18cb90d.json; empirical validation sha256:ef30582b4adcc27a061c4c1f53dafc306fd144b7f3d0da2fdda718fbee0ef391; measurement receipt sha256:b6121ca9be3efcfdea534009debc7982d58154d474e27907ba17eac057b4615; isolation sha256:9b3a1a5224c4dacf767cf438ecbae4065682fefe679f87ef9cdda13a118c6e3d; custody sha256:e89d44de4b24180fffe9afd082483ad17c54feee960bc30fad6ece7eda5a639a.

## 79. Solidification

Claim identity: SFT-MAT-PROC-SOLIDIFICATION-001

**Question and exact theorem.** Solidification is the ordered nucleation and growth path from mobile to position-retaining organization with complete energy, constituent rejection and interface provenance.

liquid-and-solid-phase-carrier\_\_nucleation-growth-and-rejection-path\_\_latent-transfer-and-composition-ledger\_\_rate-gradient-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | liquid-and-solid-phase-carrier           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | nucleation-growth-and-rejection-path     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | latent-transfer-and-composition-ledger   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | rate-gradient-bounded                    | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: liquid-and-solid-phase-carrier\_\_nucleation-growth-and-rejection-path\_\_latent-transfer-and-composition- ledger\_\_rate-gradient-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 28b0e07297d86f968e940b1b2ac43f3094e57aeb51ca1cf3957be002d4d3c900.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 1d9abadcefe83f3d19ed07b99fe3c194097758fd2c19ca09640c2d36d329b587.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 25a0e1d82af8173155367a44fdafa6c03a8a457875f6d6237beb39a838f16497.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 97e19a857b778351e4607a1e2b02197c20cd922c00b6ab7a535f726ea471fec4.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 9008ceb7d860deb437d0278efb1db506b4282e8994c079cb8310927d30804be8; certificate

sha256: ca3d3ad1afbfbf9780ad9132a1260095507c75475a0390dfe859deddde190e2b3; external-validation

hash sha256: 8442dec51ae104811a14c830ef71370b48550b273d30be09822d7dcd9ec6d08f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: 0c9f870f7f230ffeb3890f9e71ae58c89c7463037ae121df10cc17d648146081. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - SOLIDIFICATION - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/publications/solidification-0>; snapshot  
experiments/external\_sources/materials/snapshots/nist-solidification.html;  
sha256: d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; scope: solidification  
transport, nucleation, growth, interfaces, segregation and processing.



### Comparison and custody records:

- sft-mat-proc-solidification-001-authority-correspondence: predicted liquid-and-solid-phase-carrier\_\_nucleation-growth-and-rejection-path\_\_latent-transfer-and-composition-ledger\_\_rate-gradient-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived liquid-and-solid-phase-carrier\_\_nucleation-growth-and-rejection-path\_\_latent-transfer-and-composition-ledger\_\_rate-gradient-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered solidification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

### Immutable evidence identities. Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:cd4d0b64eac45c1d3a28447a50aa223d74e3c5531439f94b4aafb0ad44ec43ff; derivation seal sha256:44d972ac53f7700b71e42f3b296ec7072c34c45c9f864c2c029cbe4af795bad3; engine receipt sha256:882454d5441763ff1ecfcdc81b6298141741d84056879c62ad3fa9b1cd7c3bf85 at receipts/engine/model\_admitted/SFT-MAT-PROC-SOLIDIFICATION-001-882454d5441763ff.json; empirical validation sha256:8f82f3fcd57f8f8f6dcdbd0b69e42bc940cd8c94021d24df8dafcd8f016a00f4f; measurement receipt sha256:e07ec032b3d672f10f84002045b8112a885dc28567393f3f27097c22abaa2d6c; isolation sha256:d2bc013d9aadeadd5d8593a793caf61908d3490425424619a807233e8dfa3da7; custody sha256:549b964885d9add4a9fb9c1b65671c42f0e813321766e22d95b1f4d9e922ea1e.

## 80. Sintering and densification

Claim identity: SFT-MAT-PROC-SINTERING-001

**Question and exact theorem.** Sintering is interface-mediated constituent transport that joins particles and reorganizes structural absence while conserving material content and retaining the processing path.

particle-and-pore-network\_\_interface-driven-material-transfer\_\_content-conserved-pore-reorganized\_\_temperature-time-load-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | particle-and-pore-network                | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | interface-driven-material-transfer       | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | content-conserved-pore-reorganized       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | temperature-time-load-bounded            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: particle-and-pore-network\_\_interface-driven-material-transfer\_\_content-conserved-pore-reorganized\_\_temperature-time-load-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 427fba8bb3e13d9e946485513932e9addb8efa06915387bb6f29c16955dd431c.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: a95765c073bde4df0dde77cbd0bba9d791e2da4a9df0f3b85ffba97e4c9a869.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: a86e1fcd6ea91d3d19e8c4ba54fe00c6aed81ef80e74c371cb01a3cf5d220e80.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 973a0bc896c8d4274acedad03ad2cbdb968478e6872d029074dcbe70d0885cb8.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 6b45355ff97dda505fb391570767e63172d3e98a3db5bc1a18561bc19b1ee990; certificate

sha256: e171069695aad4a42995390f5a4d33efa475bcf7ab4de7d4e49caa9013ac2cfa; external-validation hash sha256: 8ad24f2106da658a5eaf4ae15e3958ee5bbdccfd4fe1e6c0753e5c4770761a6a.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256: 7892b6461c84d32bd9487a05f441b0e38c9d4640a478edd10b49c14a6cb2ace7. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - CERAMIC - ADDITIVE - 2026 - 07 - 24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot  
experiments/external\_sources/materials/snapshots/nist-ceramic-additive.html;  
sha256: 540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; scope: ceramic  
feedstock, defects, sintering, densification, phases and microstructure.

### Comparison and custody records:

- sft-mat-proc-sintering-001-authority-correspondence: predicted particle-and-pore-network\_\_interface-driven-material-transfer\_\_content-conserved-pore-reorganized\_\_temperature-time-load-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived particle-and-pore-network\_\_interface-driven-material-transfer\_\_content-conserved-pore-reorganized\_\_temperature-time-load-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered sintering and densification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

### Immutable evidence identities. Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:3a48aa91150d2edb674a0ea59f30e9d6dcff091178031b3568c20f1bfe7608f4; derivation seal sha256:6dcdbf2eb9b8b5012bad74a050454b19eca9f1a608ddefc30e364e667083445; engine receipt sha256:2405176e959c34190b77cd5b5acfb77dedd92245de91133d652c24ae73e097da at receipts/engine/model\_admitted/SFT-MAT-PROC-SINTERING-001-2405176e959c3419.json; empirical validation sha256:5954fe3b5f12ec2660cfb6021c0d0517f40add8351bdeb0d8a388492b1e1028c; measurement receipt sha256:6db1b0cc6f2ba17d3ce2852251728a164746c4d146429002f978129e6e66979a; isolation sha256:a13f7d3ad965c5aa3dcaaf81c675099da3d320214db342f785bf416fcf7506eb; custody sha256:06ec2311f8604a307fcde247724ac7551f9fa3cb88f4b52cd156ca063fe0f795.

## 81. Heat treatment

Claim identity: SFT-MAT-PROC-HEAT-TREATMENT-001

**Question and exact theorem.** Heat treatment is a declared thermal/time path whose phase, defect and microstructure transitions are retained and whose property consequence is path-bound.

history-labelled-phase-network\_\_thermal-path-to-microstructure-map\_\_time-temperature-transition-record\_\_cooling-and-specimen-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | history-labelled-phase-network           | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | thermal-path-to-microstructure-map       | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | time-temperature-transition-record       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | cooling-and-specimen-bounded             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: history-labelled-phase-network\_\_thermal-path-to-microstructure-map\_\_time-temperature-transition-record\_\_cooling-and-specimen-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:b9bccf53662d1f29124adbd3570acc88fe4777fd0a652a1e1919e532d8ed6c2e.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:54c522c6df12bc9faa22bd4f2e7cb3e5466453da6c0e58ea1133735b6fbc9b0.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256:955e631bff387114e5252f19d68542f07a446c13563573d355fc803804d8d9d4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:e69374df1be210f621e0f3e134dc23dc86ba34914fdd90ad287d39530861f637.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:d5c8b9fd2403519162e8c42d68270b6519c09b147cb35d04e29d194cbd536948; certificate

sha256:2009f709ca9587b3ee9090d9fe6638b2040f5762ac7c979215d8cb8ec2821dce; external-validation

hash sha256:18935d07883ce48e19d9973db302e753092903a41dd892d13d8a9f6384a3fe4b.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:f4f5c6df4ece53d108f79157d9e39adee7ab3f6e995d38609ad491640f1c285b. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MULTISCALE-MATERIALS-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot  
experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html;  
sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale  
microstructure, phase evolution, heat treatment, porous materials and in-situ processing.



- **NIST-METAL-ADDITIVE-CORROSION-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot <https://www.nist.gov/programs-projects/additive-manufacturing-metals/snapshots/nist-metal-additive-corrosion.html>; sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; scope: phase transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-proc-heat-treatment-001-authority-correspondence: predicted history-labelled-phase-network\_\_thermal-path-to-microstructure-map\_\_time-temperature-transition-record\_\_cooling-and-specimen-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived history-labelled-phase-network\_\_thermal-path-to-microstructure-map\_\_time-temperature-transition-record\_\_cooling-and-specimen-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered heat treatment observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:137c5dcf16efb098c8ba33efc12ca790c51997d030f971ed5ee21538263c3e0e; derivation seal sha256:b48749c0b7e8c1220f3d810e0fe1c7315f214c907330a1c10b73fa00ced47eab; engine receipt sha256:71c2de533e261b753919a23d2dda9c99717dc6b0893a99806067e734e9ef5081 at receipts/engine/model\_admitted/SFT-MAT-PROC-HEAT-TREATMENT-001-71c2de533e261b75.json; empirical validation sha256:48550300b7b1bc18d7b5a89db1d438340b5f3332fa47c775af37b4a7e93a9529; measurement receipt sha256:ea6182376624969f7747697ca792534ee7b5b7583dc9a85812fd0f0bcfeb5341; isolation sha256:3990d3430a57e033ce1931a8557cb3f20880d2e069e3da1e186471eebee159f6; custody sha256:5136a32f8fdbfbdbbe1df064160bc81eb12efd63dcdc487f609789aaaeccff8d.

## 82. Corrosion

Claim identity: SFT-MAT-DEGR-CORROSION-001

**Question and exact theorem.** Corrosion is an interfacial reaction/transport process that transforms a material and environment jointly while retaining complete constituent, charge, product and boundary provenance.

material-environment-joint-carrier\_\_interfacial-redox-and-transport-path\_\_mass-charge-and-product-ledger\_\_environment-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-environment-joint-carrier       | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | interfacial-redox-and-transport-path     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | mass-charge-and-product-ledger           | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | environment-time-bounded                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |

| Axis       | Eliminated form                        | Forced form                           | Exact elimination / retention basis                                                                                                                                                      |
|------------|----------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-or<br>-instance-lookup | positive-finite-successor<br>-closure | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension  | free-fit-exception<br>-or-extra-rule   | no-extra-rule                         | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: material-environment-joint-carrier\_\_interfacial-re  
dox-and-transport-path\_\_mass-charge-and-product-ledger\_\_environment-time-bounded\_\_compl  
ete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-succes  
sor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 9d15174b2bb1a9c07c26c8ba5711f9cdd9bb2f8a11a30aa320ce3f46bf1fe60d.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 66247b21764c92c8fd72734d44cdcdba2d4db4e84b4faaa730823ee7a5db3327.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 6c13de46c871b855dc4deb0ad522cb2e7c51962bf71905b9717e9d62e1dec18.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 7cf442abda319bc3a9c74bd4d226b43396730c350c7f6f90566a15f15c27c7db.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: bc99ee0b1e7dd4c872735e6e261e18d1b1a6c045e6165dba52feb87aba82c3b0; certificate  
sha256: 322b3fc3c7a431f278d9a3f9d46a2322315693e645c3c955a86a8a9358fa0a8e; external-validation  
hash sha256: c1b0a450d3020e0da5e0c078254ae93c3e1f4b760c4df000c6e1da4857be82d7.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: 12b98b8c7184be61af10b2461cf587f811865f7d0f89a232f61fe44e983ae817. Target content opened after the derivation seal: **true**. All rows preserved: **true**. Exact comparison passed: **true**. The deliberately changed unfavorable

record was rejected.

Measurement-body sources:

- NIST-METAL-ADDITIVE-CORROSION-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot experiments/external\_sources/materials/snapshots/nist-metal-additive-corrosion.html; sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; scope: phase transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-degr-corrosion-001-authority-correspondence: predicted material-environment-joint-carrier\_\_interfacial-redox-and-transport-path\_\_mass-charge-and-product-ledger\_\_environment-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-environment-joint-carrier\_\_interfacial-redox-and-transport-path\_\_mass-charge-and-product-ledger\_\_environment-time-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered corrosion observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:cbe464ce8433a734a64bd9b7432682d3246b2a27c2e1dfdf8fa429a58d8d2513; derivation seal sha256:dd05f442eff002b7acbae6f66aed2befa009ad9fc2ef5adada01cdaa20bf3838; engine receipt sha256:433102762a4f8027717c9429dddf7428947bf0bb850b88acb4929dd5d6b5078c at receipts/engine/model\_admitted/SFT-MAT-DEGR-CORROSION-001-433102762a4f8027.json; empirical validation sha256:7397c246db07236a4bebe94e39a5390cbe1a75c43d58add67e2e87aca429bfff; measurement receipt sha256:bdb80356d1b616ae5ea6fb262d58658987549ab20ca590623ae3d8c4b2906183; isolation sha256:e79ebce2c1fddc766f90acbd9028cfed588fe6beb296c5bfc4cd88d717a42323; custody sha256:72c2ef2583c1c8b4bbe902480e18bf0cb56cca88d710ecb969a1f8347cf9f995.

### 83. Wear and tribological material transfer

Claim identity: SFT-MAT-DEGR-WEAR-001

**Question and exact theorem.** Wear is retained material transfer or surface reorganization produced by a declared contact/load/motion path; frictional response and material loss remain separately recorded.

contacting-surface-carriers\_\_load-motion-interface-transition\_\_debris-transfer-and-surface-ledger\_\_contact-history-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | contacting-surface-carriers              | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | load-motion-interface-transition         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | debris-transfer-and-surface-ledger       | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | contact-history-bounded                  | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |

| Axis       | Eliminated form                        | Forced form                           | Exact elimination / retention basis                                                                                                                                                      |
|------------|----------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| generality | single-specimen-or<br>-instance-lookup | positive-finite-successor<br>-closure | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension  | free-fit-exception<br>-or-extra-rule   | no-extra-rule                         | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: `contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 0efa487d64b1eee6cd7ed3a8527673f988a9ec16ecd45ae8220ccc0757354b1f.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 8229d0af59b1f56d1ea82fa4cdbbbd8e7ede783433fe74e9419d3f2bb8f61fca.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 4347b34ae82985dfadd12d02f667736bf31f2ea92d6347f510d65e4f5162297f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 4b02a5f8e1ee1b658b888159069c30bfcca22de1c25516ba7278faf493cbb6d3.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256: 247df125f5c11760b072521e72106849cde1ca0ddf036aed5c47f6b5100eb78f; certificate  
sha256: 46c5627398f978342d464c3c65e7c094a0b903406e797f56d2b374c14effe281; external-validation  
hash sha256: a2426bbe0d4a4c3e36b80076f60ad0181b8c4e09889767de1e1dbd2bc78b985f.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256: 6b5d358984148115f3731a4f470093239e7a3bde5e29e0180f773b33385efa39. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable



record was rejected.

Measurement-body sources:

- NIST - NANOTRIBOLOGY - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived>; snapshot experiments/external\_sources/materials/snapshots/nist-nanotribology.html; sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523; scope: interacting surfaces, relative motion, friction, adhesion, lubrication and wear.

Comparison and custody records:

- sft-mat-degr-wear-001-authority-correspondence: predicted contacting-surface-carriers\_\_load-motion-interface-transition\_\_debris-transfer-and-surface-ledger\_\_contact-history-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived contacting-surface-carriers\_\_load-motion-interface-transition\_\_debris-transfer-and-surface-ledger\_\_contact-history-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered wear and tribological material transfer observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:99ac90f50feb5f2af8a5030ff952e87e3438f02da297fb6bee8c61f4489fb672; derivation seal sha256:2bc4d5b3e790c3d5ea3c51b06600f9aa2470632ff440c74aacd6b3be89fa1f11; engine receipt sha256:92c996a0f5bfa825783a55376275b47e3635f67bc54552647a07899572b75f73 at receipts/engine/model\_admitted/SFT-MAT-DEGR-WEAR-001-92c996a0f5bfa825.json; empirical validation sha256:b834c07064aee6699c3fcdc85825d0c731978cca680844c88aa92ea00f0d4ab5; measurement receipt sha256:6a0798c98fc0d49a3d347d1f9652267352ca1c9c64617f6609f52c626d2f9f82; isolation sha256:857edb5a43adcea4878ef787f307a6ffd5010ed00e07ec12712474e3c5f8c7a1; custody sha256:48e96ae55cb515efe98c53c5d32c53794e7a66ffa1bb72dd0a6e1b02e1f06021.

## 84. Radiation damage

Claim identity: SFT-MAT-DEGR-RADIATION-001

**Question and exact theorem.** Radiation damage is the complete source-bound path from probe energy transfer to retained displacement, excitation or transmutation defects and their subsequent migration/recovery.

material-and-probe-carrier\_\_energy-transfer-defect-generation\_\_displacement-transmutation-and-recovery\_\_dose-rate-spectrum-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-and-probe-carrier               | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | energy-transfer-defect-generation        | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | displacement-transmutation-and-recovery  | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | dose-rate-spectrum-bounded               | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |

| Axis       | Eliminated form                    | Forced form                       | Exact elimination / retention basis                                                                                                                                                      |
|------------|------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| provenance | authority-or-target-selected-law   | root-bound-forward-forcing        | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.          |
| generality | single-specimen-or-instance-lookup | positive-finite-successor-closure | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.              |
| extension  | free-fit-exception-or-extra-rule   | no-extra-rule                     | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present. |

**Unique survivor, base and successor.** Sole survivor: material-and-probe-carrier\_\_energy-transfer-defect-generation\_\_displacement-transmutation-and-recovery\_\_dose-rate-spectrum-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 9038baa8e438b1ac2c472866e42bf6bf72685d5b5fccccafff338d1022d68b1fc.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 41a6bac5d899d6b3dc3443ea2f8ad03f200a69dc058c1c064bb1692905e50d94.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: c5a8b28f1713b2c745128747d78b44c51d07371449023bddd94d0cb6f48d0518.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 0eae8804b4960020b1078c44aeb79a2bcf0b3b24bd45b92fee471ed6121b6014.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 4f95022daaa462fc02c4d5629e2dce03074ed2a6402f7db3fe88c45f732ebe14; certificate  
sha256: 2a3ea836c86f60a0cd147b73185a1047da7c534efca61fa7978d850f566f80b3; external-validation  
hash sha256: 8e49be83d0c7ef29a09413543caed1843c57a721e1441a22efdf106c9a749ebb.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256: f6da78675cb2d1530cb7a3007c0e23d3a8e7ad00b3a3bd09478b36b62572f8ae. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - SURFACE - DAMAGE - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot experiments/external\_sources/materials/snapshots/nist-surface-damage.html; sha256: 7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; scope: surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- NIST - MULTISCALE - MATERIALS - 2026 - 07 - 24 - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256: 2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-degr-radiation-001-authority-correspondence: predicted material-and-probe-carrier\_\_energy-transfer-defect-generation\_\_displacement-transmutation-and-recovery\_\_dose-rate-spectrum-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-and-probe-carrier\_\_energy-transfer-defect-generation\_\_displacement-transmutation-and-recovery\_\_dose-rate-spectrum-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered radiation damage observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing\_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256: 6cd407968904d1edcba930dfcd606074c06ccc12d24464f820a04bf357931e8d; derivation seal sha256: 698effffcc699b74190a413a43a06ba91aa6b1d1dd5c490766e0a6d22b501ca42; engine receipt sha256: e6e36bc645b877f36b9c969dd7bcbbda391a9f6e765e806c80d5fc11e214de42 at receipts/engine/model\_admitted/SFT-MAT-DEGR-RADIATION-001-e6e36bc645b877f3.json; empirical validation sha256: ee06521307cfbf5c880200c3b4c53e271fa83da81341ee32e8eb79fba8b1e43c; measurement receipt sha256: 8b764f65f5eb032dc6e38b45a153ac5030f775a4900fc2c5164e8c19bcc87faa; isolation sha256: 72088d354ffef387f18fefdddef742e2a4eae7b9cac2a1b92dd65903dd7638b17; custody sha256: e03d344139946255498886f2ef0be18c7ccf242d053080f8aea0062802a9381a.

## 20. Advanced Functional Sustainable

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

85. Piezoelectric coupling

Claim identity: SFT-MAT-FUNC-PIEZOELECTRIC-001

**Question and exact theorem.** Piezoelectricity is a reversible cross-relation between mechanical deformation and held electric polarization permitted only when the retained material symmetry does not close the orientation distinction.

noncentrosymmetric-structure-carrier\_\_mechanical-electric-cross-response\_\_orientation-and-charge-ledger\_\_field-stress-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis     | Eliminated form                    | Forced form                          | Exact elimination / retention basis                                                                                                                                           |
|----------|------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier  | carrier-erased-or-answer-only      | noncentrosymmetric-structure-carrier | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                        |
| relation | relation-imported-fitted-or-erased | mechanical-electric-cross-response   | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| organization | organization-collapsed                 | orientation-and-charge-ledger            | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | field-stress-condition-bounded           | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256 : 8ab926031b7ddd6f9ae6723610d89598c05082b6a6aae33124e1485e2155a550.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256 : 46b8080073955bdaff587c54bb623525c892ff9137218f8adc2ce6d36b3c9278.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256 : 769c539ede54297da90196ce94b20c57bbbfa5cea6148c8d4b022abc7c36e8a9.



- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:178a33c9bd0f4b6d3e6f7842ba574a87781b2a3174b4f3c77ea99f725c82198f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash  
sha256:0edb98b42db8eb35abf9129bf24b204bde6a7b1f7723f398d61319711f842fec; certificate  
sha256:0eb2407afb1deb0d42da9134ce6101bc3eb518446e460f33ed4948cea367c142; external-validation  
hash sha256:7ea355f4b6500f3743e60bfa36f6f943be05e13ae2c759ce168cc4085e9b94ae.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is  
sha256:0199fe3b4329ace3bce2d717af5ca80589160d0648df85bf0b5b56d1a2e622b0. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24 - National Institute of Standards and Technology;  
<https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html;  
sha256:355b4893ea3808dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-piezoelectric-001-authority-correspondence: predicted noncentrosymmetric-structure-carrier\_\_mechanical-electric-cross-response\_\_orientation-and-charge-ledger\_\_field-stress-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived noncentrosymmetric-structure-carrier\_\_mechanical-electric-cross-response\_\_orientation-and-charge-ledger\_\_field-stress-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered piezoelectric coupling observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest  
sha256:cdfe88675f774917ec9e902154853bc89336ea93db2feda54d87a9ac4e61fb8d; derivation seal  
sha256:4f680e305121b953aa576a0d8655400d3fff5a059be9a6c72c1b8035da333bc0; engine receipt  
sha256:7f640ec82c5c73c47f5749e38393747843704417ad8bd9279f671aab42365d26 at  
receipts/engine/model\_admitted/SFT-MAT-FUNC-PIEZOELECTRIC-001-7f640ec82c5c73c4.json;  
empirical validation sha256:37594ac98d49f7968d62437760964d340829aacbf4ae3c9f628d17f905f8c83d;

measurement receipt sha256:87d6d92549f3c144716efa44e1286f3c5f188352f6208e71998bf895a634c516; isolation sha256:ef1082ab87a9d58062773893911c04c5a836d5e6cc67300969b03531e0478173; custody sha256:e55d04e7bca4b730de4be5419299976028216ef5ec38114ceadd743681279539.

86. Ferroelectric order and switching

Claim identity: SFT-MAT-FUNC-FERROELECTRIC-001

**Question and exact theorem.** Ferroelectricity is a material phase with stable opposed polarization recurrence classes and a retained field-driven switching path between them.

polar-domain-network\_\_stable-opposed-polarization-classes\_\_field-driven-domain-path-retained\_\_hysteresis-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves polar-domain-network, stable-opposed-polarization-classes, field-driven-domain-path-retained and hysteresis-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis    | Eliminated form               | Forced form          | Exact elimination / retention basis                                                                                                    |
|---------|-------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| carrier | carrier-erased-or-answer-only | polar-domain-network | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained. |

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relation     | relation-imported-fitted-or-erased     | stable-opposed-polarization-classes      | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | field-driven-domain-path-retained        | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | hysteresis-condition-bounded             | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: polar-domain-network\_\_stable-opposed-polarization-classes\_\_field-driven-domain-path-retained\_\_hysteresis-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:c8c27f9a90d8b0fdf120aa63794dad5be1b2554f0f0e6cd36b647d26c59abc87.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8162c61883e1b3fe67fef308ebacf2e9ed54670ae52ee7067edf8d80ca467aa7.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:d63ee927baa6c8404ff75eb6407622cdfafe2ad7d894e0556ae9b7572cd4bab7.

- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256:1f47c545d9bf91517601315bcd5136409a1a228df3c7edab9920e71bc1afb39d.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:5467315283f5f64b807d962309ddcf27cd5f1c65a26683e6d316faf565008499; certificate

sha256:e8f91344b1dc05fb3c8e81712d23b08b06cb1612c30f64a3a83a49a66b4e595f; external-validation

hash sha256:f6cc2ab5195e78512f9c6a49d2fc79945b2b79a484890d38ade80e7a93576c82.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:950537b9a096411ed993b95bfaeb1b77308569ce1b70e1834e2c82c09f6a60c2. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`;  
sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- `sft-mat-func-ferroelectric-001-authority-correspondence: predicted polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-for-ward-forcing__positive-finite-successor-closure__no-extra-rule`; source-derived `polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered ferroelectric order and switching observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered `advanced_functional_sustainable` grammar that preserves `polar-domain-network`, `stable-opposed-polarization-classes`, `field-driven-domain-path-retained` and `hysteresis-condition-bounded`.

**Immutable evidence identities.** Pre-source branch seal

`experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

sha256:da506701f72bb70612bca1f2478c1b91e30dbc5c121e71ccb91e601b080acb0d; derivation seal

sha256:8611bbfd8a418797896a462e54d5cfd398aa65d0e3bd39effa88daa38add9c80; engine receipt

sha256:36580c9520ef5c2f5e57824b854c8e16716aacb4b2165fe37a23e6a3d14fbd55 at

receipts/engine/model\_admitted/SFT-MAT-FUNC-FERROELECTRIC-001-36580c9520ef5c2f.json;  
 empirical validation sha256:74498fd9ace1ce0b625e2b44cc82a6c1ebd9faec3e735e2f19fca20778d3a743;  
 measurement receipt sha256:d6c8aaadf1d41547c53d2d042b751edc04af4f75c71c14f1bb8dd0b08fed2817;  
 isolation sha256:1d266587e8d483f234e7469691b423578536738ae724f1795770a9a17601a8cf; custody  
 sha256:4ae27b63ebc319dcd721ece0a9d989885ce280368273a53f8e1a402036bc0ad9.

## 87. Thermoelectric coupling

Claim identity: SFT-MAT-FUNC-THERMOELECTRIC-001

**Question and exact theorem.** Thermoelectric response is the joint conserved relation between energy-gradient transport and held-charge transport on the same material network.

charge-and-energy-carrier-network\_\_temperature-gradient-charge-transfer\_\_joint-energy-charge-ledger\_\_direction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | charge-and-energy-carrier-network        | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | temperature-gradient-charge-transfer     | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | joint-energy-charge-ledger               | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | direction-condition-bounded              | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: charge-and-energy-carrier-network\_\_temperature-gradient-charge-transfer\_\_joint-energy-charge-ledger\_\_direction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256:5b2e498cf9e75a44a5afbc0345af47f4deb173f28fe5dc65f0ae665ee2fe21.



- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:d7fd528b32ee51d1a154d5a48b34ed9216d7534090f6e967ecc4f8a632ad9f69.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1b3c38b099cd9b9ec5998f66e220e1435bbe784b5c2e30a5f0d2835f7280ed36.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:7c8fad072572679744edac6081724e4c31ee8f824747cf90d31041aa70856b65.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:52d94c68bc0c817a04d1992ddf14e2f8e7371008376f88e58b6b2212c7c9659c; certificate

sha256:e2af0809eb20acbef176b000e81b81f08d1d24274e74136cc94a9af4ceb83c5d; external-validation

hash sha256:3a69e07d1138a3073f22b46120a349145fdd7034830be55e85a4b4271c1f9a61.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:49468d618ab6361c0f75798af8c71ae5516ac5325bf6b6c287bd154616fb8059. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - TRANSPORT - THERMOELECTRIC - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccf34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- **sft-mat-func-thermoelectric-001-authority-correspondence**: predicted charge-and-energy-carrier-network\_\_temperature-gradient-charge-transfer\_\_joint-energy-charge-ledger\_\_direction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived charge-and-energy-carrier-network\_\_temperature-gradient-charge-transfer\_\_joint-energy-charge-ledger\_\_direction-condition-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered thermoelectric coupling observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:95f83d012fde1ad0ca6942fe7be129412ba9e0e36dbeeb433f5c4d778a9daec4; derivation seal sha256:1c5893f35485b76b913f5de741de5cf87349e4cb7eb89f62b4687a7902a62dc7; engine receipt sha256:c861a777de27b860ffce46411542047fe44e88ddd68fcae21cc9d48d658184cd at receipts/engine/model\_admitted/SFT-MAT-FUNC-THERMOELECTRIC-001-c861a777de27b860.json; empirical validation sha256:6c525fdf2f2a8591cbb0fe07899e059cc14b5446ff5d88a1b60538aa0f4d771b; measurement receipt sha256:827f8e8fa21a88f168df8e3788a9549165f2e074652d33693846f2b23bb5e6df; isolation sha256:81aad8c3ba087051b0eca6244280ea87263116cda7f57708077f8efdcc15aba9; custody sha256:edaa008f449455c72fd96ef412f4bab31a0314fc502f8bea8e75a80e3280d417.

## 88. Photonic material organization

Claim identity: SFT-MAT-FUNC-PHOTONIC-001

**Question and exact theorem.** A photonic material organizes electromagnetic recurrence support so that allowed propagation classes and structural gaps follow from complete generated geometry and boundary relations.

electromagnetic-cell-network\_\_wave-recurrence-and-gap-support\_\_geometry-frequency-and-boundary-retained\_\_spectral-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | electromagnetic-cell-network             | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | wave-recurrence-and-gap-support          | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | geometry-frequency-and-boundary-retained | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | spectral-scale-bounded                   | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: electromagnetic-cell-network\_\_wave-recurrence-and-gap-support\_\_geometry-frequency-and-boundary-retained\_\_spectral-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:c32943d095b89193d400a5b2631eca6463a05ce83ff023c092e18e7ba3433f6f.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:50acfe0be40722d399943981a25721636a151897789b475bafcd1015ad34e87.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:61b3cdacdb55eb4a710cc30fadeb2ad4d1adc5a38bc277fcf964e0cb41b326b6f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ee7373f233a08e9471fb6686dcb5b89984f07cb5f74d32b13cd25379e37c4c9b.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:22bfd80dd14b7366e546e51abc799a76a5ab463562e49218bb9c373fce71cd15; certificate

sha256:f38b1b5ef15f21b39d7a46ca08c82d87709a63df3623fc8d55d110c32d94ea34; external-validation

hash sha256:65befe7fa71ec4b61efbb9c9025b153fa0b8b56543006349263735440208c414.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:8f8bf66262959ed2118bcb56bb0a0fa706206788b7efc3da91b17c7d057f1c92. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-OPTICAL-MATERIALS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot [experiments/external\\_sources/materials/snapshots/nist-optical-materials.html](https://www.nist.gov/programs-projects/theory-optical-properties-materials/experiments/external_sources/materials/snapshots/nist-optical-materials.html); sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- **sft-mat-func-photonic-001-authority-correspondence:** predicted electromagnetic-cell-network\_\_wave-recurrence-and-gap-support\_\_geometry-frequency-and-boundary-retained\_\_spectral-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived electromagnetic-cell-network\_\_wave-recurrence-and-gap-support\_\_geometry-frequency-and-boundary-retained\_\_spectral-scale-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered photonic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:b8359125c415f70252bad5e8662f32a4a0411fd4ecab90b210f238e5d4eff3f7; derivation seal sha256:d5719938752bfff0b363ba54eb5c0394c4b5fa74ca27361fee3d5493208f569e9; engine receipt sha256:b63114cadba712eeb0a65b991315d9035f61024e44e181fc9dbb26f93b2947c7 at receipts/engine/model\_admitted/SFT-MAT-FUNC-PHOTONIC-001-b63114cadba712ee.json; empirical validation sha256:d70953f2830575c85ac5367761e35a0a7daa230e39683dbb25e951db3f69a00b; measurement receipt sha256:076b47a085e881f1daba570530905b12afa40b4cfd62cb8dd1c26da66228211a; isolation sha256:95a9d5ee8ac6b7e943fc57db2431acee31ac9e4115a590b4eae9b3b301dc8a94; custody sha256:6a0be96588221e8b3bbc5e6b3e84565643db68aecf32f4b3ee6a428c2135b7bb.

## 89. Functional magnetic material response

Claim identity: SFT-MAT-FUNC-MAGNETIC-001

**Question and exact theorem.** A functional magnetic response is the complete relation among applied field, domain transition path, retained orientation, loss and remanence at declared frequency and temperature.

moment-domain-and-field-carrier\_\_field-history-response-map\_\_domain-wall-loss-and-remanence\_\_frequency-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | moment-domain-and-field-carrier          | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | field-history-response-map               | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | domain-wall-loss-and-remanence           | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | frequency-temperature-bounded            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt



sha256:db19f1a2cc9d5f45d18837e5d0172d17a4f6e0b27b08b0096dc0622a8dfec16.

- **tampered\_source**: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5ee5c3258afeb367bb0cd4365b9db18fdb8207765c46dad0c76f440d2f2a0614.
- **tampered\_artifact**: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:ae310bd447bd0da6fef8f21bcd1ce0c0df36b3523c38cc3dff2dd5e5be48d62f.
- **boundary**: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:334673e317625efa1bbb433ce8d7cdec742cf726cb5311cb9c5b0b1bbd9bd85f.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:cd8911fe8913dbfca7c344e88e4917190bbc07e7d04e1403ca7f513d06fce2f7; certificate

sha256:a0b1af464cab9fc6344d8193cfe38a2651617f9c887720c78ad0eae55b49a065; external-validation

hash sha256:d28ec86f9c2ea983d46ed613c328135ead66046bc3f0baf7bb15769aeafd3b8.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:8ab5d335d373302646984a94eb1f45dccc0a137eea7206abf17029cac4627ff1. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026 - 07 - 24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- **sft-mat-func-magnetic-001-authority-correspondence**: predicted moment-domain-and-field-carrier\_\_field-history-response-map\_\_domain-wall-loss-and-remanence\_\_frequency-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived moment-domain-and-field-carrier\_\_field-history-response-map\_\_domain-wall-loss-and-remanence\_\_frequency-temperature-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered functional magnetic material response observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:6d22b16de46e8b5d63aa64b2a61e80240e18216a5feb89964e7099a2854d017c; derivation seal sha256:395244dc506ff152b9c09887ab33883310f04880d049893f0e8d4cd625914bed; engine receipt sha256:1ed29896c9091b1948b2d855f0c0d51738df688e864db22fcde9f344da2dfca4 at receipts/engine/model\_admitted/SFT-MAT-FUNC-MAGNETIC-001-1ed29896c9091b19.json; empirical validation sha256:b117f217af70207aaacd9b09bb785b027e22a059780fa54a291a85793545b6cc; measurement receipt sha256:5d4519d974b7c1ab4790a14e7d29d2343daa1b3b423846555bf212c4f5e7716a; isolation sha256:58661800655851db8c9f5dc1c1717efb27bcdcf0624e4d19967f68b2400e53f2; custody sha256:2b92d674f030c9acc3cb3072a05f64cf6665bb62637a7c91f52e7aad01229652.

## 90. Nanoscale material boundary

Claim identity: SFT-MAT-FUNC-NANOMATERIAL-001

**Question and exact theorem.** A nanomaterial regime begins when boundary and finite-support distinctions remain in the material state/property census and cannot be closed by the bulk observation quotient.

finite-constituent-boundary-carrier\_\_boundary-support-comparable-to-interior\_\_discrete-state-and-interface-retained\_\_size-shape-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and size-shape-resolution-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | finite-constituent-boundary-carrier      | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | boundary-support-comparable-to-interior  | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | discrete-state-and-interface-retained    | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | size-shape-resolution-bounded            | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: `finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:31c76c3f386bed3a498183b0f457baf72566e4999e3bbfde6be9cf01415a227d.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:dc40a0ec44efafc6e16bc5818d4e5876e2169c4a4d15e75a7c44b057a6232ce2.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:42da32bcf34d016e4fb04da37d8e1a77bed1c2942845d0bbdb58abb5d23c657f.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e95636207c13c2d934d5cc7503c8af418875ef0267cb94376f56979200802167.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:69caf8202ee5680eb52cb0ab81104a75f12ccba87e08f079578e9953f1ba93e; certificate

sha256:77995cabcd5179d50722c7318e402d37e31b3cbe2ac7ad4afa6a53c415d9547; external-validation

hash sha256:5a9764c3798b5bc1028e1d681d140e937971b971a0207684cd14a01d4c057987.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 2 claim-specific source discriminators were required; their ordered identity is

sha256:453e167dbba6de5524a9410a9889cf4b02c39e7b2c962716c0a287338c24ee3e. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology; <https://www.nist.gov/mml/nano-measurement-protocols>; snapshot [experiments/external\\_sources/materials/snapshots/nist-nano-protocols.html](https://www.nist.gov/mml/nano-measurement-protocols/experiments/external_sources/materials/snapshots/nist-nano-protocols.html); sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-func-nanomaterial-001-authority-correspondence: predicted finite-constituent-boundary-carrier\_\_boundary-support-comparable-to-interior\_\_discrete-state-and-interface-retained\_\_size-shape-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived finite-constituent-boundary-carrier\_\_boundary-support-comparable-to-interior\_\_discrete-state-and-interface-retained\_\_size-shape-resolution-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered nanoscale material boundary observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and

size-shape-resolution-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:a772561a12e9f1e33384ed3b074205b589ac789c8348c602e27c320ce8c2b925; derivation seal sha256:2cc9bc4362cbc4dee60c592336a2df482696e5330953579100fc5e1f15ceb9b1; engine receipt sha256:a815d0fcf264919778c652d76e2b46dd60f5f75df6e79c325e98bc7192cc85b3 at receipts/engine/model\_admitted/SFT-MAT-FUNC-NANOMATERIAL-001-a815d0fcf2649197.json; empirical validation sha256:092002f7236e19aaf27f182f5f1ae15d177a80a287affd058a15195835790be3; measurement receipt sha256:3d6f316b710b89615ba3a69f4c6b1bfff427dcfdbf675756a2f8359d144c7f311; isolation sha256:35d270615bc1e69314b08304d5ffda3838eaf6b7e856a0a86c3d8d89b2c1b02c; custody sha256:7d55c2da7e6abd7970c8a0ba99fd51def81599e28c42cc318be5f4e2b0bcaa3a.

## 91. Biomaterial interface boundary

Claim identity: SFT-MAT-FUNC-BIOMATERIAL-001

**Question and exact theorem.** A biomaterial claim is a joint material/biological interface relation with explicit exposure, transfer, response and function boundaries; biological outcomes cannot select the material law.

material-biological-interface-carrier\_\_bidirectional-response-and-transfer\_\_material-and-biological-identities-retained\_\_exposure-function-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                                 | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | material-biological-interface-carrier       | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | bidirectional-response-and-transfer         | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | material-and-biological-identities-retained | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | exposure-function-bounded                   | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace    | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing                  | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure           | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                               | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: material-biological-interface-carrier\_\_bidirectional-response-and-transfer\_\_material-and-biological-identities-retained\_\_exposure-function-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed **true**.
- **relation:** the generated relation and organization are separately retained; passed **true**.
- **observation:** the observation boundary is distinct from the material organization; passed **true**.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt



sha256:c77ecdcc4cee7390b374f11dac9a22f81f990b7789c59a924c07fe55f79a889e.

- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:cb1ea4682cb262aad2796a49a64cedd62acb16f9534428567b7cd8dc5a118235.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:1d101e88432f57335048d4d9f0a2b09fe305b1cf62dd6fb95fe211598e2553a4.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:da528078953f494ae3b594ae054f42104f1e21c78ec26b6bae2a906ca4dc1689.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256:01f66c16a5ba692961fbabd31a4335924f06b60bb42767fddf09c7682a57a104; certificate

sha256: eaa0c22c327dabffe543439a4565a94849959aebfa0e4fe2e9c48504b4de0085; external-validation

hash sha256:4f1fda699d5d3bb69d2a10eac5fcdeb43aa1257bb54bf96f88d3f002556b7209.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256:bf5574013b2ace3aec4970b1b058bd142529651a68815439d9d26d155166b6f2. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-NANO-PROTOCOLS-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/mml/nano-measurement-protocols>; snapshot [experiments/external\\_sources/materials/snapshots/nist-nano-protocols.html](https://www.nist.gov/mml/nano-measurement-protocols/experiments/external_sources/materials/snapshots/nist-nano-protocols.html); sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.
- **NIST-POLYMER-COMPOSITES-2026-07-24** - National Institute of Standards and Technology; <https://www.nist.gov/programs-projects/polymer-composites>; snapshot [experiments/external\\_sources/materials/snapshots/nist-polymer-composites.html](https://www.nist.gov/programs-projects/polymer-composites/experiments/external_sources/materials/snapshots/nist-polymer-composites.html); sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- **sft-mat-func-biomaterial-001-authority-correspondence:** predicted material-biological-interface-carrier\_\_bidirectional-response-and-transfer\_\_material-and-biological-identities-retained\_\_exposure-function-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived material-biological-interface-carrier\_\_bidirectional-response-and-transfer\_\_material-and-biological-identities-retained\_\_exposure-function-bounded\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered biomaterial interface boundary observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

**Explicit exclusions.**

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule

- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:1f6cf609fdd14e3ef9884853b588680c81246b326cb6628497bb7524f680b695; derivation seal sha256:e04744de5df085dc908548847c8701d22343f9735c7e28a67250206beeddc22f; engine receipt sha256:5ed0cf28440ad78632017718d57debfcdda937fdc7acb83e53a782aefad24d3 at receipts/engine/model\_admitted/SFT-MAT-FUNC-BIOMATERIAL-001-5ed0cf28440ad786.json; empirical validation sha256:9810925c7e30dd79c38a3577eca33c9fbd681aded4f9d2526db42d8ebbf8359c; measurement receipt sha256:b61bf887f8ab6b53a09748fbffeff3dad5477ea07a50a081cc25e46e1899ac14; isolation sha256:d4685e1d38fb5855d3c4519cb602ff7498c3fc36794645bd703176a74392c664; custody sha256:c8acfc9ecf4861523b5dcfff4402a370a9b10fde0758517ffbf0bd42fbd9ed50.

## 92. Material lifecycle and recyclability

Claim identity: SFT-MAT-SUST-LIFECYCLE-001

**Question and exact theorem.** A material lifecycle is the complete source-to-processing-to-use-to-recovery flow ledger; recyclability requires an explicit identity-preserving recovery path and records every transformation, loss and quality boundary.

source-process-use-recovery-carrier\_\_complete-material-flow-path\_\_loss-transformation-and-quality-ledger\_\_system-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule

**Rooted dependency chain.** The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

**Generated grammar and boundary.** Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

| Axis         | Eliminated form                        | Forced form                              | Exact elimination / retention basis                                                                                                                                                               |
|--------------|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| carrier      | carrier-erased-or-answer-only          | source-process-use-recovery-carrier      | Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.                                                            |
| relation     | relation-imported-fitted-or-erased     | complete-material-flow-path              | An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.                     |
| organization | organization-collapsed                 | loss-transformation-and-quality-ledger   | Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held. |
| observation  | observation-boundary-unrecorded        | system-boundary-declared                 | An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.                          |
| record       | result-without-state-transition-record | complete-state-transition-boundary-trace | A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.                     |
| provenance   | authority-or-target-selected-law       | root-bound-forward-forcing               | Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.                   |
| generality   | single-specimen-or-instance-lookup     | positive-finite-successor-closure        | One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.                       |
| extension    | free-fit-exception-or-extra-rule       | no-extra-rule                            | A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.          |

**Unique survivor, base and successor.** Sole survivor: source-process-use-recovery-carrier\_\_complete-material-flow-path\_\_loss-transformation-and-quality-ledger\_\_system-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth\_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

#### Operational witnesses.

- **carrier:** the generated carrier label is held and nonempty; passed `true`.
- **relation:** the generated relation and organization are separately retained; passed `true`.
- **observation:** the observation boundary is distinct from the material organization; passed `true`.

**Detailed mathematical and physical meaning.** The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

#### Adverse controls.

- **false\_premise:** passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt  
sha256: 42532e43e708387eab368ef316263c2ed870195ef765bab6a76e7b16dd4a334b.
- **tampered\_source:** passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256: 1360d7af1f0267dfc482885e6b6289c9a952d85d977db53826e457f0fe3d337a.
- **tampered\_artifact:** passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt  
sha256: 599f2eca8225f1d225f32ee678bb4237582333e452455db62034e12fdbfc6981.
- **boundary:** passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt  
sha256: 29bef3fbd7454e3857b3b0be39cd76cdc6725365e836fe943c4c620e0cc2da2d.

**Independent implementation.** The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash

sha256: 8c2a8e8674efd78cc9aeb6e6ff178591c51fdae5c0f7573769bd763ad4c69bca; certificate

sha256: 7271f451ff5b28ae59d344e9b829463616b6e8225c85562520df817c592ef606; external-validation

hash sha256: a34eca90ac5590c6a6444768b73d44224b164109b923c033cce4da264b285775.

**Post-seal empirical correspondence.** This claim belonged either to the original complete 84-law seal or the complete eight-law V1/V2 reconciliation seal; in both cases its entire prediction family was fixed before its external source identities were selected.

For this claim, 4 claim-specific source discriminators were required; their ordered identity is

sha256: d1aa7f6fa9c4241ced6e2ef6f2f5900b28d79e8a963460e50d55a0fabda4f634. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- **NIST-MATERIALS-DIVISION-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot  
experiments/external\_sources/materials/snapshots/nist-materials-division.html;  
sha256: fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; scope: materials  
measurement, composition, interfaces, processing, damage, classes and lifecycle.
- **NIST-NANO-PROTOCOLS-2026-07-24** - National Institute of Standards and Technology;  
<https://www.nist.gov/mml/nano-measurement-protocols>; snapshot  
experiments/external\_sources/materials/snapshots/nist-nano-protocols.html;  
sha256: 366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial  
property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- **sft-mat-sust-lifecycle-001-authority-correspondence:** predicted source-process-use-recovery-carrier\_\_complete-material-flow-path\_\_loss-transformation-and-quality-ledger\_\_system-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; source-derived source-process-use-recovery-carrier\_\_complete-material-flow-path\_\_loss-transformation-and-quality-ledger\_\_system-boundary-declared\_\_complete-state-transition-boundary-trace\_\_root-bound-forward-forcing\_\_positive-finite-successor-closure\_\_no-extra-rule; exact match `True`
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

**Falsification boundary:** The claim fails if a complete registered material lifecycle and recyclability observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

#### Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced\_functional\_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

**Immutable evidence identities.** Pre-source branch seal

experiments/sealed\_predictions/materials\_complete\_branch\_pre\_source.json; source manifest sha256:ae6eca4c43b073a3ed7eda20aa9836c009c9bb3b05a6f443ab2d24738fd48e57; derivation seal sha256:3e9a7308036c4ef8d2b7fc56f68e6b33cfff28eab1ed004ae9f243609dc70434e; engine receipt sha256:0046611012fa957e6166fede124a4a43e24de052957566bb7411afb54ed2bc43 at receipts/engine/model\_admitted/SFT-MAT-SUST-LIFECYCLE-001-0046611012fa957e.json; empirical validation sha256:06b1e6c1f4a57c40bb00be257aff0e6c57a0e88208a7fb723ec9fb26800cfbc4; measurement receipt sha256:4feb710f4285d4f1a27cabaeca9e8b34fe808c3da8ccd7e9033c74c077225620; isolation sha256:b568d984120dc112f697754f4fb3c44d6f111e8acca6c1dc953243ff61c6f1; custody sha256:33b0dc3bcf6a48ff0c1dd145f2729a0d5e168d02649d0ced23e6809078094545.

## 21. Integrated reconciliation table

- Step 47: cubic lattice and six nearest neighbours -> SFT-MAT-CRYST-CUBIC-COORDINATION-001
- Step 49: crystalline order and crystallographic restriction -> SFT-MAT-CRYST-TRANSLATION-001, SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-SYSTEMS-001, SFT-MAT-CRYST-BRAVAIS-001
- Step 52: paired-carrier superconducting response -> SFT-MAT-SC-PAIR-001, SFT-MAT-SC-ZERO-RESISTANCE-001, SFT-MAT-SC-MEISSNER-001, SFT-MAT-SC-FLUX-QUANTIZATION-001, SFT-MAT-SC-JOSEPHSON-001
- Step 54: electronic bands, gaps and transport classes -> SFT-MAT-ELEC-BAND-GAP-001, SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- Step 72: three acoustic lattice-excitation branches -> SFT-MAT-CRYST-PHONON-001
- Step 74: aligned and opposed magnetic order -> SFT-MAT-MAG-FERROMAGNETISM-001, SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- Step 75: semiconductor carrier types, doping and junction balance -> SFT-MAT-ELEC-CARRIER-DUALITY-001, SFT-MAT-SEMI-DOPING-001, SFT-MAT-SEMI-PN-TYPE-001, SFT-MAT-SEMI-JUNCTION-001, SFT-MAT-SEMI-TRANSPORT-001
- Step 133: quasicrystalline order without periodic translation -> SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-RECIPROCAL-001, SFT-MAT-CRYST-QUASICRYSTAL-001
- Step 137: superfluid flow and quantized circulation -> SFT-MAT-SF-SUPERFLUID-001, SFT-MAT-SF-CIRCULATION-001
- Step 143: topological class and bulk-boundary protection -> SFT-MAT-TOPO-INVARIANT-001, SFT-MAT-TOPO-BULK-BOUNDARY-001
- Step 193: elastic, plastic, slip, strength, fracture, fatigue and creep response -> SFT-MAT-MECH-STRESS-STRAIN-001, SFT-MAT-MECH-ELASTICITY-001, SFT-MAT-MECH-PLASTICITY-001, SFT-MAT-MECH-SLIP-001, SFT-MAT-MECH-MODULUS-001, SFT-MAT-MECH-STRENGTH-HARDNESS-001, SFT-MAT-MECH-FRACTURE-001, SFT-MAT-MECH-FATIGUE-CREEP-001

- Step 291: quasicrystal component of the mixed materials-and-astrophysics step -> SFT-MAT-CRYST-QUASICRYSTAL-001; routed remainder: Planetary and Tully-Fisher claims are not Materials results and remain assigned to Astronomy and Cosmology.

## 22. External-source ledger

- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24 - <https://jcg.m.bipm.org/vim/en/2.41.html>; snapshot experiments/external\_sources/materials/snapshots/bipm-vim-traceability.html; sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4; measurement result, reference chain, calibration and uncertainty.
- NIST-REFERENCE-MATERIALS-2026-07-24 - <https://www.nist.gov/reference-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-reference-materials.html; sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; material identity, property characterization, uncertainty and traceability.
- NIST-MATERIALS-DIVISION-2026-07-24 - <https://www.nist.gov/mml/materials-science-and-engineering-division>; snapshot experiments/external\_sources/materials/snapshots/nist-materials-division.html; sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; materials measurement, composition, interfaces, processing, damage, classes and lifecycle.
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - [https://www.ctcms.nist.gov/wulffman/docs\\_1.2/](https://www.ctcms.nist.gov/wulffman/docs_1.2/); snapshot experiments/external\_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; lattice generation, rotation, seven crystal systems and fourteen Bravais classes.
- NIST-BRAVAIS-CLASSIFICATION-2026-07-24 - <https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification>; snapshot experiments/external\_sources/materials/snapshots/nist-bravais-classification.html; sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; diffraction classification across all fourteen Bravais lattices.
- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24 - <https://www.nist.gov/nist-and-nobel/dan-shechtman>; snapshot experiments/external\_sources/materials/snapshots/nist-shechtman-quasicrystals.html; sha256:911c06e466a95d58749422c56cd4e33fa1fbee04ec06977af428d7dac23ae0e; periodic rotation orders and fivefold quasicrystal observation.
- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - <https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external\_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffbf1e8d93ea5c0043249376c4d6ff18; microscope confirmation, nonrepeating order and fivefold symmetry.
- NIST-POINT-DEFECTS-2026-07-24 - <https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides>; snapshot experiments/external\_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eae5a5a29d99ea4436b691f77c17c74a457acb6acf88335a; vacancies, interstitials, substitutions, diffusion, composition and defect metrology.
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - <https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction>; snapshot experiments/external\_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; grains, phase fractions, texture, deformation and uncertainty.
- NIST-MULTISCALE-MATERIALS-2026-07-24 - <https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.



- NIST - SEMICONDUCTORS - 2026-07-24 - <https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors>; snapshot experiments/external\_sources/materials/snapshots/nist-semiconductors.html; sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; semiconductor structural, nanomechanical, thermal and spectroscopic measurement.
- NIST - FUNCTIONAL - ELECTRONIC - MATERIALS - 2026-07-24 - <https://www.nist.gov/programs-projects/genomics-electronic-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.
- NIST - SUPERCONDUCTING - GRAIN - BOUNDARIES - 2026-07-24 - <https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature>; snapshot experiments/external\_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; grain boundaries, strain, dislocations, current flow and superconducting response.
- NIST - SUPERCONDUCTING - FLUX - 2026-07-24 - <https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting>; snapshot experiments/external\_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; zero resistance, Meissner expulsion, quantized flux and vortices.
- NIST - JOSEPHSON - STANDARD - 2026-07-24 - <https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard>; snapshot experiments/external\_sources/materials/snapshots/nist-josephson.html; sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; superconductive tunnel junctions and quantized Josephson voltage.
- NIST - SUPERFLUID - CIRCULATION - 2026-07-24 - <https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid>; snapshot experiments/external\_sources/materials/snapshots/nist-superfluid-circulation.html; sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; superfluid flow, vortices and discrete quantized circulation.
- NIST - TOPOLOGICAL - INSULATORS - 2026-07-24 - <https://www.nist.gov/programs-projects/topological-insulators>; snapshot experiments/external\_sources/materials/snapshots/nist-topological-insulators.html; sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; topological invariants, insulating bulk, band gap and metallic boundaries.
- NIST - FATIGUE - FRACTURE - 2026-07-24 - <https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture>; snapshot experiments/external\_sources/materials/snapshots/nist-fatigue-fracture.html; sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; processing-structure-property-performance, defects, fatigue and fracture.
- NIST - POLYMER - COMPOSITES - 2026-07-24 - <https://www.nist.gov/programs-projects/polymer-composites>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.
- NIST - TRANSPORT - THERMOELECTRIC - 2026-07-24 - <https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716ccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST - CERAMIC - ADDITIVE - 2026-07-24 - <https://www.nist.gov/programs-projects/additive-manufacturing-ceramics>; snapshot experiments/external\_sources/materials/snapshots/nist-ceramic-additive.html; sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; ceramic feedstock, defects, sintering, densification, phases and microstructure.

- **NIST - POLYMER - PROCESSING - 2026 - 07 - 24 -**  
<https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-processing.html; sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18; polymer processing, crystallization, rheology, kinetics and recycling.
- **NIST - SURFACE - DAMAGE - 2026 - 07 - 24 -**  
<https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project>; snapshot experiments/external\_sources/materials/snapshots/nist-surface-damage.html; sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- **NIST - NANO - PROTOCOLS - 2026 - 07 - 24 -** <https://www.nist.gov/mml/nano-measurement-protocols>; snapshot experiments/external\_sources/materials/snapshots/nist-nano-protocols.html; sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.
- **NIST - SOLIDIFICATION - 2026 - 07 - 24 -** <https://www.nist.gov/publications/solidification-0>; snapshot experiments/external\_sources/materials/snapshots/nist-solidification.html; sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c; solidification transport, nucleation, growth, interfaces, segregation and processing.
- **NIST - METAL - ADDITIVE - CORROSION - 2026 - 07 - 24 -**  
<https://www.nist.gov/programs-projects/additive-manufacturing-metals>; snapshot experiments/external\_sources/materials/snapshots/nist-metal-additive-corrosion.html; sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322; phase transformation, heat treatment, corrosion and environmental cracking.
- **NIST - NANOTRIBOLOGY - 2026 - 07 - 24 -**  
<https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived>; snapshot experiments/external\_sources/materials/snapshots/nist-nanotribology.html; sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523; interacting surfaces, relative motion, friction, adhesion, lubrication and wear.
- **NIST - OPTICAL - MATERIALS - 2026 - 07 - 24 -** <https://www.nist.gov/programs-projects/theory-optical-properties-materials>; snapshot experiments/external\_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; index of refraction, absorption, reflectance, dielectric response and photonic crystals.
- **NIST - SIX - NEIGHBOUR - MESH - 2026 - 07 - 24 -**  
[https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard\\_Oxs\\_Ext\\_Child\\_Clas.html](https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard_Oxs_Ext_Child_Clas.html); snapshot experiments/external\_sources/materials/snapshots/nist-six-neighbour-mesh.html; sha256:505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2; six nearest neighbours as forward/backward positions on three coordinate axes.
- **NIST - POLYMER - INTERFACE - CONSORTIUM - 2026 - 07 - 24 -**  
<https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium>; snapshot experiments/external\_sources/materials/snapshots/nist-polymer-interface-consortium.html; sha256:2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635; surface damage, interfacial adhesion and interface-property measurement.
- **NIST - SEMICONDUCTOR - CARRIERS - 2026 - 07 - 24 -**  
<https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light>; snapshot experiments/external\_sources/materials/snapshots/nist-semiconductor-carriers.html; sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; electron and hole carrier signs, doping, transport and Hall measurement.
- **NIST - ANTIFERROMAGNETIC - COUPLING - 2026 - 07 - 24 -**  
<https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding>; snapshot experiments/external\_sources/materials/snapshots/nist-antiferromagnetic-coupling.html; sha256:b8ce5105622c02b84bfef643693c05c199df57beae79bd9c5e29f26c4360e85e; polarized-neutron

measurement of magnetic structure and antiferromagnetic coupling.

- NIST-GLASS-TRANSITION-2026-07-24 - <https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics>; snapshot experiments/external\_sources/materials/snapshots/nist-glass-transition.html; sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777; thermodynamic and kinetic glass-transition measurement in glass-forming materials.
- NIST-QUASICRYSTAL-APERIODIC-2026-07-27 - <https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal>; snapshot experiments/external\_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; post-seal claim-specific authoritative comparison.
- NIST-PHONON-THERMAL-LIMITS-2026-07-27 - <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication960-11.pdf>; snapshot experiments/external\_source s/materials/snapshots/nist-phonon-thermal-limits-2026-07-27.txt; sha256:208d3ff48584c6548ccb00d3b1559ba63d66b1fabe027af9bb499504d5cad64b; post-seal claim-specific authoritative comparison.
- NIST-PN-RECTIFICATION-2026-07-27 - <https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir4414.pdf>; snapshot experiments/external\_sources/materials/snapshots/nist-pn-rectification-2026-07-27.txt; sha256:ca301b8fc47ba509909982ca2b077c238b92f3ff4978ed7377d086071c519ba4; post-seal claim-specific authoritative comparison.
- NIST-SUPERCONDUCTING-ISOTOPE-2026-07-27 - [https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147\\_A1b.pdf](https://nvlpubs.nist.gov/nistpubs/jres/094/jresv94n3p147_A1b.pdf); snapshot experiments/external\_sources/materials/snapshots/nist-superconducting-isotope-2026-07-27.txt; sha256:48fa89f9f63d467e9259e766f81fd8f854924ad479ec0006d14075cfe48d3ae3; post-seal claim-specific authoritative comparison.
- NIST-FERRIMAGNETIC-SUBLATTICES-2026-07-27 - <https://www.nist.gov/publications/thickness-dependent-magnetism-variation-ferrimagnetic-rare-earth-transition-metal-fe1>; snapshot experiments/external\_sources/materials/snapshots/nist-ferrimagnetism-2026-07-27.html; sha256:685872920e5663b2ec71636b953d0e7ddec3d1cb679fb4f17bb94b6d6d811795; post-seal claim-specific authoritative comparison.
- NIST-HALL-QUANTIZATION-2026-07-27 - <https://www.nist.gov/programs-projects/integer-and-fractional-quantum-hall-effect>; snapshot experiments/external\_sor ces/materials/snapshots/nist-hall-quantization-2026-07-27.html; sha256:4536e34ab401e72216f8b9010bafc9f1cacdbbfc47ca8fc998b781fa9125994c; post-seal claim-specific authoritative comparison.
- NIST-WATER-THERMOPHYSICAL-2026-07-27 - <https://webbook.nist.gov/cgi/cbook.cgi?ID=C7732185&Mask=FFF&Units=CAL>; snapshot experiments/external\_sources/materials/snapshots/nist-water-complete-2026-07-27.html; sha256:ba9fc5225bc8d0d89cbe15703bff2d9dc06ef64244598c420c7666931f1befbf; post-seal claim-specific authoritative comparison.
- NIST-LIQUID-WATER-DENSITY-2026-07-27 - <https://www.nist.gov/document/jpcrd38200921ppdf>; snapshot experim ents/external\_sources/materials/snapshots/nist-liquid-water-properties-2026-07-27.txt; sha256:b3af256bcffd7ff87fbf2aa00f545910a531d5a1b285fa9bb9f926f7afad9e24; post-seal claim-specific authoritative comparison.
- NIST-ICE-DENSITY-2026-07-27 - [https://tsapps.nist.gov/publication/get\\_pdf.cfm?pub\\_id=926353](https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=926353); snapshot experiments/external\_sources/materials/snapshots/nist-ice-properties-2026-07-27.txt; sha256:7073b61255cf73163595c9ca25646dcf2c323dd2aca942d797f9f0e4dd6c178e; post-seal claim-specific authoritative comparison.

## 23. Reproducibility and falsification

The repository's one-command validation route parses every schema, verifies snapshot and source hashes, checks the frozen inventory, exercises the engine and replays every admitted execution manifest entry. The Materials publication gate separately requires all 92 evidence packages, complete candidate/decision parity, one survivor each, four passing controls each, 92 empirical validations, paper inclusion of each receipt identity, both pre-source seals, atomic V1/V2 reconciliation, evidence maps and a rendered PDF.

The branch fails if any registered source hash changes without an explicit new version, any required discriminator is absent, any prediction opens a target before sealing, any row is omitted, any tampered record passes, any dependency no longer resolves to the root, any census is incomplete, any claim has other than one survivor, or an independent validator fails to reconstruct the result.

## 24. Limitations

- Closure is relative to the declared positive-finite grammars and question surface, not every conceivable future material.
- The external correspondence records are independent tests, not derivational premises.
- Property laws are conditional on specimen, preparation, direction, method, scale, history and environment; this paper does not claim universal numerical moduli, conductivities or thresholds.
- Biomaterial claims stop at the material/biological interface. Biological functions and clinical outcomes require their own branches.
- Industrial performance can test a law but cannot select it.

## 25. Conclusion

The Materials foundation is current-evidence closed and extension-open at its frozen V3 successor boundary: 92 required laws, 23,552 exact candidates and decisions, 92 unique survivors, 368 adverse controls, 92 independent reconstructions, 92 post-seal authority checks and 92 root traces. The result is an openly inspectable computational account of material identity, organization, defects, collective response, classes, processing, degradation and function. Later lawful discoveries may extend or falsify this boundary; they may not be excluded merely because the present census is complete. Its evidential authority lies in the complete public chain from the premise-free root theorem to immutable receipts and falsifiable external checks.

## 26. Publication and repository links

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Materials release: <https://github.com/MettaMazza/ernos-labs-sft-platform/releases/tag/materials-v1.0.0>
- Zenodo DOI: <https://doi.org/10.5281/zenodo.21629306>
- Author: Maria Smith, Ernos Labs
- Contact: [Maria.Smith.Sftoe@gmail.com](mailto:Maria.Smith.Sftoe@gmail.com)
- Submissions: <https://discord.gg/ucwGryVxGr>

## 27. References

- Bureau International des Poids et Mesures / Joint Committee for Guides in Metrology, International Vocabulary of Metrology.
- National Institute of Standards and Technology, Materials Science and Engineering Division records listed in the external-source ledger.
- Smith, Maria. *From Nothing to Fold*. Ernos Labs Foundation Branch Paper 001. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *After Turing: The Fold Machine*. doi:10.5281/zenodo.21518311.
- Smith, Maria. *The Quantum Fold Machine*. doi:10.5281/zenodo.21518313.

- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Chemistry*. doi:10.5281/zenodo.21531455.